

The logarithmic deficit of 120-avoiding ascent sequences (OEIS A202061)

From the order of growth to every fixed inverse-logarithmic order

A merged research report built from five AI-assisted manuscripts
(batch 77, manuscripts 39, 41, 40, 42 and 38)

Manuscripts dated 1–2 October 2026; merged 2 October 2026

Abstract

Let a_n count ascent sequences avoiding the classical pattern 120 (OEIS A202061), let $\mu = 7.2958969432\dots$ be the largest root of $\mu^3 - 8\mu^2 + 5\mu + 1$, and put $D_n = n \log \mu - \log a_n$, $L = \log n$ and $F = n^{1/3}L^{2/3}$. This report prints five manuscripts, each refining the one before, as five Parts. Part I proves an exact positive height-walk representation, $D_n = \Theta(F)$, the order of the inverse correction, and that the generating function is not D-finite. Part II proves $D_n/F \rightarrow C_\star = (3\pi^2\alpha^2/(2v))^{1/3} = 2.2326253076\dots$. Part III adds the term $(7C_\star/3)F \log L/L$ with an $O(F/L)$ remainder and an explicit finite-height constant. Part IV identifies the next constant, $C_\star\kappa F/L$ with $\kappa = -1.9401481831\dots$, proving a conjecture stated in Part III. Part V proves, for every fixed J , $D_n/(C_\star F) = \sum_{j \leq J} P_j(\kappa + \frac{7}{3} \log L)L^{-j} + o(L^{-J})$ with universal polynomials P_j , and the matching inverse expansion. No multiplicative equivalent, amplitude, power-law prefactor, exponentially complete transseries or effective threshold is claimed. The report is AI-assisted and unrefereed, and none of it is formalized.

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Guide to this report

This guide was written when the five manuscripts were merged. Everything after it, from Part I on, is the manuscripts’ text, except for the passages marked **[write, 2 October 2026]** and the changes listed in the subsection “Conventions of the merge” below.

What the report proves

- **Part I** (source 39). An exact normalized state recurrence, a positive height-walk representation with a positive binomial block formula, and Theorem 1.1: $\mu^n e^{-C\Phi(n)} \leq a_n \leq C\mu^n e^{-c\Phi(n)}$ with $\Phi(n) = F$. Corollary 1.2: no pure stretched equivalent with exponent $\geq 1/3$ (which rules out the pure $n^{3/8}$ stretch estimated numerically in [2]); $N(Y) = \log Y / \log \mu + \Theta((\log Y)^{1/3}(\log \log Y)^{2/3})$; and $A(x)$ is not D-finite.
- **Part II** (source 41). Theorem 10.1: $D_n/F \rightarrow C_\star$; the finite-height radius shift $\log(\rho_H/\rho) \sim (\alpha/2) \log H/H$; the leading inverse constant $C_\star \lambda^{-4/3} = 0.8935682576 \dots$.
- **Part III** (source 40). Theorem 16.1: $D_n = C_\star F + (7C_\star/3)F \log L/L + O(F/L)$; the inverse Corollary 20.1; the finite-height radius with explicit constant $c_\star = 0.8667962458 \dots$, equation (21.1); and a conditional conjecture (23.1) for the next constant.
- **Part IV** (source 42). Theorem 24.1: the term $C_\star \kappa F/L + o(F/L)$, which proves that conjecture; Corollary 28.1: the inverse constant $\kappa_{\text{inv}} = -2.3980035405 \dots$; a variational (action) statement at that precision.
- **Part V** (source 38). Theorem 30.1: every fixed inverse-logarithmic order, with universal polynomials P_j of degree j , $[B^j]P_j = \binom{2/3}{j}(3/2)^j$, a finite beta-moment recurrence and explicit $P_1, \dots, P_4$; Theorem 32.1 (reduction to a classical action); Theorem 35.1 (every fixed order of the inverse).

Sources and provenance

The report was built from five manuscripts, all delivered in the arrival commit `096ee7b87` of 2 October 2026 as batch 77, manuscripts 39, 41, 40, 42 and 38, and placed in this directory by commit `f76fcb566`. Each author line reads “A proof and reproducible research report” (source 39) or “A proof and reproducible research companion” (sources 41, 40, 42, 38); no tool or person is named.

Part	Source	Archive	Main file	Dated	Labels
I	39	<code>oeis-a202061-report.zip</code>	<code>a202061-report.tex</code>	2 Oct ^a	<code>a61:fd:</code>
II	41	<code>oeis-a202061-sharp-deficit.zip</code>	<code>a202061-sharp-deficit.tex</code>	2 Oct	<code>a61:sh:</code>
III	40	<code>oeis-a202061-second-order-report.zip</code>	<code>a202061-second-order.tex</code>	2 Oct	<code>a61:o2:</code>
IV	42	<code>oeis-a202061-third-order-report.zip</code>	<code>a202061-third-order.tex</code>	2 Oct	<code>a61:o3:</code>
V	38	<code>oeis-a202061-allorders-report.zip</code>	<code>a202061-allorders.tex</code>	2 Oct	<code>a61:ao:</code>

^aIts derivation notes are dated 1 October.

Pins. No source pins a ProveIt commit or claims to continue a ProveIt path. Source 39’s scoped source note (shipped as `39-foundation-derivation-source_scope.md`) records a repository search whose results referred to commits `63b9d6840` and `b4d1ba280` (1 October 2026); it found no A202061 material and made no repository-wide absence claim. That was true then; this report is now the repository’s A202061 material. Sources 40, 42 and 38 each pin their predecessor’s package by the SHA-256 of its checksum manifest (40 pins 41’s, 42 pins 40’s, 38 pins 42’s); source 41 checks its embedded copy of source 39 against 39’s own checksum manifest.

Nesting. Each archive embeds its predecessors byte for byte: 41 contains 39, 40 contains 41, 42 contains 40, and 38 contains 42, so archive 38 alone carries all five papers. The placement staged each paper’s files once, from its own archive.

Where the merge had to choose. (i) The Parts follow the chain order 39, 41, 40, 42, 38, so that every Part cites only earlier Parts. (ii) Source 41 restated four displays of source 39 verbatim (the walk equation, the jump quadratic, the block formula and the first-hit bound); they are printed once, in Part I, and Part II cites them. Sources 40, 42 and 38 restate the same inputs, and further inputs from sources 41 and 40, in their own notation and with their own labels, which their proofs cite; those restatements are kept, each with a note naming where the input is proved. (iii) Part V implies the coefficient and inverse theorems of Parts I–IV (see below), but all five are printed: each later Part imports inputs proved only in earlier ones, and Part IV’s proof is a genuinely different route to the $J = 1$ case of Part V. Nothing of any source is omitted except its title block, its bibliography (merged below) and source 38’s table of contents.

How the Parts depend on one another

Taking $J = 0$ and $J = 1$ in Theorem 30.1 gives the coefficient theorems of Parts I, II, III and IV ($P_1(B) = B$ reproduces equation (24.2)), and Theorem 35.1 gives their inverse corollaries. Part V does *not* imply three results:

1. the non-D-finiteness of $A(x)$, Corollary 1.2(3);
2. the exact positive height-walk representation of Part I, Sections 2–3, which Part V imports;
3. the finite-height radius theorems, equation (10.3) and its refinement (21.1); Part V uses the constant c_* but states no finite-height result.

Part V is not self-contained and says so (Section 31). It imports the walk, block formula, first-hit bound and monotonicity of Part I; the local Gaussian lower bound (Lemma 11.1) and the

expansion (11.3) of Part II; the uniform square-root transfer with four parameter derivatives and the amplitude a_* , c_* of Part III; and the constants of Part IV. It re-proves both coefficient bounds at its precision by a new reduction (Section 32), not by citing Parts III or IV. Likewise each of Parts II–IV proves its own coefficient theorem from the kernel inputs rather than from the previous theorem.

The family: three ascent-sequence reports

A202061 is one of the three classes of ascent sequences avoiding a length-3 pattern that Duncan and Steingrímsson introduced and Conway, Conway, Elvey Price and Guttmann studied together [2]. Batch 77 opened a report for each, in the same collection directory: **a202058-ascent-000-growth** (000-avoiding, OEIS A202058; factorial growth), **a202062-ascent-201-enumeration** (201-avoiding, OEIS A202062; algebraic generating function, $a_n \sim C\mu^n n^{-9/2}$), and this report (120-avoiding; exponential growth with a stretched logarithmic deficit). A202061 and A202062 share the growth constant μ , the largest root of $\mu^3 - 8\mu^2 + 5\mu + 1$. Their regimes and methods are otherwise disjoint, which is why the three are separate reports. The A202062 report gives a second route to μ for A202061; see the note after Corollary 1.2.

Notation

No symbol of any source was renamed; each Part keeps its source’s letters. The five sources use one letter for several notions, and the same notion under several letters. The table fixes the meaning in each Part and prints the tempting false reading beside the true one. In this guide, the constants are written with $\star$ ($C_\star$, $z_\star$, $m_\star$, ...); the Parts write $\star$ or $*$ as their source did.

Symbol	Meaning and where	Do not read it as
n , a_n , $A(x)$	Word length; number of 120-avoiding ascent sequences of length n , with $a_0 = 1$; $A(x) = \sum a_n x^n$. All Parts.	
μ , ρ , λ	$\mu = 7.2958969432\dots$, largest root of $\mu^3 - 8\mu^2 + 5\mu + 1$; $\rho = 1/\mu$, the radius; $\lambda = \log \mu$ (Parts I, II, III, V), written λ_0 in Part IV.	In Part II’s Lemma 12.1 and Part III’s upper bound, λ is a small length tilt, not $\log \mu$.
$z_\star$, $t_\star$	$z_\star = 0.1980622641\dots$, smallest positive root of $z^3 - 5z^2 + 6z - 1$; $t_\star = 1/z_\star$. Part I: $z_\star$, $t_\star$. Parts II–III: z , t (Part III also $z_\star$, $t_\star$). Parts IV–V: $z_\star$, $t_\star$.	In Part I, z and t are <i>variables</i> (jump and shift variables). In Parts II–V, z and t are the constants, and the variables are u (jump) and T (shift).
α	$\alpha = 0.5100033746\dots$, the critical ratio q/ℓ : root of $7\alpha^3 + 14\alpha^2 - 7\alpha - 1$ (Part I), $= (8z^2 - 29z + 9)/7$ (Parts II–V).	In Part I, Section 8, α and β are exponents in a local power-log term.
κ	Parts I–II: $\kappa = 0.2830305201\dots$, third coordinate of the critical entropy point $((-21\alpha^2 + 17\alpha + 5)/29$ in Part I, $(17z^2 - 59z + 13)/7$ in Part II; the same number). Part III: $\kappa(r) = \frac{1}{2} \log r + \log \log r - C_0/2$, a row threshold in the upper bound, and $\kappa_\star$ in the conjecture. Parts IV–V: $\kappa = \log Y_0 - 2 \log 3 + 2c_\star = -1.9401481831\dots$	$\kappa \approx 0.283$ of Parts I–II is <i>not</i> the third-order constant $\kappa \approx -1.940$ of Parts IV–V. Part III’s $\kappa_\star$ is the latter.
κ_{inv}	$\kappa - \frac{2}{3} \log \lambda = -2.3980035405\dots$ (Parts IV, V).	

Symbol	Meaning and where	Do not read it as
v	$v = (2z^2 - 7z + 2)/2 = 0.3460107358\dots$, the height-variance parameter (Parts II–V).	In Part I, Section 2, v is a letter value (the largest lower endpoint of an increasing pair already present). In $L(v) = v \log v$ (Parts I, II) it is a dummy; v_j in Part II are increment offsets.
$\Phi(n), F, L$	$\Phi(n) = F = n^{1/3}(\log n)^{2/3}$: Φ in Parts I–II, F in Parts II–V. $L = \log n$ in Parts II–V.	$L(v) = v \log v$ (Parts I, II); L_K a total length (Part I); L a macro-degree in Part II's $k(L, q, d)$ and $L = \log Y$ in Part II's inverse section.
H	$H = n^{2/3}L^{1/3}$, the height scale (Parts II–V); $F = H^2/n$.	In Part I, H is a free height cutoff, and $H(\cdot)$ an analytic function in Section 8.
D_n, D, D_0	$D_n = n \log \mu - \log a_n$ (Parts III–V). D (Parts I, II) $= D_0$ (Parts III–V): the constant of the first-hit bound (7.2).	Part I's Lemma 5.1 uses another fixed constant D .
$C_\star$	$(3\pi^2\alpha^2/(2v))^{1/3} = 2.2326253076\dots$. Written $C_\star$ (Parts II, III), $C_\star$ (Part III's conjecture section and Part IV), C (Part V).	In Part I, c and C are unspecified positive constants. In Part V, C is the specific constant $C_\star$.
Y_0	$(2v\alpha/(3\pi^2))^{1/3} = 0.2284321390\dots$, so $C_\star = \alpha/Y_0$ (Parts IV, V). Written Y in Part III's conjecture section.	Not the threshold target Y of $N(Y)$.
$c_\star$	$\log((1 - m_\star)/(2a_\star)) = 0.8667962458\dots$, the finite-height constant (Parts III, IV, V; written $c_\star$).	Not Part I's generic c .
$m_\star$	$\rho/((1 - \rho)z_\star + \rho) = 0.4450418679\dots$, the defective critical row mass. Part I: $m_\star$; Part II: m , (11.1); Part III: $m_\star$; Parts IV–V: $m_\star$.	Part II also uses m for the number of large blocks and for elapsed degree.
$a_\star, C_q$	Row amplitude $a_\star = \rho t_\star C_q/(1 - \rho)$, $C_q = \sqrt{-16z_\star^2 + 55z_\star - 10}/(2\sqrt{\pi})$ (Parts III–V).	Not a coefficient a_n .
$N(\cdot)$	$N(Y) = \min\{n : a_n \geq Y\}$ (Parts I–IV); $N(p)$ in Part V.	In Part V, $N = K + 3$ is also a truncation order.
Inverse variables	$\log Y$ is called L (Part II), S (Part III), T (Part IV); Part V writes $T = \log p$ and $S = \log T$.	Not the shift T or the entropies S .
B	Part I (and II, IV, V): $B(\ell, q, r) = [x^\ell t^r]Q_q(x, t)$, a block coefficient. Part II: a calibration bound $0 < B < \alpha/3$. Part III: B_n (19.5) and the Kepler-arch parameter B . Part IV: $B_\star = C_\star \lambda_0^{-4/3} = 0.8935682576\dots$, the leading inverse coefficient. Part V: $B = \kappa + \frac{7}{3} \log L$, the argument of P_j ; $B_{\text{inv}} = \kappa_{\text{inv}} + \frac{7}{3} \log S$.	Part V's B is <i>not a constant</i> : it grows like $\frac{7}{3} \log \log n$. Part IV's $B_\star$ is not a value of Part V's B .
P_j, p_j	Part V: universal polynomials $P_j(B)$ and local polynomials $p_j(A)$.	Part I's operator series $P_d(T)$; Part II's bound $f(1, y) \leq Py$.
T	Shift operator $(Tf)_h = f_{h+1}$ (all Parts; in Part I also the variable t).	Inverse variables above; in Part V, Section 32, the argument T of $k_N(T)$ is $\log h$.
S	Part I: set of seen values, and the entropy $S(a, d, u)$. Part II: entropy $S(a, b, u)$ and the sine sum S . Part III: $S = \log Y$. Part V: $S = \log T$, and the series $\mathcal{S}$.	
M, M_0	Number of large macro-blocks (Parts II, III); Kepler amplitude M_0 (Part II) and M (Part III's arch); $M_0 = F/L$ (Part V).	

Symbol	Meaning and where	Do not read it as
J, K	Part I: staircase indices $J < K$. Part II: $J = \lceil L^3 \rceil$. Part V: the order J , the beta moments $J_{\pm}$, and the reduction order K .	
Ψ	The tilt exponent $\Psi(s, \theta) = (s + v\theta^2/2)/\alpha + \dots$ (Parts II–V).	

Status, scope and non-claims

The five manuscripts are AI-assisted and unrefereed. Each says its audits are research verification, not journal peer review or formal proof-assistant certification. Their finite computations (exact certificates, symbolic checks, quadratures) supplement the proofs and are not curve fits. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts, is as follows.

- No multiplicative equivalent for a_n : after any fixed order of Part V the remainder in $\log a_n$ may diverge. No amplitude, power-law prefactor or fluctuation determinant.
- No convergent series, no exponentially complete transseries, no uniformity in J (Part V fixes J first).
- No effective constants, computable thresholds, finite-data crossover threshold, or exact integer inverse rounding rule.
- Priority is not established. The sources’ literature checks are scoped. The gap reduction of Part I may overlap an unpublished polynomial-time enumeration mentioned in [2].
- The pure $n^{3/8}$ stretch of [2] was a numerical estimate there; Corollary 1.2(1) rules it out as an asymptotic form and says nothing about finite-range fits.
- Part III’s conjecture (23.1) was stated as conditional; it is now proved in Part IV.

Source 41 ships a kernel note with stronger local theorems, which its own header places outside the audit boundary. This report neither prints nor claims them; see the README.

Instances of results elsewhere in the repository

Several steps are instances of results proved in the repository’s transseries volume, **Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/**. No novelty is claimed for these methods.

- Part V’s local cutoff equation (32.2), $k - T/2 - \log k + \dots = 0$, is the linear–logarithmic equation $aX + b \log X = L$ with $a = 1$, $b = -1$, perturbed by $\log \mathcal{S}(1/k)$. Its peak equation (35.2), $L = 3k - \frac{7}{2} \log k + \dots$, has $a = 3$, $b = -\frac{7}{2}$. Both are solved on the large branch by the volume’s Lambert core, theorem `p0:thm:lambert-core`.
- The inverse corollaries of all five Parts invert $\lambda n - D_n$ about $n = \log Y/\lambda$. This is the reversion of a perturbed core, as in theorem `p0:thm:perturbed-inversion`. They then bracket the integer threshold $N(Y)$ between two rounded values by monotonicity, which is the separation step of theorem `p0:thm:staircase`.

Conventions of the merge

- Every label carries the report prefix **a61:** and the Part prefix of the table above, for example **a61:o3:up3:section**. Labels that were already unique in their source (**low:**, **upp:**, **rad:** in source 40; **up3:**, **lo3:**, **inv3:** in source 42) also carry the Part prefix. Guide labels are **a61:guide:**, Part labels **a61:part:**.
- Sections are numbered through the report. Source 39's Section k is Section k here (Sections 1–9); source 41's is $k + 9$ (Sections 10–15); source 40's is $k + 15$ (Sections 16–23); source 42's is $k + 23$ (Sections 24–29); source 38's is $k + 29$ (Sections 30–37). Theorems are numbered within sections, as in every source, so source 41's Theorem 1.1 is Theorem 10.1. Equations are now numbered within sections; the sources numbered them through the paper.
- Citations of the earlier papers of the chain (bibliography keys **Foundation**, **Sharp**, **Second**, **Third**) are replaced by references to the Parts, and those bibliography entries are dropped. Sources 40, 42 and 38 gave source 39 a title it does not carry (“A202061 positive height-walk and logarithmic deficit foundation”); its title is “Logarithmic deficit of 120 avoiding ascent sequences”.
- Sentences in the Parts about “this package”, “the accompanying archive”, “the companion package”, “the README”, “**dependency/...**” or files being “included unchanged” describe the source's delivered archive, which survives in commit **096ee7b87**. Which of those files this report ships, and under which names, is listed in the README. PDFs, checksum manifests and nested copies are not shipped.
- Each Part opens with its source's abstract as delivered. Source 38's table of contents is dropped in favour of the one above.

Part I

Logarithmic deficit of 120 avoiding ascent sequences

Source. Batch-77 manuscript 39, archive **oeis-a202061-report.zip**, main file **a202061-report.tex** (517 lines, no `\input`), dated 2 October 2026; its derivation notes are dated 1 October. Author line: “A proof and reproducible research report”. The base of this report: its text is printed in full, with label prefix **a61:fd:**.

Abstract of the source.

For the number a_n of classical 120-avoiding ascent sequences, we derive an exact positive height-walk representation and prove

$$n \log \mu - \log a_n = \Theta\left(n^{1/3}(\log n)^{2/3}\right), \quad \mu = 7.295896943239772\dots$$

This rules out the previously estimated pure $n^{3/8}$ stretched correction. The proof uses a positive binomial block formula, a critical jump tail, a finite-height radius bound, and a legal all-length staircase construction. We also obtain the order of the inverse correction and prove that the generating function is not D-finite. No exact stretched constant, asymptotic equivalent, or all-orders transseries is claimed. Exact-arithmetic certificates and independent finite diagnostics accompany the argument.

1 Results and scope

An ascent sequence is a word $w_1 \cdots w_n$ of nonnegative integers with $w_1 = 0$ and

$$w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

It avoids the classical pattern 120 if there are no $i < j < k$ with $w_k < w_i < w_j$. Let a_n count these words, including $a_0 = 1$, and let $A(x) = \sum_{n \geq 0} a_n x^n$. This is OEIS A202061 [1].

Theorem 1.1. *Let μ be the largest root of*

$$\mu^3 - 8\mu^2 + 5\mu + 1 = 0, \quad \mu = 7.29589694323977237457 \dots,$$

and put $\Phi(n) = n^{1/3}(\log n)^{2/3}$. There exist $c, C > 0$ and n_0 such that, for every integer $n \geq n_0$,

$$\mu^n e^{-C\Phi(n)} \leq a_n \leq C\mu^n e^{-c\Phi(n)}. \quad (1.1)$$

In particular,

$$n \log \mu - \log a_n = \Theta(\Phi(n)), \quad \frac{\log(n \log \mu - \log a_n)}{\log n} \rightarrow \frac{1}{3}.$$

The radius of convergence is $\rho = 1/\mu$.

Corollary 1.2. *The following consequences hold.*

1. *No equivalent $C_0 \mu^n \exp(-dn^\gamma) n^g$, with $C_0, d > 0$ and fixed $g \in \mathbb{R}$, is possible for $\gamma > 1/3$ or for $\gamma = 1/3$.*
2. *For $N(Y) = \min\{n \geq 0 : a_n \geq Y\}$ and $Y \rightarrow \infty$,*

$$N(Y) = \frac{\log Y}{\log \mu} + \Theta\left((\log Y)^{1/3}(\log \log Y)^{2/3}\right).$$

3. *$A(x)$ is not D -finite over $\mathbb{C}(x)$; equivalently, (a_n) is not P -recursive over $\mathbb{C}[n]$.*

Conway, Conway, Elvey Price and Guttmann [2] proved equality of the exponential growth rates of the 120- and 201-avoiding classes. Their numerical analysis estimated a pure stretched power $\sigma = 3/8$, with other fitted parameters depending on that choice. Their final paper says the later 500-term computation reinforced the estimate; it does not prove that stretched power. Theorem 1.1 determines a different asymptotic order. It does not supply the leading constant in the deficit, prove convergence of the normalized deficit, or furnish a full expansion. The proof of its exponential rate is independent of any generating-function theorem for A202062.

[write, 2 October 2026] *Status of this paragraph in the merged report.* The leading constant and the convergence of the normalized deficit are proved in Part II (Theorem 10.1); the next three terms in Parts III and IV; every fixed inverse-logarithmic order in Part V. A full expansion in the sense of an exponentially complete transseries is still not obtained.

A second route to the growth rate. The batch-77 report on 201-avoiding ascent sequences, `a202062-ascent-201-enumeration`, which sits beside this report in `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`, in its section “A consequence for 120 avoidance”, obtains $\lim a_n^{1/n} = \mu$ for the present sequence. It combines Theorem 4 of [2] (the 120- and 201-avoiding classes have equal exponential growth rates) with its own proof of the Guttmann–Kotešovec cubic generating function for A202062. That route gives the growth rate only. Theorem 1.1 proves the rate directly, without any generating-function theorem for A202062, together with the order of the deficit. The two routes give the same μ .

The original derivation and the additional independent audit are included in the accompanying package. They are research verification, not journal peer review or formal proof-assistant certification. A scoped source check does not establish comprehensive publication priority; in particular the gap reduction may overlap an unpublished polynomial-time enumeration mentioned in [2].

2 An exact normalized recurrence

For an avoiding prefix, let v be the largest lower endpoint of an already occurring increasing pair, or 0 if there is none. A future letter smaller than v would complete a 120 pattern, and every letter at least v obeying the ascent bound is permitted. Delete the seen values smaller than v and translate the surviving alphabet by $-v$. The normalized state is (a, l, S) , where a is the number of previous ascents minus v , l is the translated last letter, and S is the set of seen translated surviving values. Write

$$S = \{0 = s_0 < s_1 < \cdots < s_k\}, \quad m = s_k, \quad h = a + 1 - m, \quad d_i = s_i - s_{i-1}.$$

Every ascent sequence satisfies $\max w \leq \text{asc}(w)$, so $h \geq 1$.

Appending $i \in \{0, \dots, a + 1\}$ sets

$$p = \max(\{s \in S : s < i\} \cup \{0\})$$

and gives the exact child

$$(a', l', S') = (a + \mathbf{1}_{l < i} - p, i - p, \{s - p : s \in S, s \geq p\} \cup \{i - p\}). \quad (2.1)$$

Indeed p is the greatest newly forbidden lower threshold, and the insertion of $i - p$ records the new letter. Thus this recurrence follows directly from the class definition. In every normalized state, the last letter is either 0 or the least positive member of S : if $i > 0$, there was no old seen value strictly between p and i .

Let $E_h(R)$ be the suffix generating function when the last letter is 0 and the list of positive gaps is R . Let $G_h(d, R)$ be the suffix generating function when the gap list is (d, R) and the last letter is d . All empty suffixes have weight 1. Define

$$D_h(d, R) = \sum_{q=1}^d G_h(q, (d - q, R)), \quad e_h = E_h(\emptyset), \quad N_h = \sum_{q=1}^h G_{h+1-q}(q, \emptyset),$$

where a zero gap is omitted. Write $(Tf)_h = f_{h+1}$.

Lemma 2.1. *Put $c = x/(1-x)$ and $\mathcal{B} = x+x^2T/(1-x)$. There are formal series $R_d(T), P_d(T), Q_d(T)$ with nonnegative integer coefficients in x, T such that*

$$\begin{aligned} D_h(d, R) &= R_d(T)E_h(R), \\ E_h((d, R)) &= P_d(T)E_h(R), \quad G_h(d, R) = Q_d(T)E_h(R), \end{aligned} \quad (2.2)$$

where

$$\begin{aligned} P_d &= 1 + cTR_d, \quad Q_d = 1 + \mathcal{B}R_d, \\ (1 - \mathcal{B})R_d &= 1 + \sum_{q=1}^{d-1} Q_q P_{d-q}. \end{aligned} \quad (2.3)$$

Proof. Classify the next letter as zero, in an old gap, or a new maximum. For $R = (d_1, \dots, d_k)$ this gives

$$(1-x)E_h(R) = 1 + xN_h + x \sum_{j=1}^k D_{h+1}(d_j, (d_{j+1}, \dots, d_k)). \quad (2.4)$$

From last letter d , selecting in the first gap gives no ascent; selecting in a later gap does. Subtraction from (2.4) therefore gives

$$G_h(d, R) = E_h((d, R)) + x(1-T)D_h(d, R).$$

Subtracting (2.4) for R from its version for (d, R) also gives

$$E_h((d, R)) = E_h(R) + cTD_h(d, R).$$

Together these imply $G_h(d, R) = E_h(R) + \mathcal{B}D_h(d, R)$. Induct on d , splitting the definition of D into $q = d$ and $q < d$. For $q < d$, both shorter-gap identities apply to the arbitrary tail. All resulting operators commute, being series in the same shift T . The $q = d$ term gives the factor $1 - \mathcal{B}$, whose inverse exists x -adically and has nonnegative coefficients. This proves the identities, existence, uniqueness and positivity. $\square$

Use an ordinary variable t for the shift and set $Q(x; z, t) = \sum_{q \geq 1} Q_q(x, t)z^q$, with similar notation for P, R . With $b = 1 - x$ and $\mathcal{B} = x + x^2t/b$, summing (2.3) yields

$$R = Q(1 + P), \quad Q = \frac{z}{1-z} + \mathcal{B}R, \quad P = \frac{z}{1-z} + \frac{xt}{b}R.$$

Elimination gives the exact jump equation

$$xt(1-z)Q^2 + [b(z-b) + xt(x-z)]Q + bz = 0, \quad (2.5)$$

with $Q(x; 0, t) = 0$ and $Q(0; z, t) = z/(1-z)$. In particular,

$$Q_1(x, t) = \frac{1}{1-x-x^2t/(1-x)}. \quad (2.6)$$

3 Positive height walks and block weights

The empty-gap case of (2.4) is

$$e_h = \frac{1}{1-x} + \frac{x}{1-x} \sum_{q=1}^h (Q_q(x, T)e)_{h+1-q}, \quad A(x) = 1 + xe_1(x). \quad (3.1)$$

Interpret a term indexed by q, r as a macro-step that first drops by $q-1$, then rises by r , has weight

$$\frac{x}{1-x} [t^r] Q_q(x, t),$$

and changes height from h to $h+1-q+r$. Its legality condition is exactly $1 \leq q \leq h$. The terminal weight is $1/(1-x)$ at every height. This is a positive formal representation of the original sequence; no additional word-level bijection is required for coefficient selection. Every nonterminal step has positive x -degree, and only finitely many shifts occur at any fixed degree. Hence the formal Neumann series is well defined and uniquely solves (3.1).

Put $\beta = xt/b$ and $\eta = x^2t/b^2$. Equation (2.5) is equivalently

$$Q = \frac{z + xW}{1 - z}, \quad W = \frac{z(1 + \beta)(1 + W)}{b(1 - \eta(1 + W))}, \quad [z^0]W = 0. \quad (3.2)$$

Lagrange inversion gives, for $j \geq 1$,

$$[z^j]W = \frac{(1 + \beta)^j}{b^j} \sum_{k \geq 0} \mathcal{C}(j, k) \eta^k, \quad \mathcal{C}(j, k) = \frac{1}{j} \binom{j + k - 1}{k} \binom{j + k}{k + 1} > 0. \quad (3.3)$$

For $\ell, r \geq 0$ and $q \geq 1$, write $B(\ell, q, r) = [x^\ell t^r]Q_q(x, t)$. Expanding (3.3) yields

$$B(\ell, q, r) = \mathbf{1}_{\ell=0, r=0} + \sum_{j=1}^q \sum_{k=0}^r \mathcal{C}(j, k) \binom{j}{r - k} \binom{j + \ell - 2}{\ell - r - k - 1}. \quad (3.4)$$

The summation includes only $0 \leq r - k \leq j$ and $\ell \geq r + k + 1$. In particular, keeping $j = q$ and one admissible k gives

$$B(\ell, q, r) \geq \mathcal{C}(q, k) \binom{q}{r - k} \binom{q + \ell - 2}{\ell - r - k - 1}. \quad (3.5)$$

4 An exact critical point

Let α be the unique positive root of $7\alpha^3 + 14\alpha^2 - 7\alpha - 1 = 0$ and define

$$\kappa = \frac{-21\alpha^2 + 17\alpha + 5}{29}, \quad \mu = \frac{1 + \alpha}{1 - \alpha - \kappa}, \quad z_\star = \frac{\kappa(1 - \alpha - \kappa)}{(\alpha - \kappa)(2\alpha + \kappa)}. \quad (4.1)$$

The following bounds and identities are certified by rational interval arithmetic and polynomial reduction in the supplied program:

$$\begin{aligned} \alpha &= 0.5100033746178888\dots, & \kappa &= 0.2830305201361994\dots, \\ z_\star &= 0.1980622641951617\dots, & z_\star^3 - 5z_\star^2 + 6z_\star - 1 &= 0, \\ 0 &< \kappa < \alpha, & \alpha + \kappa &< 1. \end{aligned}$$

The value of μ satisfies the cubic and interval in Theorem 1.1, which identify its largest real root. Descartes' rule gives uniqueness of the positive α root, isolated in

$$\frac{51000337}{10^8} < \alpha < \frac{51000339}{10^8}.$$

For $L(v) = v \log v$, let

$$\begin{aligned} S(a, d, u) &= 2L(a + u) - L(a) - 2L(u) - L(d - u) - L(a - d + u) \\ &\quad + L(1 + a) - L(1 - d - u) - L(a + d + u). \end{aligned}$$

This is the factorial entropy of (3.5) divided by ℓ , for $a = q/\ell$, $d = r/\ell$ and $u = k/\ell$. At (α, α, κ) ,

$$S_u = 0, \quad S_a + S_d = 0, \quad S_d = \log z_\star, \quad S = \log \mu. \quad (4.2)$$

For clarity, the two stationary identities reduce to

$$\begin{aligned} (\alpha + \kappa)^2(\alpha - \kappa)(1 - \alpha - \kappa) &= \kappa^3(2\alpha + \kappa), \\ (\alpha + \kappa)^2(1 + \alpha)(1 - \alpha - \kappa) &= \alpha(\alpha - \kappa)(2\alpha + \kappa)^2. \end{aligned}$$

The displayed expression for z_* gives S_d . Euler's identity applied to the homogeneous factorial entropy then gives $S = \log((1 + \alpha)/(1 - \alpha - \kappa))$. These are finite algebraic verifications; global maximality of S is not needed.

Set $\rho = 1/\mu$, $b_* = 1 - \rho$, $t_* = 1/z_*$, and

$$\begin{aligned} \beta_* &= \frac{\rho}{b_* z_*}, & \eta_* &= \frac{\rho^2}{b_*^2 z_*}, & Q_* &= \frac{b_* z_*}{b_* z_* + \rho}, \\ W_* &= \frac{(1 - z_*)Q_* - z_*}{\rho}, & m_* &= \frac{\rho}{b_*} t_* Q_* = \frac{\rho}{b_* z_* + \rho}. \end{aligned} \quad (4.3)$$

The exact certificate proves the fixed-point identity (3.2) at these values, and

$$W_* = 1 + \beta_* > 0, \quad \eta_*(2 + \beta_*)^2 = 1, \quad 0 < \eta_*(1 + W_*) < \frac{1}{2}, \quad \frac{2}{5} < m_* < \frac{1}{2}. \quad (4.4)$$

Numerically, $W_* = 1.8019377358\dots$ and $m_* = 0.4450418679\dots$

Lemma 4.1. *The jump series converges at (ρ, z_*, t_*) , with $Q(\rho; z_*, t_*) \leq Q_*$. Moreover,*

$$e_h(\rho) \leq \frac{t_*^h}{b_*(1 - m_*)}, \quad A(\rho) \leq 1 + \frac{\rho t_*}{b_*(1 - m_*)} < \infty. \quad (4.5)$$

Proof. Expand the denominator on the right of (3.2) geometrically. Iteration starting from 0 is nonnegative and bounded by the fixed-point supersolution W_* , by (4.4). Monotone convergence proves the assertion about Q . For the height weight t_*^h , the sum of an unrestricted transition row divided by its starting weight is at most

$$\frac{\rho}{b_*} \sum_{q \geq 1, r \geq 0} [t^r] Q_q(\rho, t) t_*^{1-q+r} = \frac{\rho}{b_*} t_* Q(\rho; z_*, t_*) \leq m_* < 1.$$

Each legal row is a subsum. The terminal weight $1/b_*$ is at most t_*^h/b_* , so the nonnegative Neumann series is bounded by a geometric series. This proves (4.5). No spectral-radius identification from a discriminant is being assumed. $\square$

5 The all length lower bound

Lemma 5.1. *For each fixed $D > 0$ there exist $c_D, C_D > 0$ and ℓ_D such that the conditions*

$$\ell \geq \ell_D, \quad |q - \alpha\ell| \leq 2, \quad |k - \kappa\ell| \leq 1, \quad |r - q| \leq D\sqrt{\ell \log \ell}$$

imply admissibility in (3.5) and

$$B(\ell, q, r) \geq c_D \ell^{-C_D} \mu^\ell z_*^{r-q}. \quad (5.1)$$

If $|r - q| \leq D\sqrt{\ell}$, one can use the more precise factor $c_D \ell^{-3}$.

Proof. All normalized factorial arguments lie in one compact subset of the positive interior, for all sufficiently large ℓ . The exact identities

$$\begin{aligned} \mathcal{C}(q, k) &= \frac{q}{(q+k)(k+1)} \binom{q+k}{k}^2, \\ \binom{q+\ell-2}{\ell-r-k-1} &= \frac{(\ell-r-k)(q+r+k)}{(q+\ell)(q+\ell-1)} \binom{q+\ell}{\ell-r-k} \end{aligned}$$

and uniform Stirling bounds give a lower bound $c\ell^{-3} \exp(\ell S(q/\ell, r/\ell, k/\ell))$. Taylor's theorem and (4.2) yield

$$\ell S = \ell \log \mu + (r - q) \log z_\star + O_D(\log \ell).$$

Indeed the first differential is exactly $(r - q) \log z_\star$, and the quadratic remainder is $O((r - q)^2/\ell + 1)$. This proves (5.1); with square-root perturbations the remainder is bounded. $\square$

Choose a sufficiently large fixed integer J . For $j \geq J$, define

$$h_j = \lfloor j^2 \log j \rfloor, \quad q_j = \lfloor h_j/2 \rfloor, \quad \ell_j = \lfloor q_j/\alpha \rfloor, \quad k_j = \text{a nearest integer to } \kappa \ell_j.$$

An ascent step from h_j to h_{j+1} uses

$$r_j^+ = q_j - 1 + (h_{j+1} - h_j),$$

and a descent step from h_j to h_{j-1} uses

$$r_j^- = q_j - 1 + (h_{j-1} - h_j).$$

Both preserve the exact increment $1 - q + r$. As $\ell_j = \Theta(j^2 \log j)$ and $h_{j+1} - h_j = O(j \log j)$, every such step satisfies Lemma 5.1 with the same D . Its down parameter obeys $q_j \leq h_j$, so it is legal. Retain just the leading x in the macro factor $x/(1 - x)$; then its length is $\ell_j + 1$ and multiplicity at least $B(\ell_j, q_j, r_j^\pm)$.

Seed the walk from height 1 to h_J using $q = 1, r = h_J - 1$. Equation (2.6) gives a coefficient 1 at its leading degree $\ell = 2(h_J - 1)$, so the macro-step has length $2h_J - 1$. The final return from h_J to 1 uses $q = h_J, r = 0, \ell = 0$, of length 1 and multiplicity 1.

Ascend from h_J to h_K and descend to h_J , then add the seeds and the original leading letter. The total length is

$$\begin{aligned} L_K &= 2h_J + 1 + \sum_{j=J}^{K-1} (\ell_j + 1) + \sum_{j=J+1}^K (\ell_j + 1) \\ &= \frac{K^3 \log K}{3\alpha} + O(K^3), \end{aligned} \tag{5.2}$$

and

$$L_{K+1} - L_K = \ell_K + \ell_{K+1} + 2 = \Theta(h_K).$$

The total increment of the large blocks is zero. If there are M such blocks, then $\sum(r - q) = -M$, so their height-tilt factors multiply to $z_\star^{-M}$. There is no uncanceled exponentially expensive endpoint factor.

To cover every sufficiently large integer n , choose maximal K with $L_K \leq n$ and let $R = n - L_K$. A bounded residual is absorbed by the terminating geometric series. Otherwise split $R = m_1 + m_2$ into two nearly equal integers, both above a fixed threshold, and insert two self-loops at height h_K . For length m choose

$$\ell = m - 1, \quad q = \text{a nearest integer to } \alpha \ell, \quad r = q - 1, \quad k = \text{a nearest integer to } \kappa \ell.$$

The loop has exactly zero increment and meets Lemma 5.1. Furthermore,

$$q \leq \frac{\alpha R}{2} + O(1) \leq \frac{h_K + h_{K+1}}{4} + O(1) < h_K$$

for all sufficiently large K , proving legality.

There are $O(K)$ large blocks and at most two repair loops, each with $\ell = O(K^2 \log K)$. Multiply (5.1). The extra factor μ^{-1} for each macro-step's leading x , and the fixed seed lengths, cost at most $\exp(O(K))$. Polynomial factors cost at most $\exp(O(K \log K))$, while the tilt telescopes as above. Thus

$$\log a_n \geq n \log \mu - O(K \log K).$$

By (5.2) and $R = O(h_K)$, one has $K = \Theta((n/\log n)^{1/3})$, hence $K \log K = \Theta(\Phi(n))$. This proves the lower half of (1.1) for all large integers, not merely a subsequence.

6 A finite height radius gain

6.1 A fixed gap derivative bound

For fixed $t = t_\star$ and variable x , put $b = 1 - x$, $\beta = xt_\star/b$ and $\eta = x^2 t_\star/b^2$. The coefficient formula (3.3) implies

$$Q_q(x, t_\star) = 1 + x \sum_{j=1}^q W_j(x), \quad W_j(x) = \left(\frac{1+\beta}{b} \right)^j \frac{N_{j-1}(\eta)}{(1-\eta)^{2j-1}}, \quad (6.1)$$

where $N_0 = 1$ and, for $j \geq 2$,

$$N_{j-1}(v) = \frac{1}{j-1} \sum_{l=0}^{j-2} \binom{j-1}{l} \binom{j-1}{l+1} v^l.$$

Here is a formal proof, with no analytic continuation needed. If $F_j(v) = \sum_{k \geq 0} \mathcal{C}(j, k) v^k$, its coefficient ratio gives

$$v(1-v)F_j'' + [2 - (2j+2)v]F_j' - j(j+1)F_j = 0.$$

Substitute $F_j = (1-v)^{1-2j}N$. The equation becomes

$$v(1-v)N'' + [2 - (4-2j)v]N' - (2-j)(1-j)N = 0.$$

With $N(0) = 1$, its unique formal coefficients satisfy

$$(l+1)(l+2)n_{l+1} = (l+2-j)(l+1-j)n_l.$$

They are exactly the displayed coefficients of N_{j-1} , and terminate after $j-2$. For $j=1$, $\mathcal{C}(1, k) = 1$ directly gives $F_1 = (1-v)^{-1}$.

Since $\eta(\rho) = 0.12737466 \dots < 1$, choose a fixed interval $[\rho, \rho + \epsilon_0]$ on which $b > 0$ and $0 < \eta < 1$. The positive polynomial N_{j-1} has degree at most j , so

$$0 \leq \frac{\eta N_{j-1}'(\eta)}{N_{j-1}(\eta)} \leq j.$$

Logarithmic differentiation of (6.1) gives a constant C_0 , independent of j , such that $0 \leq (\log W_j)' \leq C_0 j$ throughout this interval. Increasing a constant C_1 therefore gives

$$Q_q(x, t_\star) \leq Q_q(\rho, t_\star) \exp(C_1(q+1)(x-\rho)) \quad (6.2)$$

for all $q \geq 1$ and x in the interval.

6.2 The critical jump tail

At $x = \rho, t = t_*$, let $b = b_*$ and $s = b^2 - \rho^2 t_* > 0$. The discriminant of (2.5) as a quadratic in Q factors, as a polynomial in z , as

$$\Delta(z) = s^2(1 - z/z_*)(1 - z/z_2), \quad z_2 = \frac{s^2}{(b + \rho t_*)^2 z_*}.$$

The exact certificate verifies $0 < z_* < z_2 < 1$, with $z_2 = 0.8817230195\dots$. The branch with $Q(0) = 0$ is

$$Q(\rho; z, t_*) = \frac{s - (b - \rho t_*)z - s\sqrt{1 - z/z_*}\sqrt{1 - z/z_2}}{2\rho t_*(1 - z)}. \quad (6.3)$$

The rational nonsingular part is analytic in $|z| < 1$, and the multiplier of $\sqrt{1 - z/z_*}$ is analytic in $|z| < z_2$. Consequently, for some fixed C_2 ,

$$0 \leq Q_q(\rho, t_*) z_*^q \leq C_2 q^{-3/2} \quad (q \geq 1). \quad (6.4)$$

For an elementary coefficient justification, rescale $z = z_* w$. The coefficients of $\sqrt{1 - w}$ have absolute value at most $C(m + 1)^{-3/2}$. The analytic multiplier has coefficients $O(R^{-m})$ for some $R > 1$. In the convolution, indices at most $q/2$ yield $O(q^{-3/2})$ by geometric summation, and the other indices yield an exponentially small bound. The rational part is exponentially smaller as well. This proves (6.4) without a transfer theorem.

6.3 The cutoff estimate

Let $a_n^{[\leq H]}$ denote the coefficients in (3.1) restricted to macro-boundary heights in $\{1, \dots, H\}$. This is a coefficientwise positive subclass of the formal walk expansion. Set

$$\delta_H = \epsilon \frac{\log(H + 1)}{H + 1}, \quad x_H = \rho + \delta_H,$$

with a sufficiently small fixed $\epsilon > 0$. Write $A_q = Q_q(\rho, t_*) z_*^q$. By (6.2), the weighted row sum of the cutoff at x_H is at most

$$\frac{x_H}{1 - x_H} t_* e^{C_1 \delta_H} \sum_{q \leq H} A_q e^{C_1 q \delta_H}. \quad (6.5)$$

Its unperturbed infinite mass is at most $m_* < 1$. Moreover, (6.4) implies

$$\sum_{q \leq H} A_q (e^{C_1 q \delta_H} - 1) \longrightarrow 0. \quad (6.6)$$

Indeed split at $q = 1/\delta_H$. The lower part is $O(\sqrt{\delta_H})$, using $e^{C_1 q \delta_H} - 1 = O(q \delta_H)$ there. The upper part is bounded by

$$C e^{C_1 H \delta_H} \sqrt{\delta_H} = O\left(H^{C_1 \epsilon - 1/2} \sqrt{\log H}\right) = o(1)$$

if $C_1 \epsilon < 1/2$. The prefactor in (6.5) tends to its critical value. Thus the row mass is uniformly at most a fixed $\bar{m} < 1$ for large H . The same weighted geometric bound as in Lemma 4.1, now in the finite cutoff, gives a uniform upper bound on its generating function at x_H . Positive coefficient extraction yields

$$a_n^{[\leq H]} \leq C x_H^{-n} \leq C \mu^n \exp\left(-c \frac{n \log(H + 1)}{H + 1}\right). \quad (6.7)$$

All constants are independent of H, n once H is sufficiently large.

7 The first hit bound and the upper estimate

Keep $W = W_*$ fixed and write the right side of (3.2) as

$$F(x, z, t; W) = \frac{z}{1-x} \frac{(1+xt/(1-x))(1+W)}{1-x^2t(1+W)/(1-x)^2}.$$

Let $f(x, \theta) = F(x, z_*e^{-\theta}, t_*e^\theta; W_*)$. By (4.4), $f(\rho, 0) = W_*$ and $f_\theta(\rho, 0) = 0$. Specifically, with $u = \eta_*(1+W_*)$,

$$\frac{f_\theta}{f}(\rho, 0) = -\frac{1}{1+\beta_*} + \frac{u}{1-u} = 0.$$

Also $f_x(\rho, 0) > 0$, since the positive factors increase with x and the denominator remains positive. Choose a fixed sufficiently large $D > 0$ and put

$$x_\theta = \rho e^{-D\theta^2}, \quad t_\theta = t_*e^\theta, \quad z_\theta = 1/t_\theta.$$

Taylor's theorem gives

$$f(x_\theta, \theta) = W_* + \left(\frac{1}{2} f_{\theta\theta}(\rho, 0) - \rho D f_x(\rho, 0) \right) \theta^2 + O(\theta^3).$$

Make the bracket negative. For $0 \leq \theta \leq \theta_0$ with sufficiently small $\theta_0 > 0$, the positive fixed-point map then has the supersolution W_* and positive denominator. Iteration from zero gives

$$Q(x_\theta; z_\theta, t_\theta) \leq \frac{z_\theta + x_\theta W_*}{1 - z_\theta}.$$

Its unrestricted weighted row mass satisfies

$$m_\theta = \frac{x_\theta}{1-x_\theta} t_\theta Q(x_\theta; z_\theta, t_\theta) \leq \frac{x_\theta}{1-x_\theta} t_\theta \frac{z_\theta + x_\theta W_*}{1-z_\theta}.$$

The explicit supersolution bound on the right is continuous at zero, where it equals $m_* < 1$. Reducing θ_0 makes all such masses at most one fixed $\bar{m} < 1$. The sum of the weights of all finite macro-prefixes starting at height 1, including the factor t_θ^h at their final height but no terminating weight, is therefore at most

$$t_\theta \sum_{k \geq 0} \bar{m}^k \leq C. \quad (7.1)$$

Let $p_{m,h}$ count the first-hit macro-prefixes of length m whose first height above H is $h > H$. The remaining unrestricted suffix has generating function e_h . By (4.5), for the complementary coefficients at fixed word length n ,

$$a_n^{[>H]} \rho^n \leq C \rho \sum_{m \leq n-1, h > H} p_{m,h} \rho^m t_*^h.$$

Here ρ accounts for the original leading letter, and the use of $e_h(\rho)$ only enlarges the permitted remaining lengths. For $0 < \theta \leq \theta_0$,

$$\rho^m \leq x_\theta^m e^{D\theta^2 n}, \quad t_*^h \leq t_\theta^h e^{-\theta H}.$$

First-hit prefixes are a subset of all prefixes, so (7.1) implies

$$a_n^{[>H]} \leq C \mu^n \exp(D\theta^2 n - \theta H).$$

When $H/(2Dn) \leq \theta_0$, use $\theta = H/(2Dn)$ to obtain

$$a_n^{[>H]} \leq C\mu^n \exp\left(-\frac{H^2}{4Dn}\right). \quad (7.2)$$

Continuation at the critical tilt in (4.5) is what cancels the final height weight. No constraint on the final height of the entire walk is imposed.

Take $H = \lfloor n^{2/3}(\log n)^{1/3} \rfloor$. It is $o(n)$, so (7.2) applies for large n . Both exponents in (6.7) and (7.2) are at least a positive constant times $\Phi(n)$. Adding the two positive subclasses proves the upper half of (1.1). Together with the lower bound, this completes Theorem 1.1.

8 Inverse order and non D finiteness

The main theorem gives $c\Phi(n) - O(1) \leq n \log \mu - \log a_n \leq C\Phi(n)$. If a pure stretched equivalent in Corollary 1.2(1) held, its deficit would be $dn^\gamma - g \log n - \log C_0 + o(1)$. This grows faster than $\Phi(n)$ for $\gamma > 1/3$ and more slowly for $\gamma = 1/3$, contradicting the two-sided bounds.

The sequence is nondecreasing: append a copy of the last letter to an avoiding ascent sequence. The ascent condition persists. A new 120 occurrence using the appended letter would already occur using its previous copy. This extension is injective. Hence $N(Y)$ is well defined for large Y .

Let $L = \log Y$ and $\lambda = \log \mu$. Applying the upper bound at $n = N(Y)$ first gives $N(Y) \geq L/\lambda - O(1)$ and then

$$\lambda N(Y) - L \geq c\Phi(N(Y)) - O(1) \geq c'\Phi(L).$$

Conversely set $n = \lceil L/\lambda + K\Phi(L) \rceil$. Since $\Phi(n)/\Phi(L) \rightarrow \lambda^{-1/3}$, any sufficiently large fixed K makes $\lambda n - C\Phi(n) \geq L$. The coefficient lower bound implies $a_n \geq Y$, hence $N(Y) \leq n$. This proves the stated inverse order. It is a positive correction of a known order, with no asserted leading coefficient or exact rounding rule.

For non-D-finiteness, the stretched upper bound gives

$$\sum_{n \geq 0} n^k a_n \rho^n < \infty \quad (k = 0, 1, 2, \dots).$$

Thus A and all its derivatives extend continuously along the real radius to ρ . Suppose A were D-finite over $\mathbb{Q}(x)$. Its coefficients are exponentially bounded nonnegative integers, their algebraic conjugates have the same bound, and all common denominators are 1. It would therefore be a Siegel G-function. The G-function regularity theorem gives a convergent finite local sum of terms

$$(x - \rho)^\alpha (\log(x - \rho))^\beta H(x - \rho),$$

with rational α , nonnegative integer β and analytic H , on a slit neighborhood of ρ ; see [3, Definition 2.2 and Theorem 2.1(c)].

Such a convergent finite power-log expansion can be smooth to every order along a radius only if it is analytic at the endpoint. To see this, combine terms with the same rational exponents modulo integers and the same logarithmic powers. The possible exponents form a discrete set bounded below. A least surviving negative integer power is unbounded; a least noninteger power fails to have a bounded derivative at some finite order; and a least nonzero logarithmic term fails similarly. The absence of all these terms leaves a convergent ordinary power series. Consequently A would be analytic at ρ , contrary to Pringsheim's theorem for a nonnegative series of finite radius. Thus A is not D-finite over $\mathbb{Q}(x)$.

Finally, a complex differential relation of bounded order and polynomial degree for a rational series produces a homogeneous linear system with rational entries and finitely many unknowns. Its rank is attained by finitely many rows. If it has a nonzero complex solution, it has a nonzero rational solution. Hence complex D-finiteness would imply rational D-finiteness here, proving Corollary 1.2(3). This argument uses G-function arithmetic regularity; it does not assert that all D-finite functions have regular singularities.

9 Reproduction and remaining questions

The exact certificate checks the algebraic identities and rational sign margins. Independent programs compare literal words, normalized states, gap operators and binomial height walks, including the OEIS terms through $n = 20$. Additional checks cover fixed-gap formulas and staircase legality, lengths and tilt cancellation. These diagnostics supplement the proofs above.

The package includes editable source, a build script, mathematical notes, a separate audit and content hashes. The checks require Python with SymPy and mpmath. The PDF build uses standard LaTeX with AMS packages, geometry, hyperref, microtype and Latin Modern fonts.

The convergence and possible limit of $(n \log \mu - \log a_n)/\Phi(n)$, and lower-order corrections, remain open here. There is no full transseries or all-orders inverse expansion. Although the $n^{3/8}$ estimate is ruled out asymptotically, the theorem gives no explicit crossover threshold for finite-data fits.

[write, 2 October 2026] Of the questions left open here, the convergence and the limit of $(n \log \mu - \log a_n)/\Phi(n)$ are settled in Part II; the lower-order corrections in Parts III, IV and, to every fixed inverse-logarithmic order, V; and the inverse expansion to every fixed order in Theorem 35.1. Still open: an exponentially complete transseries, a multiplicative equivalent, and any explicit crossover threshold (Section 37).

Part II

The leading logarithmic deficit of 120 avoiding ascent sequences

Source. Batch-77 manuscript 41, archive `oeis-a202061-sharp-deficit.zip`, main file `a202061-sharp-deficit.tex` (429 lines), dated 2 October 2026. Author line: “A proof and reproducible research companion”. Printed in full except for four displays restated from source 39, which are cited from Part I instead (see the note in Section 11). Label prefix `a61:sh:`.

Abstract of the source.

Let a_n count classical 120-avoiding ascent sequences. We refine the previously established two-sided logarithmic-deficit order to an exact leading constant:

$$n \log \mu - \log a_n \sim C_* n^{1/3} (\log n)^{2/3}, \quad C_* = 2.232625307612844922 \dots$$

The lower coefficient bound sums positive blocks around a sine-square height profile while imposing exact total length and endpoint constraints. The upper coefficient bound uses a time-dependent exponential calibration whose transformed renewal rows remain strictly below one. We also determine the leading finite-height radius shift and the exact first inverse correction. No multiplicative equivalent, second-order deficit coefficient, or all-orders transseries is claimed.

10 Results and the shared foundation

An ascent sequence is a nonnegative integer word with $w_1 = 0$ and $w_i \leq 1 + \#\{j < i - 1 : w_j < w_{j+1}\}$ for $i \geq 2$. It avoids classical 120 if no $i < j < k$ satisfy $w_k < w_i < w_j$. The counting sequence, including $a_0 = 1$, is OEIS A202061 [1].

Let z be the smallest positive root of $z^3 - 5z^2 + 6z - 1 = 0$, and set

$$\begin{aligned} \rho &= 3z^2 - 10z + 2, & \mu &= \rho^{-1}, & t &= z^{-1}, \\ \alpha &= \frac{8z^2 - 29z + 9}{7}, & \kappa &= \frac{17z^2 - 59z + 13}{7}, & v &= \frac{2z^2 - 7z + 2}{2}. \end{aligned} \quad (10.1)$$

Here μ is the largest root of $\mu^3 - 8\mu^2 + 5\mu + 1 = 0$ and

$$\alpha = 0.510003374617888819\dots, \quad v = 0.346010735815047935\dots$$

All logarithms are natural.

Theorem 10.1. *With $\Phi(n) = n^{1/3}(\log n)^{2/3}$,*

$$\frac{n \log \mu - \log a_n}{\Phi(n)} \longrightarrow C_\star = \left(\frac{3\pi^2 \alpha^2}{2v} \right)^{1/3} = 2.232625307612844922\dots \quad (10.2)$$

If $a_n^{[\leq H]}$ denotes the positive height-walk expansion restricted to boundary heights at most H , and ρ_H is its generating-function radius, then

$$\log(\rho_H/\rho) \sim \frac{\alpha \log H}{2H}. \quad (10.3)$$

Finally, put $\lambda = \log \mu$ and $N(Y) = \min\{n : a_n \geq Y\}$. Then

$$\begin{aligned} N(Y) &= \frac{\log Y}{\lambda} + C_\star \lambda^{-4/3} (\log Y)^{1/3} (\log \log Y)^{2/3} \\ &\quad + o\left((\log Y)^{1/3} (\log \log Y)^{2/3}\right), \end{aligned} \quad (10.4)$$

where $C_\star \lambda^{-4/3} = 0.893568257650894909\dots$

This companion uses the exact recurrence, positive block formula and first-hit estimate proved in the separately supplied foundation report, printed as Part I of this report. That report is reproduced, with its source, checks and independent audit, in this package; it is unchanged. The shared statements needed here are restated below.

[write, 2 October 2026] In this report the four displays that the source restated verbatim from its foundation (the walk equation, the jump quadratic, the block formula and the first-hit bound) are not reprinted. Section 11 cites them from Part I, and every later reference to them in this Part points there.

The new proof and its independent review are research verification, not journal peer review or formal proof-assistant certification.

Conway, Conway, Elvey Price and Guttmann [2] proved the exponential growth-rate identification and numerically estimated a pure $n^{3/8}$ stretched correction. The earlier foundation report proved the different order $\Theta(\Phi(n))$. Equation (10.2) now determines its leading constant. A scoped source comparison accompanies the package; it does not establish comprehensive priority, and the original gap reduction may overlap the unpublished polynomial-time enumeration mentioned in [2].

11 The exact positive walk and the required kernel estimates

Part I establishes formal suffix series $e_h(x)$, $h \geq 1$, satisfying the positive walk equation (3.1), with $(Tf)_h = f_{h+1}$ and $A(x) = 1 + xe_1(x)$. The positive series $Q(x; u, T) = \sum_{q \geq 1} Q_q(x, T)u^q$ is the root with $Q(x; 0, T) = 0$ of the jump equation (2.5). This Part writes the jump variable as u (Part I: z) and the shift variable as T (Part I: t), because z and t denote the critical constants here. A macro-step with parameters q, r changes height by $d = 1 - q + r$, has weight $x[t_0^r]Q_q(x, t_0)/(1 - x)$, and is legal precisely when $q \leq h$ at its start. Its endpoint is automatically positive. All weights are nonnegative.

The exact coefficient formula for $B(\ell, q, r) := [x^\ell T^r]Q_q(x, T)$ is (3.4), with $\mathcal{C}(j, k)$ as in (3.3); only admissible binomial terms are included. Let $k(L, q, d)$ be the coefficient of total macro-degree $L \geq 1$ for a step with increment d , and define the critically tilted weight

$$w(L, q, d) = k(L, q, d)\rho^L t^d.$$

The unrestricted row mass is

$$m = \sum_{L, q, d} w(L, q, d) = \frac{\rho}{(1 - \rho)z + \rho} = 0.4450418679 \dots < 1. \quad (11.1)$$

Part I also gives a constant $D > 0$ such that $a_n^{[>H]} \rho^n \leq C \exp[-H^2/(4Dn)]$ for $H = o(n)$: this is the first-hit bound (7.2), multiplied by ρ^n . The terminal factor t^{1-h} at final height h must be retained in all critically tilted coefficient sums.

11.1 Uniform transfer at fixed down parameter

Let $u_c(x, T)$ be the smaller positive discriminant root in u in (2.5). Near $(x, T) = (\rho, t)$ the two discriminant roots are distinct, positive and below 1. The explicit square-root branch therefore gives uniformly

$$Q_q(x, T) = c(x, T)u_c(x, T)^{-q}q^{-3/2}(1 + O(q^{-1})), \quad c(x, T) > 0,$$

where c is analytic. In particular, the same expression without $1 + O(q^{-1})$ is an upper bound after increasing a constant, for all $q \geq 1$.

Define

$$\Psi(s, \theta) = -\log(u_c(\rho e^s, t e^\theta) t e^\theta).$$

Implicit differentiation of the quadratic discriminant gives the exact identities

$$\Psi(0, 0) = \Psi_\theta(0, 0) = 0, \quad \Psi_s(0, 0) = \alpha^{-1}, \quad \Psi_{\theta\theta}(0, 0) = v/\alpha. \quad (11.2)$$

Consequently

$$\Psi(s, \theta) = \frac{s + v\theta^2/2}{\alpha} + O(s^2 + |s\theta| + |\theta|^3). \quad (11.3)$$

The fixed- q height variance is $\nu = v/\alpha = (1 + 2z - z^2)/2 > 0$. Analytic transfer also proves its Gaussian limit; only a positive mass on fixed intervals is used in Section 14. All identities in (11.2) are checked exactly by the supplied symbolic certificate.

11.2 A local positive lower bound

For $L(v) = v \log v$, the factorial entropy of the $j = q$ summand in (3.4) is

$$\begin{aligned} S(a, b, u) &= 2L(a + u) - L(a) - 2L(u) - L(b - u) - L(a - b + u) \\ &\quad + L(1 + a) - L(1 - b - u) - L(a + b + u). \end{aligned}$$

At $(a, b, u) = (\alpha, \alpha, \kappa)$,

$$S = \log \mu, \quad S_u = 0, \quad S_a + S_b = 0, \quad S_b = \log z.$$

The Hessian $\mathcal{H}$ of $S(a, a + s, u)$ there is negative definite. Its inverse covariance $(-\mathcal{H})^{-1}$ has s, s entry v . The upper-left 2×2 block, in coordinates (a, s) , is

$$\Sigma = \begin{pmatrix} -(11z^2 - 37z + 4)/7 & 2(5z^2 - 17z + 2)/7 \\ 2(5z^2 - 17z + 2)/7 & (2z^2 - 7z + 2)/2 \end{pmatrix}. \quad (11.4)$$

Its conditional q -center has shift $\beta(d - 1)$, where $\beta = \Sigma_{12}/v$.

Lemma 11.1. *Fix $C > 0$ and $K > 3$. For all sufficiently large n , uniformly for $\ell \geq (\log n)^K$ and $|d| \leq C\sqrt{\ell \log n}$, summing (3.4) over a fixed-width $O(\sqrt{\ell})$ window centered at $q = \alpha\ell + \beta(d - 1)$ gives*

$$\rho^\ell t^d \sum_q B(\ell, q, q + d - 1) \geq c\ell^{-2} \exp[-d^2/(2v\ell) - o(1)]. \quad (11.5)$$

All retained q satisfy $|q - \alpha\ell| \leq C'\sqrt{\ell \log n}$.

Proof. Retain only the positive $j = q$ terms, and restrict (q, k) to fixed-width $O(\sqrt{\ell})$ windows about their joint Gaussian conditional centers at fixed $d - 1$. Uniform Stirling gives each summand a factor at least $c\ell^{-3}$. Taylor expansion of the tilted entropy gives

$$-\frac{(d - 1)^2}{2v\ell} - O(1) - O\left(\frac{(\log n)^{3/2}}{\sqrt{\ell}}\right).$$

All factorial arguments remain in a fixed positive interior set. The last remainder is $o(1)$, and the number of retained integer pairs is at least $c\ell$. Replacing $d - 1$ by d changes the exponential by $1 + o(1)$ and the tilt by a fixed factor. This proves (11.5). It requires no global coefficient-tail estimate or full local asymptotic equivalent. $\square$

12 A sharp upper coefficient bound

Set

$$H = n^{2/3}(\log n)^{1/3}, \quad F = H^2/n = \Phi(n), \quad a = F/n, \quad b = F/H,$$

so $b^2 = a$ and $aH = \log n$.

Lemma 12.1 (Uniform row calibration). *Fix $R, \epsilon > 0$, $0 < B < \alpha/3$, and a continuously differentiable function f on $[0, 1] \times [0, R]$ with uniformly continuous derivatives, satisfying*

$$-f_s(s, y) + \frac{v}{2}f_y(s, y)^2 \leq \frac{B}{y + \epsilon}. \quad (12.1)$$

For elapsed degree m , current height h , and legal next step (L, q, d) , restrict to $m + L \leq n$ and $1 \leq h + d \leq RH$. Then the row sum of

$$w(L, q, d) \exp \left(F \left[f \left(\frac{m}{n}, \frac{h}{H} \right) - f \left(\frac{m+L}{n}, \frac{h+d}{H} \right) \right] \right) \quad (12.2)$$

is at most a fixed $\bar{m} < 1$ for all sufficiently large n , uniformly in m, h .

Proof. For $L/n, |d|/H \leq \eta$, uniform differentiability gives the upper linear majorant

$$a[-f_s + \omega]L - bf_y d + b\omega|d|,$$

where ω is arbitrarily small when the fixed η is small. For either sign of d this is $\lambda L + \theta d$ with

$$\lambda = a[-f_s + \omega], \quad \theta = b[-f_y \pm \omega].$$

Choose ω so that, for some $B < B' < \alpha/3$,

$$\lambda + v\theta^2/2 \leq aB'/(y + \epsilon).$$

The bounded positive macro prefactor and (11.3) show uniformly for $q \leq h \leq RH$ that

$$q\Psi(\lambda, \theta) \leq \frac{B'}{\alpha} \frac{q \log n}{h + \epsilon H} + o(1),$$

since $H(a^2 + ab + b^3) = O(n^{-1/3}(\log n)^{4/3}) = o(1)$.

For the tail $q > Q$, split at $H/(\log n)^2$. Below this threshold the exponential factor is $1 + o(1)$, giving $O(Q^{-1/2})$. Above it the total is at most

$$C \left(\frac{H}{(\log n)^2} \right)^{-1/2} n^{B'/\alpha} = o(1).$$

For $q \leq Q$, fixed- q generating functions provide exponential domination on a fixed neighborhood of (ρ, t) . First restricting L, d to a finite set and then controlling its complement shows a uniform limsup at most the unrestricted critical row. Equality is not asserted near $m = n$, where the degree cutoff can remove terms. Splitting the two signs of d is needed only in the vanishing $q > Q$ tail. Thus these local steps have row limsup at most $m < 1$.

For the remaining steps, global Lipschitz continuity bounds the exponent in (12.2) by $CaL + Cb|d|$. If $L > \eta n$, an additional small fixed length tilt $s_0 > 0$ bounds its sum by $\exp[-s_0 \eta n + O(H)] = o(1)$; fixed- q transfer at ρe^{s_0} has at most exponential growth in $q \leq RH$. If $|d| > \eta H$, use an additional signed height tilt θ_0 . Equation (11.3) at vanishing length tilt gives $\Psi \leq C_1 \theta_0^2 + o(1)$. Choosing θ_0 small enough that $C_1 R \theta_0^2 < \eta \theta_0/2$ gives an $\exp[-cH + o(H)]$ bound. Two signs cover both height tails. These bounds are uniform in m, h and complete the proof. $\square$

Suppose also $f(1, y) \leq Py$ for a finite P . Summing all finite transformed prefixes of degree at most $n - 1$ gives at most $(1 - \bar{m})^{-1}$. The exponential transformations telescope. The exact terminal coefficient then gives

$$a_n^{[\leq RH]} \rho^n \leq C \exp[-Ff(0, 1/H)]. \quad (12.3)$$

Indeed a prefix of degree m ending at h leaves the factor

$$\rho^{n-1-m} t^{1-h} \exp[Ff(m/n, h/H)],$$

which is uniformly bounded: $f(s, y) \leq Py + C(1 - s)$, whereas $F/H, F/n \rightarrow 0$. The original leading letter contributes only the fixed factor ρ . This accounts for every possible unused terminal geometric degree and for arbitrary final height.

Combine (12.3) with (7.2), choosing R so large that $R^2/(4D) > f(0, 0)$. It follows that

$$\liminf_{n \rightarrow \infty} \frac{n \log \mu - \log a_n}{\Phi(n)} \geq f(0, 0). \quad (12.4)$$

12.1 Explicit calibrations attaining the constant

Take a smooth bounded increasing function p on $[0, 1]$ with $p' > 0$. Set

$$f(s, y) = g(s) + p(s)y, \quad g(1) = 0, \quad g' = \frac{v}{2}p^2 - 2\sqrt{Bp'} + \epsilon p'.$$

The arithmetic-geometric mean inequality gives

$$-f_s + \frac{v}{2}f_y^2 = 2\sqrt{Bp'} - p'(y + \epsilon) \leq \frac{B}{y + \epsilon}.$$

Thus (12.4) applies, and

$$f(0, 0) = \int_0^1 \left(2\sqrt{Bp'} - \frac{v}{2}p^2 \right) ds - \epsilon[p(1) - p(0)]. \quad (12.5)$$

To evaluate attainable values, put $M_0 = (2vB/\pi^2)^{1/3}$ and parametrize

$$s = \frac{u - \sin u \cos u}{\pi}, \quad y = M_0 \sin^2 u, \quad 0 < u < \pi.$$

Then $y' = \pi M_0 \cot u$, $y'' = -vB/y^2$, and $p_\star = -y'/v$ is increasing with $p'_\star = B/y^2$. Both $p_\star^2$ and $\sqrt{p'_\star}$ are integrable at the endpoints. Restrict $p_\star$ to $[\delta, 1 - \delta]$ and linearly reparametrize to $[0, 1]$. The resulting smooth bounded p makes the integral in (12.5) converge, as $\delta \rightarrow 0$, to

$$\int_0^1 \left(\frac{2B}{y} - \frac{y'^2}{2v} \right) ds = \frac{3B}{M_0} = 3 \left(\frac{\pi^2 B^2}{2v} \right)^{1/3}.$$

Here $y'^2/(2v) = B(1/y - 1/M_0)$ and $\int_0^1 B/y ds = 2B/M_0$. First let $\epsilon \rightarrow 0$ for each fixed truncation, then $\delta \rightarrow 0$, and finally $B \uparrow \alpha/3$. This proves

$$\liminf \frac{n \log \mu - \log a_n}{\Phi(n)} \geq C_\star. \quad (12.6)$$

13 A sharp lower coefficient bound at every integer length

Write $L = \log n$ and choose

$$\begin{aligned} M &= \text{round} \left[\left(\frac{12\pi^2 \alpha^2 n}{vL} \right)^{1/3} \right], & J &= \lceil L^3 \rceil, & \eta &= L^{-1/4}, \\ s_j &= \sin^2(\pi j/M), & S &= \sum_{j=J}^{M-J-1} s_j, & H_0 &= \frac{\alpha n}{(1-\eta)S}, & h_j &= \lfloor H_0 s_j \rfloor. \end{aligned}$$

There are $m = M - 2J$ large blocks. Put $h_0 = h_J = h_{M-J}$ and $n' = n - m - 2h_0 - 1$. Elementary sine estimates give

$$S \sim M/2, \quad H_0 \sim 2\alpha n/M, \quad h_0 = \Theta(L^7), \quad \min_j h_j = \Theta(L^7), \quad H_0/M^2 = \Theta(L).$$

Choose positive integers $\ell_j = (n'/S)s_j + O(1)$ whose sum is exactly n' by rounding down and distributing the fewer than m remaining units. Let $g_j = h_{j+1} - h_j$. Then

$$\sum_j g_j = 0, \quad \ell_j = \frac{(1-\eta)h_j}{\alpha} [1 - o(\eta)] + O(1), \quad |g_j| \leq C\sqrt{\ell_j L}, \quad \ell_j \geq cL^7. \quad (13.1)$$

For each block independently allow

$$\ell = \ell_j + u_j, \quad |u_j| \leq w_j = \lfloor \ell_j / L^2 \rfloor, \quad d = g_j + v_j, \quad |v_j| \leq b_j = \lfloor \sqrt{\ell_j} \rfloor,$$

and retain choices satisfying

$$\sum_j u_j = 0, \quad \sum_j v_j = 0. \quad (13.2)$$

For every pair (ℓ, d) use the shifted q -window from Lemma 11.1.

13.1 Legality under both global constraints

The actual starting height of block j is $h_j + E_j$, where E_j is the partial sum of the v_i . By using both this prefix and its complementary suffix in (13.2),

$$|E_j| \leq \min \left(\sum_{i < j} b_i, \sum_{i \geq j} b_i \right) \leq C h_j / \sqrt{L}. \quad (13.3)$$

On the ascending half, $b_i \leq C \sqrt{n/M} \sin(\pi i/M)$, and

$$\sum_{i=J}^{j-1} \sin(\pi i/M) \leq CM[1 - \cos(\pi j/M)] \leq CM s_j.$$

Divide by $h_j \asymp (n/M)s_j$ and use $M^{3/2}/\sqrt{n} = \Theta(L^{-1/2})$. The descending half follows from the complementary suffix and symmetry. Rounding errors are smaller because $h_j \geq cL^7$.

All selected q satisfy

$$q \leq \alpha \ell_j (1 + O(L^{-2})) + C \sqrt{\ell_j L}.$$

The margin ηh_j dominates (13.3), the square-root term and all rounding errors: $\eta = L^{-1/4}$, while $|E_j|/h_j = O(L^{-1/2})$ and $\sqrt{L/h_j} = O(L^{-3})$. Thus $q \leq h_j + E_j$, and the entire selected walk is legal and positive.

13.2 Exact constraints and exact degree

The coefficients of the generating polynomial of an integer interval $[-w, w]$ are symmetric and unimodal. Convolution preserves those properties, so the number of length choices with zero total displacement is at least

$$\frac{\prod_j (2w_j + 1)}{2 \sum_j w_j + 1}.$$

The analogous bound holds separately for the increment choices. Hence the number of pairs satisfying (13.2) is at least

$$\frac{\prod_j (2w_j + 1)(2b_j + 1)}{(2n + 1)^2}. \quad (13.4)$$

For clarity, convolution preserves symmetric unimodality because each such sequence is a nonnegative linear combination of centered interval indicators, whose pairwise convolutions are symmetric unimodal.

Seed from height 1 to h_0 using $q = 1, r = h_0 - 1$, of degree $2h_0 - 1$ and multiplicity at least one. After the large blocks return to h_0 , return to height 1 using $q = h_0, r = 0$, of degree one and

multiplicity one. These seeds follow from $Q_1(x, T) = 1/(1 - x - x^2T/(1 - x))$ and $B(0, q, 0) = 1$. Including the original leading letter and the leading x of every macro-step, the total degree is exactly

$$2h_0 + 1 + m + \sum_j \ell_j = n.$$

The large-block height increments sum to zero. Thus all height tilts cancel, the fixed seed costs $O(h_0) = O(L^7) = o(\Phi(n))$, and fixed per-block factors cost $O(M) = o(\Phi(n))$. This holds at every sufficiently large integer n .

13.3 The kinetic and polynomial costs

Uniformly throughout the rectangles,

$$\frac{d^2}{\ell} \leq \frac{g_j^2}{\ell_j} + O(\sqrt{L}) + O(1), \quad \log \ell = \log \ell_j + o(1).$$

Combine Lemma 11.1, (13.4), $w_j \sim \ell_j/L^2$ and $b_j \sim \sqrt{\ell_j}$ to obtain

$$\begin{aligned} \log(a_n \rho^n) &\geq - \sum_j \frac{g_j^2}{2v\ell_j} - \frac{1}{2} \sum_j \log \ell_j \\ &\quad - O(M\sqrt{L} + M \log L + \log n + L^7). \end{aligned} \quad (13.5)$$

The final line is $o(\Phi(n))$.

The sine profile gives

$$\sum_j \frac{g_j^2}{\ell_j} = (1 + o(1)) \frac{4\pi^2 \alpha^2 n}{M^2}. \quad (13.6)$$

Indeed, before rounding, it is $(H_0^2 S/n') \sum_j (s_{j+1} - s_j)^2 / s_j$. Taylor expansion with an absolute remainder, followed by summation, gives

$$\sum_j \frac{(s_{j+1} - s_j)^2}{s_j} = (1 + o(1)) \frac{4\pi^2}{M^2} \sum_j \cos^2(\pi j/M) = (1 + o(1)) \frac{2\pi^2}{M}.$$

No pointwise relative expansion at the vanishing central derivative is used. The endpoint removal has $J/M \rightarrow 0$ and $J \rightarrow \infty$, and roundings are negligible since $\ell_j \geq cL^7$.

Also

$$\frac{1}{2} \sum_j \log \ell_j = (1 + o(1)) \frac{M}{3} L. \quad (13.7)$$

Here $\log(n'/S) = \frac{2}{3}L + O(\log L)$ and $\sum_j \log s_j = O(M) + o(M)$. The latter follows from

$$\prod_{j=1}^{M-1} \sin(\pi j/M) = M/2^{M-1}.$$

The omitted endpoint terms are $O(JL) = o(M)$.

Equations (13.5)–(13.7) give the leading cost

$$\frac{2\pi^2 \alpha^2 n}{vM^2} + \frac{M}{3} L.$$

Our choice of M makes the first term asymptotic to $ML/6$, so the total is asymptotic to $ML/2 = C_\star \Phi(n)$. Therefore

$$\limsup \frac{n \log \mu - \log a_n}{\Phi(n)} \leq C_\star.$$

Together with (12.6), this proves (10.2).

14 The finite height radius

Let $K_H(x)$ be the finite matrix of exact macro-transition series on heights $1, \dots, H$. For fixed q, r ,

$$[T^r]Q_q(x, T) = \mathbf{1}_{r=0} + \sum_{j=1}^q \sum_{k=\max(0, r-j)}^r \mathcal{C}(j, k) \binom{j}{r-k} \frac{x^{r+k+1}}{(1-x)^{j+r+k}}.$$

Hence every entry is rational and analytic for $0 < x < 1$. It is positive: $q = h, r = h' - 1$ connects any h to any h' . The strictly increasing Perron eigenvalue starts at zero and diverges at one, so it crosses one at a unique ρ_H . The initial vector and the positive terminal vector have positive Perron projections, so the resulting pole is noncancelled. This is the restricted radius. Diagonal similarity by t^h does not change it.

Fix $0 < \delta < 1$ and put $s_- = (1 - \delta)\alpha \log H / (2H)$. Uniform transfer gives

$$Q_q(\rho e^{s_-}, t) z^q \leq C q^{-3/2} \exp(q s_- / \alpha + O(q s_-^2)).$$

The $q \leq H$ row mass differs from its critical value by $o(1)$, splitting at $H / (\log H)^2$ as before. The upper tail is bounded by $C H^{-\delta/2} \log H = o(1)$. Thus all rows are below one and $\rho_H > \rho e^{s_-}$.

For the reverse bound set $s_+ = (1 + \delta)\alpha \log H / (2H)$ and choose $\max(1/2, 1 - \delta/2) < \gamma < 1$. Put $w = \lfloor H^\gamma \rfloor$ and restrict to heights $[H - w, H]$. In every row retain $q \in [H - 2w, H - w]$, all legal. Uniform transfer gives

$$Q_q(\rho e^{s_+}, t) z^q \geq c H^{-3/2 + (1 + \delta)/2}.$$

For these q , the tilted actual increment $d = 1 + r - q$ has mean $O(\log H)$ and variance asymptotic to νq , where $\nu = v/\alpha > 0$. The transfer formula and its derivatives give, for fixed real u ,

$$\mathbb{E} \exp(ud/\sqrt{q}) \longrightarrow \exp(\nu u^2/2)$$

uniformly in this range: the mean term is $O(\log H / \sqrt{H}) = o(1)$ and the cubic remainder vanishes. Hence weak Gaussian convergence places a fixed positive mass in either inward increment interval $[0, \sqrt{H}]$ or $[-\sqrt{H}, 0]$, according to the half of the strip containing the start height. Since $w/\sqrt{H} \rightarrow \infty$, these increments remain in the strip. Each restricted row sum is therefore at least

$$c H^{\gamma - 1 + \delta/2} \longrightarrow \infty.$$

The Perron row bound implies $\rho_H < \rho e^{s_+}$. Letting $\delta \downarrow 0$ proves (10.3). Only weak Gaussian convergence is needed here, not a lattice local limit theorem.

[write, 2 October 2026] Part III, equation (21.1), refines (10.3) to

$$\log(\rho_H/\rho) = \frac{\alpha}{H} \left\{ \frac{1}{2} \log H + \log \log H + c_* + o(1) \right\}$$

with an explicit constant c_* .

15 Inverse asymptotics and remaining questions

Appending the final letter gives an injection from avoiding words of length n to those of length $n + 1$, so a_n is nondecreasing. Set $L = \log Y$ and insert

$$n = \frac{L}{\lambda} + \left(C_* \lambda^{-4/3} \pm \epsilon \right) L^{1/3} (\log L)^{2/3}$$

into (10.2). Since $\Phi(n) \sim \lambda^{-1/3} L^{1/3} (\log L)^{2/3}$, the two choices fall below and above the threshold for sufficiently large Y . Integer rounding costs $O(1)$. Letting $\epsilon \downarrow 0$ proves (10.4).

The theorem is an exact first-order equivalent for the logarithmic deficit. It is not a multiplicative equivalent for a_n : the $o(\Phi(n))$ remainder may diverge. No amplitude, power-law prefactor, second-order coefficient, transseries, exact integer inverse rule, or finite-data crossover threshold is established. The leading proof discards integrated block-length factors, endpoint layers, the defective critical mass and fluctuations about the minimizing profile at lower order. Recovering these is separate work.

[write, 2 October 2026] In this report the second-order coefficient is proved in Part III (Theorem 16.1), the next constant in Part IV, every fixed inverse-logarithmic order in Part V, and the inverse expansion to the same orders in Corollary 20.1, Corollary 28.1 and Theorem 35.1. No amplitude, power-law prefactor, transseries, exact integer inverse rule or crossover threshold is established anywhere in the report.

The accompanying archive contains the editable source and PDF, the shared foundation report, exact symbolic certificates, independent checks and audit reports, verification scripts and file hashes. The numerical constants are evaluations of exact algebraic expressions, not curve fits. The scoped source check and prior unpublished enumeration caveat remain applicable.

Part III

The second logarithmic deficit of 120 avoiding ascent sequences

Source. Batch-77 manuscript 40, archive `oeis-a202061-second-order-report.zip`, main file `a202061-second-order.tex` (141 lines) with five `\input` files (`proofs/lower-section.tex`, `upper-section.tex`, `corollaries-section.tex`, `finite-height-section.tex`, `conjectural-outlook.tex`), inlined here in the order of the source; dated 2 October 2026. Author line: “A proof and reproducible research companion”. Printed in full; label prefix `a61:o2:`.

Abstract of the source.

Let a_n count classical 120-avoiding ascent sequences, OEIS A202061. We refine the sharp leading logarithmic deficit to

$$n \log \mu - \log a_n = C_\star n^{1/3} (\log n)^{2/3} \left[1 + \frac{7}{3} \frac{\log \log n}{\log n} + O\left(\frac{1}{\log n}\right) \right],$$

where $C_\star = 2.232625307612844922\dots$ is the previously established leading constant. The coefficient lower bound sums balanced length and increment rectangles and uses Jensen’s inequality to preserve cancellation. The matching upper bound uses a finite- n affine calibration for the full logarithmic potential, with quantitative endpoint and transformed-row estimates. An inverse correction and a finite-height radius refinement are also proved. The remainder is additive in the logarithm; no multiplicative equivalent or next constant in the global deficit is claimed.

16 Statement and dependency boundary

An ascent sequence is a nonnegative integer word with $w_1 = 0$ and $w_i \leq 1 + \#\{j < i - 1 : w_j < w_{j+1}\}$ for $i \geq 2$. It avoids classical 120 if no $i < j < k$ satisfy $w_k < w_i < w_j$. Its counting sequence, with $a_0 = 1$, is OEIS A202061 [1]. Let z be the smallest positive root of $z^3 - 5z^2 + 6z - 1 = 0$, and set

$$\begin{aligned} \rho &= 3z^2 - 10z + 2, & \mu &= \rho^{-1}, & t &= z^{-1}, \\ \alpha &= \frac{8z^2 - 29z + 9}{7}, & v &= \frac{2z^2 - 7z + 2}{2}. \end{aligned}$$

Thus μ is the largest real root of $\mu^3 - 8\mu^2 + 5\mu + 1 = 0$, $\alpha = 0.510003374617888819\dots$ and $v = 0.346010735815047935\dots$. Set $z_* = z$ and $t_* = t$. All logarithms are natural, and $L = \log n$. Our principal scales and constants are

$$H = n^{2/3}L^{1/3}, \quad F = \frac{H^2}{n} = n^{1/3}L^{2/3}, \quad C_* = \left(\frac{3\pi^2\alpha^2}{2v}\right)^{1/3}. \quad (16.1)$$

Theorem 16.1. *For all sufficiently large integers n ,*

$$D_n := n \log \mu - \log a_n = C_* F + \frac{7C_*}{3} \frac{F \log L}{L} + O\left(\frac{F}{L}\right). \quad (16.2)$$

Equivalently,

$$\frac{D_n - C_* F}{F \log \log n / \log n} \rightarrow \frac{7C_*}{3}.$$

The constants in the O terms are independent of n . The theorem improves the independently reviewed leading result $D_n \sim C_* F$ of Part II. It is not obtained by fitting finite sequence coefficients. [write, 5 October 2026] *A second instance of the method.* The report `a279551-inversion-log-deficit` in this directory (batch 104) proves results of the same shape as Parts I–III for two inversion-sequence classes of Britt and Beaton, OEIS A279551 and A279556, through a different combinatorial model (commitment trees): the order $F = n^{1/3}L^{2/3}$ of the deficit, its sharp constant $\sigma_D = (3\pi^2/(2D))^{1/3}$, and the second term, $\sigma_D F + \frac{7\sigma_D}{3} F \log L/L + O(F/L)$ (its Theorem 29.1, `ivd:11:thm:main`), the counterpart of Theorem 16.1. There the critical local root has the form $(\frac{1}{2} \log p + \log \log p + c)/p$; here it is the row threshold $\kappa(r)/r = (\frac{1}{2} \log r + \log \log r - C_0/2)/r$ of Section 19, measured in the exponent Ψ , which carries a factor $1/\alpha$. Its global potential $\frac{1}{3} + \frac{7}{6} \log L/L$ corresponds to $B_n = \frac{\alpha}{3} + \frac{7\alpha}{6} \frac{\log L}{L} + O(1/L)$ of (19.5). Its Remark 1.2 (`ivd:sc:rem:bb`) proves that Britt and Beaton’s numerically estimated forms with a stretched exponent $3/8$ (their (4.12)–(4.13), made with A. J. Guttmann’s assistance) fail, as Corollary 1.2 does for the estimate of Conway, Conway, Elvey Price and Guttmann. There $D = v/\alpha$ is written with that report’s own α , v and height normalization, against $C_* = (3\pi^2\alpha^2/(2v))^{1/3}$ here; the sequences and the constants are different, and no relation between them is asserted. Neither source manuscript cites the other; that report’s Remark 29.2 (`ivd:11:rem:a202061`) records the parallel from its side. That report does not determine its next constant: the analogue of Part IV’s κ is the first open question of its Section 39, for which Parts IV and V are the model.

The previous sharp report, including its unchanged foundation report, kernel checks and audit records, is copied byte-for-byte into the accompanying package. The exact positive representation, local kernel estimates and first-hit inequality used here are inputs from those reports; we restate their needed forms next.

[write, 2 October 2026] This section restates inputs proved earlier in this report, in this Part’s notation and with this Part’s labels, which its proofs cite: the walk equation and jump quadratic, (3.1) and (2.5) (with u, T for Part I’s z, t); the row mass m_* , (11.1); the uniform transfer and the expansion of Ψ , Section 11 and (11.3); the local lower bound, Lemma 11.1; and the first-hit bound (7.2) (D_0 here is D there). “The foundation” is Part I and “the sharp report” Part II.

The new two-sided argument and its scope have a separate audit in this package. This is mathematical research verification, not journal peer review or formal proof-assistant certification. No new priority claim is made relative to the source comparisons already recorded in the dependency reports.

17 Positive kernel inputs

The foundation gives formal suffix series

$$e_h = \frac{1}{1-x} + \frac{x}{1-x} \sum_{q=1}^h (Q_q(x, T)e)_{h+1-q}, \quad (Tf)_h = f_{h+1}, \quad A(x) = 1 + xe_1(x). \quad (17.1)$$

The positive series $Q(x; u, T) = \sum_{q \geq 1} Q_q(x, T)u^q$ is the root vanishing at $u = 0$ of

$$xT(1-u)Q^2 + [(1-x)(u - (1-x)) + xT(x-u)]Q + (1-x)u = 0. \quad (17.2)$$

A macro-step with parameters q, r changes height by $d = 1 - q + r$, has generating weight $x[T_0]Q_q(x, T_0)/(1-x)$, and is legal exactly when $q \leq h$ at its start. Its endpoint is then positive. Let $k(e, q, d)$ be its multiplicity at macro-length e , and set

$$w(e, q, d) = k(e, q, d)\rho^e t^d.$$

The unrestricted critical row has defective total mass

$$m_\star = \sum_{e, q, d} w(e, q, d) = \frac{\rho}{(1-\rho)z + \rho} < 1. \quad (17.3)$$

For small real (s, θ) write

$$W_q(s, \theta) = \sum_{e, d} w(e, q, d)e^{se + \theta d}.$$

Uniform fixed- q square-root transfer and the exact root derivatives give

$$W_q(s, \theta) \leq Cq^{-3/2} \exp\{q\Psi(s, \theta)\}, \quad (17.4)$$

$$\Psi(s, \theta) = \frac{s + v\theta^2/2}{\alpha} + O(s^2 + |s\theta| + |\theta|^3). \quad (17.5)$$

These estimates hold in one fixed real neighborhood of zero, uniformly in $q \geq 1$; each fixed W_q is analytic there. The underlying transfer also has a positive asymptotic amplitude, used only for the auxiliary finite-height theorem.

For the coefficient lower construction the audited local positive estimate says that, for any fixed $K > 3$, $e \geq L^K$ and $|d| \leq C\sqrt{eL}$, summing over a fixed sufficiently wide $O(\sqrt{e})$ window about the conditional center

$$q = \alpha e + \beta(d-1), \quad \beta = -\frac{4(z-3)(2z-1)}{7},$$

gives tilted block weight at least

$$ce^{-2} \exp\left\{-\frac{d^2}{2ve} - \epsilon_n\right\}, \quad \epsilon_n = o(1), \quad (17.6)$$

uniformly in that range. Every retained q obeys $|q - \alpha e| \leq C'\sqrt{eL}$. The leading x of $x/(1-x)$ adds exactly one to the macro-length and a fixed factor to its tilted weight.

The exact residual terminal factor for a prefix of macro-length k ending at height h , at total word length n , is

$$\rho^{n-1-k} t^{1-h}, \quad (17.7)$$

apart from the fixed leading factor ρ . The foundation's first-hit bound supplies a fixed $D_0 > 0$ such that, for fixed R and all sufficiently large n ,

$$a_n^{[>RH]} \rho^n \leq C \exp\{-R^2 F/(4D_0)\}. \quad (17.8)$$

All these inputs are proved and checked in the unchanged dependency; none is inferred from a discriminant alone.

18 Coefficient lower bound by balanced rectangles

We prove the one-sided refinement

$$D_n \leq C_* F + \frac{7C_*}{3} \frac{F \log L}{L} + O(F/L), \quad L = \log n. \quad (18.1)$$

Its new ingredient is a symmetric set of balanced increment bridges. It allows a height margin of order $1/L$, while Jensen's inequality removes the otherwise excessive linear variation of the kinetic cost.

18.1 The local positive kernel estimate

Let $B(\ell, q, r)$ denote the exact positive block multiplicity in the frozen height-walk representation, before the leading macro multiplier $x/(1-x)$. We use only its audited local lower bound: for every fixed $C > 0$, uniformly when $\ell \geq cL^7$ and $|d| \leq C\sqrt{\ell L}$, summing over a fixed-width conditional Gaussian window in q gives

$$\sum_{q \text{ in window}} \rho^\ell t_*^d B(\ell, q, q+d-1) \geq c_0 \ell^{-2} \exp\left\{-\frac{d^2}{2v\ell} - \epsilon_n\right\}, \quad \epsilon_n \rightarrow 0. \quad (18.2)$$

Every q in this window satisfies $|q - \alpha\ell| \leq C_0\sqrt{\ell L}$. The estimate follows by retaining the $j = q$ summand in the exact binomial formula and summing q and its inner saddle variable in fixed $O(\sqrt{\ell})$ windows around their conditional centers. Uniform Stirling and the audited Hessian identity give the exponent $d^2/(2v\ell)$, while the cubic remainder is $O(L^{3/2}/\sqrt{\ell}) = o(1)$. Thus no global kernel equivalent or multiplicative amplitude is needed here. We retain only the leading x in $x/(1-x)$, at a fixed cost per macro-block.

18.2 A sine-square path with a small legality margin

Put

$$A_0 = \frac{2\pi^2 \alpha^2}{v}, \quad M = \text{round}\left[\left(\frac{6A_0 n}{L}\right)^{1/3}\right], \quad J = \lceil L^3 \rceil, \quad \eta = 4/L, \quad m = M - 2J.$$

For $J \leq j \leq M - J$ define

$$s_j = \sin^2(\pi j/M), \quad S = \sum_{j=J}^{M-J-1} s_j, \quad H_0 = \frac{\alpha n}{(1-\eta)S}, \quad h_j = \lfloor H_0 s_j \rfloor.$$

Then $h_J = h_{M-J} =: h_0$. Set $n' = n - m - 2h_0 - 1$ and choose positive integers ℓ_j such that

$$\sum_{j=J}^{M-J-1} \ell_j = n', \quad \ell_j = \frac{n'}{S} s_j + O(1).$$

Flooring followed by distributing the rounding residual gives this choice for every sufficiently large integer n . Let $g_j = h_{j+1} - h_j$, so $\sum g_j = 0$. Uniformly over these indices,

$$h_j \asymp (n/M)s_j, \quad \ell_j \asymp (n/M)s_j, \quad \min_j(h_j, \ell_j) \asymp LJ^2 \asymp L^7, \quad (18.3)$$

$$|g_j| \leq C\sqrt{\ell_j L}, \quad \alpha \ell_j \leq (1 - 4/L)h_j + O(1). \quad (18.4)$$

The last inequality uses $n'/n < 1$; replacing $H_0 s_j$ by its floor changes its right side by only $O(1)$.

For each block allow integer deviations

$$|u_j| \leq w_j := \lfloor \ell_j/L \rfloor, \quad |v_j| \leq b_j := \lfloor \sqrt{\ell_j} \rfloor.$$

Impose the two exact constraints

$$\sum_j u_j = 0, \quad \sum_j v_j = 0. \quad (18.5)$$

The actual internal block length and increment are $\ell_j + u_j$ and $g_j + v_j$, respectively.

18.3 Many balanced increment bridges stay inside the margin

First take independent random variables V_j uniform on the integer intervals $[-b_j, b_j]$. A centered interval has symmetric unimodal coefficients, and convolution preserves this property. Therefore its central coefficient is at least the average coefficient, giving

$$\mathbb{P}\left\{\sum_j V_j = 0\right\} \geq \frac{1}{2\sum_j b_j + 1} \geq \frac{1}{2n + 1}. \quad (18.6)$$

For an interior boundary k let $E_k = \sum_{j=J}^{k-1} V_j$. Define the event

$$G = \{|E_k| \leq h_k/L \text{ for every } J \leq k \leq M - J\}.$$

For $J \leq k \leq M/2$, elementary sine bounds and $n/M^3 \asymp L$ give

$$\sum_{j=J}^{k-1} b_j^2 \leq CLk^3, \quad \max_{J \leq j < k} b_j \leq C\sqrt{L}k, \quad h_k/L \geq ck^2.$$

Bernstein's inequality consequently yields

$$\mathbb{P}\{|E_k| > h_k/L\} \leq 2 \exp\left\{-\frac{ck^4}{Lk^3 + \sqrt{L}k^3}\right\} \leq 2 \exp(-ck/L) \leq 2 \exp(-cL^2). \quad (18.7)$$

For $k > M/2$, use the complementary suffix, which equals $-E_k$ when the total sum vanishes. The same unconditional estimate applies with $M - k$ in place of k . Dividing by (18.6) and taking a union bound over at most M boundaries proves

$$\mathbb{P}\left(G^c \mid \sum_j V_j = 0\right) \leq CnM \exp(-cL^2) \leq \exp(-c'L^2). \quad (18.8)$$

At the two endpoints the bridge error is zero. In particular, at least half of the balanced increment vectors are good for large n . No independence after conditioning is asserted or needed.

Both balance and G are invariant under the simultaneous sign reversal $V_j \mapsto -V_j$. The uniform distribution on good balanced increment vectors therefore has

$$\mathbb{E}V_j = 0, \quad \mathbb{E}V_j^2 \leq b_j^2 \leq \ell_j. \quad (18.9)$$

Independently, choose the length vector uniformly among the balanced vectors in the u -rectangles. It too is centrally symmetric, so

$$\mathbb{E}U_j = 0, \quad |U_j| \leq \ell_j/L. \quad (18.10)$$

Independence here is between the two complete vectors, not between coordinates of a conditioned vector.

Every rectangle choice so retained is legal. At the beginning of block j its actual height is at least $h_j - h_j/L$. For every q in the local window of (18.2),

$$\begin{aligned} q &\leq \alpha(\ell_j + u_j) + C\sqrt{\ell_j L} \\ &\leq (1 - 4/L)(1 + 1/L)h_j + O(1) + C\sqrt{\ell_j L} \leq h_j - 2h_j/L \end{aligned}$$

for all large n . The last step uses $h_j \geq cL^7$, which makes $\sqrt{\ell_j L} = o(h_j/L)$. Thus q does not exceed the actual height; all macro-boundary heights stay positive. The perturbed lengths and increments still satisfy the uniform hypotheses of (18.2).

The same central-coefficient argument for the length balance, together with (18.8), gives the following lower bound for the cardinality of the retained rectangle set $\mathcal{R}$:

$$|\mathcal{R}| \geq \frac{\prod_j (2w_j + 1)(2b_j + 1)}{2(2n + 1)^2}. \quad (18.11)$$

18.4 Jensen's inequality removes the linear fluctuations

Let expectation be the product of the two uniform balanced-vector measures just constructed. Define

$$X = \sum_j \left\{ \frac{(g_j + V_j)^2}{2v(\ell_j + U_j)} + 2\log(\ell_j + U_j) \right\}.$$

By independence between the complete vectors and the centered moments in (18.9)–(18.10),

$$\mathbb{E} \frac{(g_j + V_j)^2}{\ell_j + U_j} = (g_j^2 + \mathbb{E}V_j^2) \mathbb{E} \frac{1}{\ell_j + U_j}. \quad (18.12)$$

The exact identity $(1 + x)^{-1} = 1 - x + x^2/(1 + x)$ gives

$$\mathbb{E} \frac{1}{\ell_j + U_j} \leq \frac{1}{\ell_j} (1 + C/L^2).$$

Since the deterministic kinetic sum is $O(ML)$, as also verified below,

$$\mathbb{E} \sum_j \frac{(g_j + V_j)^2}{\ell_j + U_j} \leq \sum_j \frac{g_j^2}{\ell_j} + O(M). \quad (18.13)$$

Concavity gives $\mathbb{E} \log(\ell_j + U_j) \leq \log \ell_j$. The useful direction of Jensen's inequality is

$$\sum_{\mathcal{R}} e^{-X} = |\mathcal{R}| \mathbb{E} e^{-X} \geq |\mathcal{R}| \exp(-\mathbb{E} X).$$

Thus the apparently large first-order fluctuation of the energy does not create an $O(M\sqrt{L})$ loss.

Combining these facts with (18.2), (18.11), and

$$\log(2w_j + 1) \geq \log \ell_j - \log L + O(1), \quad \log(2b_j + 1) \geq \frac{1}{2} \log \ell_j + O(1),$$

leaves the logarithmic cost

$$D_n \leq \sum_j \frac{g_j^2}{2v\ell_j} + \frac{1}{2} \sum_j \log \ell_j + m \log L + O(M + L + h_0). \quad (18.14)$$

For clarity, the exact seeds are unchanged from the audited construction: the ascent from height 1 to h_0 has degree $2h_0 - 1$, the final descent has degree 1, and the original first letter has degree 1. Together with the m leading x factors and $\sum \ell_j = n'$, the total degree is exactly n . The total height tilt cancels because the full walk returns to 1. Seed costs are $O(h_0) = O(L^7) = o(M)$; fixed block constants cost $O(M)$.

18.5 Deterministic sums at the required precision

We record enough precision to justify the $O(M)$ remainder. First,

$$S = M/2 + O(J^3/M^2 + 1), \quad n' = n - O(M + L^7).$$

Writing $\delta = \pi/M$, the exact identity

$$\frac{s_{j+1} - s_j}{\sqrt{s_j}} = \sin \delta \left[2 \cos(\pi j/M) \cos \delta + \frac{\cos(2\pi j/M)}{\sin(\pi j/M)} \sin \delta \right]$$

and elementary summation give

$$\sum_j \frac{(s_{j+1} - s_j)^2}{s_j} = \frac{2\pi^2}{M} + O\left(\frac{J + \log M}{M^2}\right). \quad (18.15)$$

Indeed, the main square is $2\pi^2/M + O(J/M^2)$, the cross term is $O(\log M/M^2)$ by $\sum 1/\sin(\pi j/M) = O(M \log M)$, and the final square is $O(1/(M^2 J))$ by $\sum 1/\sin^2(\pi j/M) = O(M^2/J)$.

Before integer rounding the kinetic sum is $H_0^2 S/n'$ times the left side of (18.15). The factor $(1 - 4/L)^{-2} = 1 + O(1/L)$ contributes $O(M)$, while the trigonometric error contributes $O((J+L)L) = O(L^4) = o(M)$. Height rounding changes the sum by at most $C \sum (|g_j|/\ell_j + 1/\ell_j) = O(L)$; length rounding changes it by at most $C \sum g_j^2/\ell_j^2 = O(ML/L^7) = o(M)$. Consequently

$$\sum_j \frac{g_j^2}{\ell_j} = \frac{4\pi^2 \alpha^2 n}{M^2} + O(M). \quad (18.16)$$

The same estimates, before seeking their leading constant, give the $O(ML)$ bound used in (18.13).

The sine product $\prod_{j=1}^{M-1} \sin(\pi j/M) = M/2^{M-1}$ yields

$$\sum_{j=J}^{M-J-1} \log s_j = -2M \log 2 + O(J \log M + \log M).$$

Length rounding changes the logarithmic sum by $O(\sum 1/\ell_j) = o(1)$. Hence

$$\frac{1}{2} \sum_j \log \ell_j = \frac{M}{2} \log(n/M) + O(M). \quad (18.17)$$

Also $(M - m) \log L = O(J \log L) = o(M)$. Substituting (18.16)–(18.17) into (18.14) proves

$$D_n \leq \frac{A_0 n}{M^2} + \frac{M}{2} \log(n/M) + M \log L + O(M). \quad (18.18)$$

Finally, the rounded choice of M satisfies

$$\frac{A_0 n}{M^2} = \frac{ML}{6} + O(L), \quad \frac{ML}{2} = C_* F + O(L), \quad \log(n/M) = \frac{2}{3} L + \frac{1}{3} \log L - \frac{1}{3} \log(6A_0) + o(1).$$

Since $L = o(M)$, (18.18) is

$$D_n \leq \frac{ML}{2} + \frac{7}{6} M \log L + O(M),$$

which is exactly (18.1). The coefficient $7/6$ per macro-block consists of $1/6$ from the optimal length scale inside $\frac{1}{2} \log(n/M)$ and 1 from the usable length-window width ℓ/L .

19 The coefficient upper bound by a finite length calibration

We prove

$$D_n \geq C_* F + \frac{7C_*}{3} \frac{F \log L}{L} - O(F/L). \quad (19.1)$$

Unlike a fixed-parameter leading calibration, the construction below retains its quantitative accuracy as n varies.

19.1 The truncated row threshold

For a fixed constant C_0 and $r \rightarrow \infty$, put

$$\kappa(r) = \frac{1}{2} \log r + \log \log r - C_0/2.$$

For fixed Q ,

$$\sum_{Q < q \leq r} q^{-3/2} e^{\kappa(r)q/r} \leq CQ^{-1/2} + o(1) + Ce^{-C_0/2}. \quad (19.2)$$

The error is uniform when the minimum allowed r tends to infinity. To see this, split at r/κ and $r/2$. The first range is $O(Q^{-1/2})$. The second is at most

$$C(\kappa/r)^{1/2} e^{\kappa/2} = O(r^{-1/4} \log r) = o(1).$$

For $q \geq r/2$, sum the geometric progression after bounding $q^{-3/2}$ by $Cr^{-3/2}$; this gives

$$Cr^{-1/2} e^{\kappa}/\kappa \leq Ce^{-C_0/2}.$$

Here $\kappa \sim \frac{1}{2} \log r$. Constants can be chosen independent of fixed C_0 once r is large enough. Integer rounding does not change these bounds.

Together with (17.4), this means the defective mass m_* can absorb the large- q tail whenever vanishing tilts satisfy

$$q\Psi(s, \theta) \leq q\kappa(r)/r + o(1), \quad q \leq h \leq r, \quad (19.3)$$

uniformly with $r \rightarrow \infty$. Indeed, fix Q large, then fix C_0 large; the finite $q \leq Q$ rows converge to their critical weights, and (19.2) makes the remainder smaller than a fixed fraction of $1 - m_*$.

19.2 The variable potential and its affine dual

Choose a sufficiently large fixed R at the end. Put $\delta = \varepsilon = L^{-12}$ and

$$V_n(y) = \frac{\alpha}{2L(y + \varepsilon)} \{ \log[H(y + \varepsilon)] + 2 \log \log[H(y + \varepsilon)] - C_0 \}, \quad (19.4)$$

$$B_n = \frac{\alpha}{2L} \{ \log H + 2 \log \log H - C_0 \}, \quad (19.5)$$

$$J_n(A) = \min_{0 \leq y \leq R} \{ Ay + V_n(y) \}, \quad A > 0. \quad (19.6)$$

All logarithms are defined for large n , as $\varepsilon H \rightarrow \infty$; B_n remains bounded away from zero and infinity.

For the Kepler arch with parameter $B = B_n$, let

$$y_*(s) = M \sin^2 u, \quad s = \frac{u - \sin u \cos u}{\pi}, \quad M^3 = \frac{2vB}{\pi^2}, \quad p_*(s) = -y'_*(s)/v. \quad (19.7)$$

Then $p'_* = B/y_*^2 > 0$. Define, for $0 \leq s \leq 1$,

$$\begin{aligned} p_n(s) &= p_*(\delta + (1 - 2\delta)s), \\ g_n(1) &= 0, \quad g'_n(s) = vp_n(s)^2/2 - J_n(p'_n(s)), \\ f_n(s, y) &= g_n(s) + p_n(s)y. \end{aligned} \quad (19.8)$$

The defining minimum gives the exact inequality

$$-\partial_s f_n + \frac{v}{2} (\partial_y f_n)^2 = J_n(p'_n) - p'_n y \leq V_n(y). \quad (19.9)$$

Near either endpoint, the arch has $p_* = O(s^{-1/3})$, $p'_* = O(s^{-4/3})$ and $p''_* = O(s^{-7/3})$, interpreted with distance to the appropriate endpoint. Uniformly in the compact range of B_n ,

$$\|p_n\|_\infty = O(L^4), \quad \|p'_n\|_\infty = O(L^{16}), \quad \|p''_n\|_\infty = O(L^{28}). \quad (19.10)$$

The potential is strictly convex for large n . In fact, writing $u = y + \varepsilon$, $x = \log(Hu)$ and $\phi(x) = x + 2 \log x - C_0$, its second derivative has the sign of

$$\phi'' - 3\phi' + 2\phi = 2x + 4 \log x - 2C_0 - 3 - 6/x - 2/x^2 > 0.$$

The minimum in (19.6) is therefore unique. Consequently J_n is C^1 with $0 \leq J'_n(A) \leq R$, including when the minimizer reaches a boundary. Since $g''_n = vp_n p'_n - J'_n(p'_n) p''_n$, we have

$$\|f_n\|_{C^2} = O(L^{28}), \quad \|\partial_s f_n\|_\infty = O(L^{16}), \quad \|\partial_y f_n\|_\infty = O(L^4) \quad (19.11)$$

on $[0, 1] \times [0, R]$. For the value bound one can use $0 < J_n(A) \leq AR + V_n(R)$.

19.3 Calibration value and endpoint losses

For A in the range of p'_n , the dual obeys

$$J_n(A) \geq 2\sqrt{B_n A} - \frac{C}{L} \sqrt{A} (1 + |\log A|) - \varepsilon A. \quad (19.12)$$

Here is a direct uniform proof. Put $a = \alpha/(2L)$, $B = B_n$ and $u = y + \varepsilon$. Since $\log u \geq -12 \log L$,

$$V_n(y) \geq \frac{B + a \log u - a}{u}, \quad B + a \log u - a \geq B/2$$

for large n ; the first inequality uses $2 \log(\log(Hu)/\log H) \geq -1$. Set $\lambda = \sqrt{B/A}$ and $r = u/\lambda$. After adding $A\varepsilon$, the expression $Ay + V_n(y)$ divided by $\sqrt{AB}$ is at least

$$r + \frac{1 + (a/B)(\log \lambda - 1) + (a/B) \log r}{r}.$$

Outside $r \in [1/8, 8]$ the cruder bound $r + 1/(2r) \geq 4$ suffices. Inside that interval, $r + 1/r \geq 2$ and the remaining term is bounded below by $-C(a/B)(1 + |\log \lambda|)$. This proves (19.12).

The endpoint singularities are integrable with the needed logarithm:

$$\int_0^1 \sqrt{p'_n} (1 + |\log p'_n|) ds \leq C, \quad \int_0^1 p'_n ds = O(\delta^{-1/3}). \quad (19.13)$$

The first comparison is with $s^{-2/3}(1 + |\log s|)$ near an endpoint. Integrating (19.12) yields

$$\begin{aligned} f_n(0, 0) &\geq \int_0^1 \{2\sqrt{B_n p'_n} - v p_n^2/2\} ds \\ &\quad - O(1/L) - O(\varepsilon \delta^{-1/3}). \end{aligned} \quad (19.14)$$

For the full arch this integral equals

$$C(B_n) = 3 \left(\frac{\pi^2 B_n^2}{2v} \right)^{1/3}.$$

For completeness, $y_*'^2/(2v) = B_n(1/y_* - 1/M)$ and $\int_0^1 B_n/y_* ds = 2B_n/M$, giving the value $3B_n/M$. Both integrands discarded by cutting off $[0, \delta]$ and $[1 - \delta, 1]$ have absolute size $O(s^{-2/3})$, so their loss is $O(\delta^{1/3})$. The linear reparametrization contributes $O(\delta)$. Therefore

$$f_n(0, 0) \geq C(B_n) - O(1/L), \quad (19.15)$$

since $\delta = \varepsilon = L^{-12}$. This is a finite- n estimate, with no interchange of unquantified limits.

19.4 Uniform contraction of transformed rows

At elapsed macro-length k and height h , retain legal transitions with $k+e \leq n-1$ and $1 \leq h+d \leq RH$. Write $s = k/n$, $y = h/H$ and transform each weight to

$$w(e, q, d) \exp\{F[f_n(s, y) - f_n(s + e/n, y + d/H)]\}. \quad (19.16)$$

Set $a = F/n$ and $b = F/H$, so $b^2 = a$ and $aH = L$. On the range

$$e \leq HL^{40}, \quad |d| \leq \sqrt{H}L^{40}, \quad (19.17)$$

Taylor's formula gives uniformly

$$F[f_n(s, y) - f_n(s + e/n, y + d/H)] = -a(\partial_s f_n)e - b p_n(s)d + o(1). \quad (19.18)$$

There is no d^2 term because f_n is affine in height. By (19.11), the absolute remainder is at most

$$CFL^{28} \left\{ (HL^{40}/n)^2 + (HL^{40}/n)(L^{40}/\sqrt{H}) \right\} = n^{-1/3}L^{O(1)} = o(1).$$

Put $\lambda = -a\partial_s f_n$ and $\theta = -bp_n$. Both tilts vanish uniformly. Multiplication of the remainder in (17.5) by $q \leq RH$ still gives $o(1)$, because the derivatives grow only polylogarithmically. By (19.9), with $r = h + \varepsilon H$,

$$\begin{aligned} q\Psi(\lambda, \theta) &\leq \frac{q}{\alpha} aV_n(y) + o(1) \\ &= \frac{q}{2r} \{\log r + 2\log \log r - C_0\} + o(1) = \frac{q}{r} \kappa(r) + o(1). \end{aligned} \quad (19.19)$$

Since $r \geq \varepsilon H \rightarrow \infty$ uniformly, the truncated-row argument bounds the contribution of (19.17) by a fixed number below one, choosing C_0 sufficiently large once and for all.

We now check all omitted steps. The global derivative bound gives

$$F[f_n(s, y) - f_n(s + e/n, y + d/H)] \leq CaL^{16}e + CbL^{16}|d|. \quad (19.20)$$

For $e > HL^{40}$, add a fixed small positive length tilt λ_0 , split the two signs of d , and apply (17.4) in its fixed neighborhood. The resulting upper bound is

$$C \exp\{-\lambda_0 HL^{40} + O(H)\} = o(1).$$

For $|d| > \sqrt{HL}^{40}$, add the corresponding signed height tilt $\theta_0 = cL^{40}/\sqrt{H}$, with $c > 0$ small and fixed. The generating row for the majorant is at most

$$C \exp\{C_R H[aL^{16} + (bL^{16} + \theta_0)^2]\}.$$

The Chernoff factor is $\exp(-cL^{80})$. Since $Ha = L$ and $\sqrt{H}b = \sqrt{L}$, the positive exponent is $C_R c^2 L^{80} + O(L^{57})$. Taking $c < 1/(2C_R)$ proves this tail is $o(1)$. The root-Taylor remainders are $n^{-1/3}L^{O(1)} = o(1)$ as well. These estimates allow unrestricted lengths and destinations and hence cover every omitted retained transition.

Leaving a fixed margin when choosing Q and C_0 above proves the uniform transformed row bound

$$\text{each row in (19.16) has sum at most } \bar{m} < 1 \quad (19.21)$$

for all sufficiently large n , uniformly in k and h .

19.5 Telescoping and the terminal factor

The sum of all transformed finite prefixes is at most $(1 - \bar{m})^{-1}$. Telescoping (19.16) therefore gives

$$a_n^{[\leq RH]} \rho^n \leq C \exp\{-Ff_n(0, 1/H)\}, \quad (19.22)$$

provided the exact terminal factor (17.7) remains bounded after undoing the transformation. It does: by (19.11),

$$f_n(s, y) \leq p_n(1)y + CL^{16}(1 - s).$$

With $r_0 = n - 1 - k \geq 0$, the residual factor is bounded by

$$\rho^{r_0} t^{1-h} \exp\{bp_n(1)h + CaL^{16}(r_0 + 1)\}.$$

Both $bp_n(1)$ and aL^{16} tend to zero, so the fixed negative exponents $\log \rho$ and $-\log t$ dominate uniformly. No endpoint height restriction is imposed, and the terminal weight is not discarded. Also

$$F[f_n(0, 1/H) - f_n(0, 0)] = bp_n(0) = o(1).$$

Choose R fixed so that $R^2/(4D_0) > C_\star + 1$. Combining (17.8), (19.15) and (19.22) gives

$$D_n \geq FC(B_n) - O(F/L).$$

Finally,

$$\begin{aligned} \log H &= \frac{2}{3}L + \frac{1}{3}\log L, \\ 2\log\log H &= 2\log L + 2\log(2/3) + O((\log L)/L), \\ B_n &= \frac{\alpha}{3} + \frac{7\alpha}{6}\frac{\log L}{L} + O(1/L), \\ C(B_n) &= C_\star \left(1 + \frac{7}{3}\frac{\log L}{L}\right) + O(1/L). \end{aligned}$$

This proves (19.1); together with the coefficient lower construction it proves Theorem 16.1.

20 The second inverse correction

Let $\lambda = \log \mu$ and $N(Y) = \min\{n : a_n \geq Y\}$. The foundation proves a_n is nondecreasing by appending its last letter. The two-sided logarithmic expansion therefore has the following inverse consequence.

Corollary 20.1. *As $Y \rightarrow \infty$, with $S = \log Y$,*

$$N(Y) = \frac{S}{\lambda} + C_\star \lambda^{-4/3} S^{1/3} (\log S)^{2/3} \left[1 + \frac{7}{3} \frac{\log \log S}{\log S} + O\left(\frac{1}{\log S}\right)\right]. \quad (20.1)$$

Here $C_\star \lambda^{-4/3} = 0.893568257650894909\dots$

Proof. Set $G(S) = S^{1/3}(\log S)^{2/3}$ and $E(S) = G(S)/\log S$. Uniformly for $n = S/\lambda + O(G(S))$, elementary differentiation gives

$$\begin{aligned} n^{1/3}(\log n)^{2/3} &= \lambda^{-1/3}G(S) + O(E(S)), \\ \frac{\log \log n}{\log n} &= \frac{\log \log S}{\log S} + O(1/\log S). \end{aligned}$$

The first error includes the constant shift $-\log \lambda$ in $\log n$; the change caused by $n - S/\lambda = O(G(S))$ is still smaller. Theorem 16.1 consequently implies

$$D_n = C_\star \lambda^{-1/3} G(S) \left[1 + \frac{7}{3} \frac{\log \log S}{\log S}\right] + O(E(S))$$

uniformly in that window. Choose integers just below and above

$$\frac{S}{\lambda} + C_\star \lambda^{-4/3} G(S) \left[1 + \frac{7}{3} \frac{\log \log S}{\log S}\right] \pm KE(S)$$

for a sufficiently large fixed K . Since $\log a_n = \lambda n - D_n$, the lower choice is below the logarithmic target S and the upper choice is above it, eventually. Monotonicity brackets $N(Y)$ between them. Integer rounding costs $O(1) = o(E(S))$, proving (20.1). $\square$

21 The explicit constant in the finite-height radius

This auxiliary refinement is independent of the matching global coefficient bounds. Let ρ_H be the radius obtained by restricting all macro-boundary heights to $1, \dots, H$. Define

$$b = 1 - \rho, \quad C_q = \frac{\sqrt{-16z_*^2 + 55z_* - 10}}{2\sqrt{\pi}}, \quad m_* = \frac{\rho}{bz_* + \rho}, \quad a_* = \frac{\rho t_* C_q}{1 - \rho},$$

$$c_* = \log \frac{1 - m_*}{2a_*} = 0.8667962458055399529886348076493374 \dots$$

We prove

$$\log(\rho_H/\rho) = \frac{\alpha}{H} \left\{ \frac{1}{2} \log H + \log \log H + c_* + o(1) \right\}. \quad (21.1)$$

The proof must address the killed boundary: the unrestricted row threshold alone is insufficient to identify the finite-height Perron threshold.

21.1 Exact tilted rows and their endpoint mass

Write $k(\ell, q, d)$ for the exact macro-transition coefficients, including the multiplier $x/(1-x)$, and put $x = \rho e^s$, $T = t_* e^\theta$. At fixed q the full tilted row is

$$A_q(s, \theta) := \sum_{\ell, d} k(\ell, q, d) x^\ell T^d = \frac{xT}{1-x} Q_q(x, T) T^{-q}.$$

Uniformly in a sufficiently small fixed complex neighborhood of $(s, \theta) = (0, 0)$, square-root coefficient extraction gives

$$A_q(s, \theta) = a(s, \theta) q^{-3/2} e^{q\Psi(s, \theta)} (1 + O(q^{-1})), \quad a(0, 0) = a_*, \quad (21.2)$$

where $\Psi(s, \theta) = -\log[z_c(\rho e^s, t_* e^\theta) t_* e^\theta]$. The expansion and its first four parameter derivatives are uniform on a smaller neighborhood. Indeed, the exact quadratic branch has separated analytic discriminant roots and nonvanishing rational prefactors near the critical point. Expanding the analytic multiplier at its smaller root, then using the binomial coefficients of the square root, gives the uniform error in (21.2). Cauchy's formula applied on a slightly larger parameter neighborhood also bounds its derivatives. Finitely many small q are handled by their analytic generating functions.

The discriminant identities give

$$\Psi(0, 0) = \Psi_\theta(0, 0) = 0, \quad \Psi_s(0, 0) = \alpha^{-1}, \quad \Psi_{\theta\theta}(0, 0) = \nu := v/\alpha > 0.$$

Thus, for $s = O(\log H/H)$,

$$\Psi(s, 0) = s/\alpha + O(s^2), \quad \Psi_\theta(s, 0) = O(s), \quad \Psi_{\theta\theta}(s, 0) = \nu + O(s). \quad (21.3)$$

For fixed $c \in \mathbb{R}$ set

$$s_H(c) = \frac{\alpha}{H} \left(\frac{1}{2} \log H + \log \log H + c \right), \quad R_H(c) = \sum_{q=1}^H A_q(s_H(c), 0).$$

We claim, locally uniformly in c , that

$$R_H(c) \longrightarrow m_* + 2a_* e^c. \quad (21.4)$$

The same limit holds when the upper cutoff is $H - W$, provided $W \log H/H \rightarrow 0$.

To prove this, put $k = H\Psi(s_H(c), 0)$, so $k = \frac{1}{2} \log H + \log \log H + c + o(1)$. Below $H/(\log H)^2$, summable $q^{-3/2}$ domination recovers the critical total mass m_* . The range $H/(\log H)^2 < q \leq H/2$ contributes at most $C(\log H)H^{-1/2}e^{k/2} = o(1)$. On the remaining range the endpoint Laplace sum is

$$\sum_{H/2 < q \leq H} q^{-3/2} e^{kq/H} = \frac{H^{-1/2} e^k}{k} (1 + o(1)) \rightarrow 2e^c. \quad (21.5)$$

For example, write $q = H - r$ and sum the geometric window $r = O(H/k)$; replacing $(1 - r/H)^{-3/2}$ by 1 has relative error $O(k^{-1})$. The more remote terms are negligible. Removing the last W terms has relative cost $O(kW/H) = o(1)$. This proves (21.4).

In particular, c_* is the unique solution of $m_* + 2a_*e^c = 1$. If $c < c_*$, every diagonally tilted row of the height- H matrix at $x = \rho e^{s_H(c)}$ has mass at most $R_H(c) < 1$. The Perron eigenvalue is therefore below 1, and

$$\rho_H > \rho e^{s_H(c)}. \quad (21.6)$$

21.2 A homogeneous kernel on a high strip

Fix $c > c_*$ and any $1/2 < \gamma < 1$. Set

$$W = \lfloor H^\gamma \rfloor, \quad Q = H - W, \quad I = \{H - W + 1, \dots, H\}, \quad x = \rho e^{s_H(c)}.$$

Every $q \leq Q$ is legal throughout I . Thus the fixed subkernel

$$a_d = \sum_{q=1}^Q \sum_{\ell} k(\ell, q, d) x^\ell t_*^d, \quad R = \sum_d a_d$$

is translation invariant on the strip before killing its exits, and

$$R \rightarrow m_* + 2a_*e^c > 1. \quad (21.7)$$

This fixed subkernel avoids any comparison of height-dependent row errors against a small endpoint test-vector entry.

The differentiated transfer formula implies the uniform estimates

$$\sum_d a_d d = O(\log H), \quad \sum_d a_d d^2 = \Theta(H), \quad \sum_d a_d |d|^3 = O(H^{3/2}), \quad (21.8)$$

$$\sum_d a_d e^{c_1 |d|/\sqrt{H}} \leq C \quad \text{for some fixed } c_1, C > 0. \quad (21.9)$$

Here, conditionally on q , the increment moment generating function is $A_q(s, \theta)/A_q(s, 0)$. Its mean is $O(1 + qs)$ and its variance is $\nu q + O(qs + 1)$. Evaluating at $\theta = \pm c_1/\sqrt{H}$ gives a uniformly bounded moment ratio, since its logarithm is $O(qs/\sqrt{H} + q/H + H^{-1/2}) = O(1)$. This proves the upper moment bounds and the mean bound after summing with the row weights. The lower second-moment bound follows from (21.5), which places a fixed positive mass at $q = H - O(H/\log H)$, where the conditional variance is at least a constant times H .

Spending half the exponential moment to dominate a polynomial gives, for each fixed integer $r \geq 0$,

$$\sum_{|d| > W} a_d |d|^r \leq C_r H^{r/2} e^{-c_2 W/\sqrt{H}} = o(1). \quad (21.10)$$

21.3 An exactly centered sine test

Put $\delta = \pi/(W+1)$ and truncate to $|d| \leq W$. Define

$$S(\tau) = \sum_{|d| \leq W} a_d e^{\tau d} \sin(\delta d).$$

Since d and $\sin(\delta d)$ have the same sign in this range, $S'(\tau) > 0$. Using $|\sin u - u| \leq |u|^3/6$ and (21.8)–(21.10) gives

$$S(0) = O(\delta \log H + \delta^3 H^{3/2}). \quad (21.11)$$

For every fixed sufficiently small $\varepsilon > 0$, uniformly on $|\tau| \leq \varepsilon/\sqrt{H}$,

$$S'(\tau) \geq c\delta H. \quad (21.12)$$

Indeed, choose a large fixed B . By the second-moment lower bound and the exponential moment, at least cH of that moment lies in $|d| \leq B\sqrt{H}$. There $e^{\tau d} \geq e^{-\varepsilon B}$ and $d \sin(\delta d) \geq \delta d^2/2$ for large H .

The intermediate value theorem, applied to (21.11)–(21.12), now gives a unique root τ_H with

$$|\tau_H| = O\left(\frac{\log H}{H} + \frac{H^{1/2}}{W^2}\right) = o(H^{-1/2}), \quad S(\tau_H) = 0. \quad (21.13)$$

Set

$$C_H = \sum_{|d| \leq W} a_d e^{\tau_H d} \cos(\delta d).$$

Then

$$C_H = R + o(1). \quad (21.14)$$

The omitted tail is $o(1)$ by (21.10); the tilt error is $O(|\tau_H|\sqrt{H}) = o(1)$ by the exponential moment; and the cosine error is $O(\delta^2 H) = o(1)$.

Index I by $i = 1, \dots, W$, and take the positive test vector $f_i = \sin(\delta i)$, extended by zero outside this interval. For all $|d| \leq W$ one has

$$f_{i+d} \geq \sin(\delta(i+d)).$$

Inside the strip this is equality. Outside, the sine on the right is nonpositive because $1 - W \leq i + d \leq 2W$. Consequently the killed, exponentially tilted subkernel B_H satisfies

$$\begin{aligned} (B_H f)_i &\geq \sum_{|d| \leq W} a_d e^{\tau_H d} \sin(\delta(i+d)) \\ &= f_i C_H + \cos(\delta i) S(\tau_H) = C_H f_i. \end{aligned}$$

Collatz–Wielandt gives $r(B_H) \geq C_H$. The positive diagonal similarities by t_*^h and $e^{\tau_H h}$ preserve spectral radii, and retaining transitions can only decrease them. Hence

$$r(K_H(x)) \geq r(B_H) \geq C_H = R + o(1) > 1.$$

It follows that $\rho_H < \rho e^{s_H(c)}$ whenever $c > c_*$ and H is large. Combining this with (21.6) and letting $c \rightarrow c_*$ proves (21.1).

The constant c_* is a local-row and finite-height constant. This argument does not identify the coefficient in the global $O(F/L)$ deficit remainder.

22 What the theorem does not determine

Equation (16.2) leaves an $O(n^{1/3}(\log n)^{-1/3})$ error in $\log a_n$, which diverges. Exponentiating it therefore does not give an asymptotic equivalent for a_n . The coefficient of that error scale, fluctuation determinants, a power-law prefactor, and an all-orders transseries are not established here. The finite-height shift is an auxiliary spectral statement; it does not by itself determine the next global deficit constant. The inverse result also has a diverging error and is not an exact integer rounding rule.

[**write, 2 October 2026**] The coefficient of the F/L scale is determined in Part IV (Theorem 24.1), and the inverse result is refined in Corollary 28.1 and Theorem 35.1. The other limitations of this paragraph hold for the whole report.

The coefficient $7/3$ has two consistent origins. In the coefficient construction, $\frac{1}{2} \log(n/M)$ contributes $\frac{1}{6} \log L$ per block and the usable length-window width contributes $\log L$, giving $\frac{7}{6} \log L$ per block. In the upper bound, the cutoff tail is of order $h^{-1/2}e^\kappa/\kappa$; filling the defective critical row requires $\kappa = \frac{1}{2} \log h + \log \log h + O(1)$. Both mechanisms lead to the same global correction.

23 A conditional next scale conjecture

This section is explicitly conjectural. It is not part of the proved coefficient or inverse asymptotics. Let c_* be the proved finite-height constant in the preceding theorem and put

$$Y = \left(\frac{2v\alpha}{3\pi^2} \right)^{1/3}, \quad \kappa_* = \log Y - 2 \log 3 + 2c_* = -1.94014818313627 \dots$$

A natural candidate for the next global term is

$$D_n \stackrel{\text{conj.}}{=} C_* F \left[1 + \frac{7}{3} \frac{\log L}{L} + \frac{\kappa_*}{L} + o(1/L) \right]. \quad (23.1)$$

Here is the conditional calculation, including its missing step. If the exact local row threshold transfers to a global action with error $o(F/L)$ and no additional contribution at that scale, its potential would be

$$\frac{\alpha}{2h} \{ \log h + 2 \log \log h + 2c_* + o(1) \}.$$

After setting $h = Hy$, the first perturbation of the leading action is

$$\frac{\alpha F}{2L} \int_0^1 \frac{\frac{7}{3} \log L + \log y + 2 \log(2/3) + 2c_*}{y} ds.$$

For the leading Kepler minimizer $y = Y \sin^2 u$ and $s = (u - \sin u \cos u)/\pi$, one has

$$\frac{ds}{y} = \frac{2 du}{\pi Y}, \quad \frac{\int_0^1 (\log y)/y ds}{\int_0^1 1/y ds} = \log Y - 2 \log 2,$$

using $\int_0^\pi \log(\sin u) du = -\pi \log 2$. Since $C_* = \alpha/Y$, the resulting constant is exactly $\log Y - 2 \log 3 + 2c_*$.

The unproved premise is the global $o(F/L)$ transfer, equivalently an $o(M)$ error at the macro-block scale. The present balanced-rectangle lower construction intentionally loses $O(M)$ through its length and increment windows, profile margin, and fixed kernel factors. The exact finite-height constant alone does not remove those losses. Thus (23.1) is a research target, not a consequence of Theorem 16.1; it cannot be used as an established coefficient estimate or inverse correction.

[write, 2 October 2026] *Proved.* Part IV, Theorem 24.1, proves (23.1) with $\kappa_* = \kappa = -1.940148183136270392\dots$. It does so by its own argument (a finite-time calibration that keeps the exact row amplitude, and a lower construction with complete normalized rows and an exact central repair), not by establishing the global transfer premise named above. Part V, at $J = 1$, is a second proof. In this section Y denotes Part IV's constant Y_0 , not the threshold target of Corollary 20.1, and C_* is the C_* of the rest of this Part.

Part IV

The third logarithmic deficit term of 120 avoiding ascent sequences

Source. Batch-77 manuscript 42, archive `oeis-a202061-third-order-report.zip`, main file `a202061-third-order.tex` (214 lines) with three `\input` files (`proofs/upper-section.tex`, `lower-section.tex`, `inverse-section.tex`), inlined; dated 2 October 2026. Author line: “A proof and reproducible research companion”. Printed in full; label prefix `a61:o3:`.

Abstract of the source.

Let a_n count classical 120-avoiding ascent sequences, OEIS A202061. We determine the constant following the established log-log correction:

$$n \log \mu - \log a_n = C_* n^{1/3} (\log n)^{2/3} \left[1 + \frac{7}{3} \frac{\log \log n}{\log n} + \frac{\kappa}{\log n} + o\left(\frac{1}{\log n}\right) \right],$$

where $C_* = 2.232625307612844922\dots$ and $\kappa = -1.940148183136270392\dots$ have exact expressions below. The coefficient upper bound retains the exact row amplitude in a quantitative finite-time calibration. The matching lower bound uses complete normalized tilted rows, including their small-jump mass, and an exact central repair with subleading cost. A one-dimensional energy minimum connects that construction to a classical path action without differentiating an asymptotic remainder. We also give the corresponding inverse constant. This is an additive expansion in the logarithm, not a multiplicative equivalent or an all-orders transseries.

24 Statement and exact constants

An ascent sequence is a nonnegative integer word with $w_1 = 0$ and $w_i \leq 1 + \#\{j < i - 1 : w_j < w_{j+1}\}$ for $i \geq 2$. It avoids classical 120 if no $i < j < k$ satisfy $w_k < w_i < w_j$. Let a_n count such words, with $a_0 = 1$; this is OEIS A202061 [1].

Let z_* be the smallest positive root of $z^3 - 5z^2 + 6z - 1 = 0$, and define

$$\begin{aligned} \rho &= 3z_*^2 - 10z_* + 2, & \mu &= \rho^{-1}, & t_* &= z_*^{-1}, \\ \alpha &= \frac{8z_*^2 - 29z_* + 9}{7}, & v &= \frac{2z_*^2 - 7z_* + 2}{2}. \end{aligned}$$

For occasional shorter notation we set $z = z_*$ and $t = t_*$. The exact row mass, square-root amplitude and local threshold constant are

$$\begin{aligned} m_* &= \frac{\rho}{(1 - \rho)z_* + \rho}, & C_q &= \frac{\sqrt{-16z_*^2 + 55z_* - 10}}{2\sqrt{\pi}}, \\ a_* &= \frac{\rho t_* C_q}{1 - \rho}, & c_* &= \log \frac{1 - m_*}{2a_*}. \end{aligned}$$

Finally put

$$Y_0 = \left(\frac{2v\alpha}{3\pi^2}\right)^{1/3}, \quad C_* = \frac{\alpha}{Y_0} = \left(\frac{3\pi^2\alpha^2}{2v}\right)^{1/3}, \quad \kappa = \log Y_0 - 2\log 3 + 2c_*. \quad (24.1)$$

Numerically,

$$\begin{aligned} c_* &= 0.8667962458055399529886348076493374\dots, \\ Y_0 &= 0.2284321390064290542681603916276110\dots, \\ C_* &= 2.232625307612844922179764700526\dots, \\ \kappa &= -1.940148183136270392093282071371174\dots \end{aligned}$$

All logarithms are natural. Write

$$L = \log n, \quad H = n^{2/3}L^{1/3}, \quad F = H^2/n = n^{1/3}L^{2/3}, \quad D_n = n \log \mu - \log a_n.$$

Theorem 24.1. *As $n \rightarrow \infty$ through all positive integers,*

$$D_n = C_*F + \frac{7C_*}{3} \frac{F \log L}{L} + C_*\kappa \frac{F}{L} + o(F/L). \quad (24.2)$$

In particular,

$$\frac{D_n - C_*F - (7C_*/3)F \log L/L}{F/L} \longrightarrow C_*\kappa = -4.33162393418911786929363375816324\dots$$

[**write, 2 October 2026**] This theorem proves the conjecture (23.1) of Part III. It is also the case $J = 1$ of Part V, Theorem 30.1, since $P_1(B) = B$. Part V proves that case by a different route, a reduction to a classical action, and imports this Part's constants. This Part's one-sided proof is therefore kept in full, as a second route.

The previous report established the first two terms with an $O(F/L)$ remainder and separately identified c_* in the local row and finite-height cutoff (Part III). The new theorem identifies the coefficient of that remainder scale. The proof is not a numerical fit. The matching new arguments have independent mathematical audits, and the complete earlier report and its sharp-constant and positive-walk foundations are included unchanged in the companion package. These are research audits, not journal peer review or formal proof-assistant certification. No new literature-priority claim is made beyond the source comparisons retained in the dependencies.

25 Positive kernel and dependency boundary

The exact positive height-walk representation has suffix series

$$e_h = \frac{1}{1-x} + \frac{x}{1-x} \sum_{q=1}^h (Q_q(x, T)e)_{h+1-q}, \quad (Tf)_h = f_{h+1}, \quad A(x) = 1 + xe_1(x).$$

Here $Q(x; u, T) = \sum_{q \geq 1} Q_q(x, T)u^q$ is the root vanishing at $u = 0$ of

$$xT(1-u)Q^2 + [(1-x)(u - (1-x)) + xT(x-u)]Q + (1-x)u = 0.$$

A macro-step with q, r changes height by $d = 1 - q + r$, has generating weight $x[T_0^r]Q_q(x, T_0)/(1 - x)$, and is legal exactly when q does not exceed its starting height. Its endpoint is then positive. Let $k(e, q, d)$ be its multiplicity at total macro degree e and set

$$w(e, q, d) = k(e, q, d)\rho^e t_*^d, \quad W_q(s, \theta) = \sum_{e, d} w(e, q, d)e^{se+\theta d}.$$

Total macro degree here includes the leading multiplier. The critical row mass is $\sum_q W_q(0, 0) = m_* < 1$.

The audited quadratic-branch analysis supplies uniform analytic square-root transfer, including its first four parameter derivatives, and the root expansion

$$\Psi(s, \theta) = \frac{s + v\theta^2/2}{\alpha} + O(s^2 + |s\theta| + |\theta|^3).$$

Section 26 restates its exact amplitude and the quantitative bounds needed for the coefficient upper estimate.

For the patch in the coefficient lower construction we use only the previously audited local positive bound. For any fixed $K > 3$, internal block degree $\ell \geq L^K$ and $|d| \leq C\sqrt{\ell L}$, summing over a fixed sufficiently wide $O(\sqrt{\ell})$ conditional q window gives

$$\sum_q \rho^\ell t_*^d B(\ell, q, q + d - 1) \geq c\ell^{-2} \exp\left[-\frac{d^2}{2v\ell} - o(1)\right]. \quad (25.1)$$

The conditional center is $q = \alpha\ell + \beta(d - 1)$, where $\beta = -4(z_* - 3)(2z_* - 1)/7$; every retained q differs from $\alpha\ell$ by $O(\sqrt{\ell L})$. Using the leading x term makes its total macro degree $\ell + 1$. The local estimate follows from a positive summand, uniform Stirling and the exact Hessian identity; a full coefficient equivalent is not assumed.

The exact residual terminal factor at elapsed macro degree k and height h is $\rho^{n-1-k} t_*^{1-h}$, apart from the original fixed leading ρ . The foundation also gives a fixed $D_0 > 0$ with

$$a_n^{[>RH]} \rho^n \leq C \exp[-R^2 F/(4D_0)]$$

for fixed R and all sufficiently large n . These inputs, and the local-row cutoff constant, are pinned to the unchanged dependency manifests in the companion package.

[write, 2 October 2026] These inputs are proved earlier in this report: the walk and jump quadratic in Part I ((3.1), (2.5)); the expansion of Ψ and the local lower bound in Part II ((11.3), Lemma 11.1); the uniform transfer with its parameter derivatives, the amplitude and the local-row cutoff constant c_* in Part III (Section 21); the first-hit bound in Part I, (7.2).

26 Upper coefficient bound at the constant scale

For brevity write $C = C_*$ in this section. Put $B_0 = \alpha/3$ and $Y_0 = (2vB_0/\pi^2)^{1/3}$. We prove the one-sided estimate

$$D_n \geq CF \left[1 + \frac{7 \log L}{3L} + \frac{\kappa}{L}\right] + o(F/L), \quad \kappa = \log Y_0 - 2 \log 3 + 2c_*. \quad (26.1)$$

The inputs are the exact positive macro-kernel and terminal representation, the uniform square-root transfer with analytic parameter derivatives, the critical root derivative identities, and the

fixed-ceiling first-hit estimate. Only the exact row mass and tail amplitude are needed from the finite-height calculation; its spectral comparison is not used.

The endpoint geometric sum has relative error $O(1/\log r)$, but converting its main term to $2e^c$ adds an error $O(\log \log r / \log r)$. Uniformly in the range below this is $O(\log L/L)$, smaller than the explicit row slack $L^{-1/4}$.

26.1 Exact analytic input and amplitude

Let z be the smallest positive root of $z^3 - 5z^2 + 6z - 1 = 0$, and set

$$\rho = 3z^2 - 10z + 2, \quad t = z^{-1}, \quad \alpha = (8z^2 - 29z + 9)/7, \quad v = (2z^2 - 7z + 2)/2.$$

For the exact critically tilted macro-step weights $w(e, q, d)$, define

$$W_q(\sigma, \theta) = \sum_{e, d} w(e, q, d) e^{\sigma e + \theta d}.$$

There is one fixed parameter neighborhood such that

$$W_q(\sigma, \theta) = a(\sigma, \theta) q^{-3/2} e^{q\Psi(\sigma, \theta)} (1 + E_q(\sigma, \theta)), \quad |\partial^j E_q| \leq C_j/q \quad (0 \leq j \leq 4), \quad (26.2)$$

uniformly for sufficiently large q ; finitely many remaining q are analytic in that same neighborhood. The amplitude is analytic and positive at the origin. The root expansion is

$$\Psi(\sigma, \theta) = \frac{\sigma + v\theta^2/2}{\alpha} + O(\sigma^2 + |\sigma\theta| + |\theta|^3). \quad (26.3)$$

The exact mass and amplitude are

$$m_* := \sum_{q \geq 1} W_q(0, 0) = \frac{\rho}{(1 - \rho)z + \rho} < 1, \quad C_q = \frac{\sqrt{-16z^2 + 55z - 10}}{2\sqrt{\pi}}, \quad a_* := a(0, 0) = \frac{\rho C_q}{(1 - \rho)z}, \quad c_* := \log \frac{1 - m_*}{2a_*}. \quad (26.4)$$

For clarity, the amplitude is that of the actual macro-kernel, including its factor $\rho t/(1 - \rho)$. Omitting the factor t would change the constant in (26.1).

These formulas can also be recovered directly from the quadratic for Q . With $b = 1 - x$,

$$\mathcal{B} = b(\zeta - b) + xT(x - \zeta), \quad \Delta = \mathcal{B}^2 - 4xT(1 - \zeta)b\zeta,$$

and smaller discriminant root z_c , the square-root singular amplitude is

$$C_q(x, T) = \frac{\sqrt{-z_c \Delta_\zeta(x, z_c, T)}}{4\sqrt{\pi} x T (1 - z_c)}.$$

Substituting $(x, T, z_c) = (\rho, z^{-1}, z)$ and reducing modulo the cubic gives (26.4). Also, at the root,

$$\frac{xT}{1 - x} Q(x, z, T) = -\frac{\mathcal{B}(x, z, T)}{2(1 - x)(1 - z)},$$

which reduces to the stated m_* . The algebraic identities here follow by polynomial reduction modulo the defining cubic; no numerical approximation is used in the proof.

26.2 Quantitative head estimate

For real (σ, θ) with $Q(|\sigma| + |\theta|) = o(1)$, differentiating (26.2) along the line segment from the origin gives

$$|\nabla W_q(\sigma', \theta')| \leq Cq^{-1/2}, \quad 1 \leq q \leq Q,$$

uniformly on that segment. Indeed, derivatives of a and E_q are bounded, derivatives of $e^{q\Psi}$ supply at most a factor Cq , and the exponential remains bounded since $|q\Psi| \leq CQ(|\sigma| + |\theta|) = o(1)$. The finitely many small indices are absorbed by analyticity. Therefore

$$\sum_{q \leq Q} W_q(\sigma, \theta) \leq \sum_{q \leq Q} W_q(0, 0) + C(|\sigma| + |\theta|)\sqrt{Q} \leq m_* + C(|\sigma| + |\theta|)\sqrt{Q}. \quad (26.5)$$

This is a quantitative bound for an increasing head cutoff. No unspecified finite-cutoff continuity is used.

26.3 Uniform endpoint-tail lemma

Put $\ell = \log L$,

$$H = n^{2/3}L^{1/3}, \quad \epsilon = L^{-12}, \quad Q = \lceil L^4 \rceil,$$

and suppose

$$\epsilon H \leq r \leq (R+1)H, \quad k_r = \frac{1}{2} \log r + \log \log r + c,$$

where R is fixed and c remains in a fixed compact interval. Uniformly in these choices,

$$\sum_{Q < q \leq r} q^{-3/2} e^{k_r q/r} = 2e^c + O\left(\frac{\ell}{L} + L^{-2} + n^{-1/6}L^5\right). \quad (26.6)$$

Only the upper-bound version of (26.6) is needed; the equality follows from the same splitting.

To prove it, note that $\log r = (2/3)L + O(\ell)$ uniformly and split the sum at r/L^2 and $r/2$. For large n , these points exceed Q .

- On $Q < q \leq r/L^2$, the exponential is at most $e^{C/L}$, so the sum is $O(Q^{-1/2}) = O(L^{-2})$.
- On $r/L^2 < q \leq r/2$, the sum is bounded by

$$C \frac{L}{\sqrt{r}} e^{k_r/2} \leq CL^{3/2} r^{-1/4} \leq Cn^{-1/6} L^5.$$

- On $r/2 < q \leq r$, write $q = \lfloor r \rfloor - j$. The geometric distribution proportional to $e^{-k_r j/r}$ has mean $O(r/k_r)$. On the retained range, the factor $(q/r)^{-3/2}$ differs from 1 by at most a constant times $(r-q)/r$. Summing this bound gives

$$\sum_{r/2 < q \leq r} q^{-3/2} e^{k_r q/r} = \frac{r^{-1/2} e^{k_r}}{k_r} \left[1 + O(k_r^{-1} + k_r/r) \right]. \quad (26.7)$$

The geometric tail beyond this range is exponentially small in k_r ; nonintegral r changes the answer by relative $O(k_r/r)$. Since

$$\frac{r^{-1/2} e^{k_r}}{k_r} = e^c \frac{\log r}{k_r} = 2e^c \left[1 + O\left(\frac{\log \log r}{\log r}\right) \right],$$

this is $2e^c + O(\ell/L)$.

In particular, the error in (26.6) is $o(L^{-1/4})$.

26.4 Refined potential and exact finite-n calibration

Use two different names for the cutoffs:

$$\delta = \epsilon = L^{-12}, \quad d_n = L^{-1/4}, \quad \tau = 1 - 2\delta, \quad a = \alpha/(2L).$$

Here d_n is a row-mass slack, not the endpoint cutoff. Define

$$V_n(y) = \frac{\alpha}{L(y + \epsilon)} \left[\frac{1}{2} \log(H(y + \epsilon)) + \log \log(H(y + \epsilon)) + c_* - d_n \right], \quad 0 \leq y \leq R, \quad (26.8)$$

and

$$B = B_0 + a \left[\frac{7}{3} \ell + 2 \log(2/3) + 2c_* - 2d_n \right]. \quad (26.9)$$

For large n , B lies in a fixed compact subset of $(0, \infty)$. Let p_* be the full Kepler momentum for this B , with

$$y_*(x) = Y \sin^2 u, \quad x = (u - \sin u \cos u)/\pi, \quad Y = (2vB/\pi^2)^{1/3}, \quad p_* = -y'_*/v, \quad p'_* = B/y_*^2.$$

Set

$$p_n(s) = p_*(\delta + \tau s), \quad A(s) = p'_n(s) > 0, \quad J_n(A) = \min_{0 \leq y \leq R} (Ay + V_n(y)),$$

and choose $g_n(1) = 0$, $g'_n = vp_n^2/2 - J_n(p'_n)$. Then

$$f_n(s, y) = g_n(s) + p_n(s)y$$

obeys the exact inequality

$$-\partial_s f_n + \frac{v}{2} (\partial_y f_n)^2 = J_n(A) - Ay \leq V_n(y). \quad (26.10)$$

The potential is strictly convex for large n : its second derivative has leading positive term proportional to $\log(H(y + \epsilon))/(L(y + \epsilon)^3)$, while all remaining terms are smaller uniformly. Hence J_n is C^1 , including any boundary regimes, and $0 \leq J'_n \leq R$. The same exact estimates as in the second-order calibration argument give

$$\begin{aligned} \|p_n\|_\infty &= O(L^4), \quad \|p'_n\|_\infty = O(L^{16}), \quad \|p''_n\|_\infty = O(L^{28}), \\ \|f_n\|_{C^2} &= O(L^{28}), \quad \|(f_n)_s\|_\infty = O(L^{16}), \quad \|(f_n)_y\|_\infty = O(L^4). \end{aligned} \quad (26.11)$$

For example, $g''_n = vp_n p'_n - J'_n(p'_n) p''_n$; no second derivative of the optimizer is required.

26.5 Uniform Legendre expansion, including the cutoff endpoints

For $u = y + \epsilon$, an exact rearrangement of (26.8) is

$$\begin{aligned} V_n(y) &= \frac{B + a \log u + R_n(u)}{u}, \\ R_n(u) &= 2a \log \left(1 + \frac{\ell + 3 \log u}{2L} \right). \end{aligned} \quad (26.12)$$

On $\epsilon \leq u \leq R + \epsilon$, the logarithm's argument is positive and uniformly tends to 1. In this range,

$$|R_n(u)| \leq C \frac{\ell + |\log u|}{L^2}, \quad |(u \partial_u)^j R_n(u)| \leq C/L^2 \quad (j = 1, 2). \quad (26.13)$$

Let $\lambda = \sqrt{B/A}$, for $A = A(s)$ along the truncated arch. Because $A = \tau B/y_*^2$,

$$\lambda = \frac{y_*(\delta + \tau s)}{\sqrt{\tau}}, \quad cL^{-8} \leq \lambda \leq C_1. \quad (26.14)$$

Choose the fixed R large enough that $\lambda \leq (R + \epsilon)/3$ for all large n . Also $\epsilon/\lambda = O(L^{-4})$. Consequently the admissible domain for $r = u/\lambda$ contains $[1/2, 2]$, and $|\log \lambda| = O(\ell)$ uniformly.

After adding ϵA and dividing the objective by $\sqrt{BA}$, the minimization becomes

$$\Phi(r) = r + \frac{1}{r} + \frac{a}{B} \frac{\log \lambda + \log r}{r} + \frac{R_n(\lambda r)}{Br}. \quad (26.15)$$

Globally on its admissible domain, $B + a \log u + R_n(u) \geq B/2$ for large n . Thus $\Phi(r) \geq r + 1/(2r)$, which is at least 4 outside $[1/8, 8]$. At $r = 1$, $\Phi(1) = 2 + o(1)$. Therefore its minimizer belongs to $[1/8, 8]$. On that fixed interval the perturbation and its first two derivatives are bounded by

$$C \left[\frac{1 + |\log \lambda|}{L} + \frac{\ell + |\log \lambda|}{L^2} \right] = o(1).$$

The strict separation of $r + 1/r$ from 2 outside $[1/2, 2]$ then puts the minimizer in $(1/2, 2)$. It is an interior minimizer of the original constrained problem. The derivative equation, and the lower bound for $2r^{-3}$ on this compact interval, give

$$r_{\min} = 1 + O((1 + |\log \lambda|)/L). \quad (26.16)$$

Taylor expansion at 1, using (26.13), now proves the genuinely uniform value estimate

$$J_n(A) = 2\sqrt{BA} - \epsilon A + a \frac{\log \lambda}{\lambda} + O \left(\frac{\sqrt{A}}{L^2} [1 + \log^2 A + \ell] \right). \quad (26.17)$$

In particular the perturbation's square, which could have mattered after endpoint integration, is retained explicitly.

For $j = 0, 1, 2$, the Kepler endpoint estimates give

$$\int_0^1 \sqrt{A(s)} [1 + |\log A(s)|^j] ds = O(1). \quad (26.18)$$

To check this uniformly, change variables to $x = \delta + \tau s$. Near either endpoint, the full-arch comparison is $x^{-2/3}(1 + |\log x|^j)$, which is integrable; reparameterization changes the bounds only by fixed factors. Therefore the integrated error in (26.17) is

$$O((1 + \ell)/L^2) = o(1/L). \quad (26.19)$$

Also

$$\epsilon \int_0^1 A(s) ds = \epsilon[p_n(1) - p_n(0)] = O(L^{-8}). \quad (26.20)$$

The complete-arch action is

$$\int_0^1 [2\sqrt{Bp'_*} - vp_*^2/2], dx = 3B/Y =: C(B).$$

Cutting off its two ends and applying the factors $\tau^{-1/2}$ and τ^{-1} changes the action by $O(\delta^{1/3}) = O(L^{-4})$. This follows by separately bounding both discarded integrands by $Cx^{-2/3}$.

Finally,

$$\begin{aligned} \int_0^1 \frac{\log \lambda(s)}{\lambda(s)} ds &= \frac{1}{\sqrt{\tau}} \int_\delta^{1-\delta} \frac{\log y_*(x) - \frac{1}{2} \log \tau}{y_*(x)}, dx \\ &= \frac{2}{Y} (\log Y - 2 \log 2) + O(\delta^{1/3}(1 + |\log \delta|)). \end{aligned} \quad (26.21)$$

Indeed, $dx/y_* = 2 du/(\pi Y)$, and $\int_0^\pi \log \sin u du = -\pi \log 2$. The omitted logarithmic endpoint integrals are $O(\delta^{1/3}(1 + |\log \delta|))$, by the same integrable comparison.

Integrating (26.17) now yields

$$f_n(0, 0) = C(B) + a \frac{2}{Y(B)} (\log Y(B) - 2 \log 2) + O\left(\frac{1 + \ell}{L^2} + L^{-4}\right). \quad (26.22)$$

Since $C'(B) = 2/Y(B)$ and $C = \alpha/Y_0$, Taylor expansion of (26.22) at B_0 gives

$$f_n(0, 0) = C \left[1 + \frac{7\ell}{3L} + \frac{\log Y_0 - 2 \log 3 + 2c_*}{L} - \frac{2d_n}{L} \right] + O\left(\frac{(1 + \ell)^2}{L^2} + L^{-4}\right). \quad (26.23)$$

The identity $2 \log(2/3) - 2 \log 2 = -2 \log 3$ is the source of the constant in (26.1). The slack costs $2Cd_n/L = 2CL^{-5/4} = o(1/L)$.

26.6 Quantitatively contracting transformed rows

Retain source and destination heights in $[1, RH]$, and legal steps satisfying $q \leq h$ and $k + e \leq n - 1$. Let $s = k/n$, $y = h/H$,

$$\eta = F/n, \quad \beta = F/H, \quad \beta^2 = \eta, \quad \eta H = L.$$

The transformed step has weight

$$w(e, q, d) \exp\{F[f_n(s, y) - f_n(s + e/n, y + d/H)]\}. \quad (26.24)$$

On $e \leq HL^{40}$, $|d| \leq \sqrt{H}L^{40}$, use (26.11) and the fact that f_n is affine in height. The Taylor remainder is at most

$$CFL^{28} \left[(HL^{40}/n)^2 + (HL^{40}/n)(L^{40}/\sqrt{H}) \right] \leq Cn^{-1/3}L^{110}. \quad (26.25)$$

The linear tilts are

$$\sigma = -\eta(f_n)_s, \quad \theta = -\beta p_n,$$

so $|\sigma| \leq Cn^{-2/3}L^{50/3}$, $|\theta| \leq Cn^{-1/3}L^{13/3}$. In particular $Q(|\sigma| + |\theta|) = o(1)$, as required for (26.5).

By (26.3) and (26.10), uniformly for $q \leq h \leq RH$,

$$q\Psi(\sigma, \theta) \leq \frac{q\eta}{\alpha}V_n(y) + Cn^{-1/3}L^{34} = \frac{q}{r} \left[\frac{1}{2} \log r + \log \log r + c_* - d_n \right] + Cn^{-1/3}L^{34}, \quad r = h + \epsilon H. \quad (26.26)$$

This estimate does not extend legality: we only enlarge the upper bound from $q \leq h$ to $q \leq r$. On $q > Q$, (26.2) gives $a(\sigma, \theta) = a_* + O(|\sigma| + |\theta|)$ and $E_q = O(Q^{-1})$. Combining (26.5), (26.6), (26.25), and (26.26), the good-step row has mass at most

$$m_* + 2a_*e^{c_*-d_n} + O\left(\frac{\ell}{L} + L^{-2} + n^{-1/6}L^5 + n^{-1/3}L^{110}\right). \quad (26.27)$$

All estimates are uniform in the source height and elapsed length. The head contributes its true critical mass only once. Since $2a_*e^{c_*} = 1 - m_*$, (26.27) is

$$1 - (1 - m_*)d_n + O(d_n^2) + o(d_n). \quad (26.28)$$

For completeness, the omitted steps have much smaller mass. The full transformed exponent is bounded by $C\eta L^{16}e + C\beta L^{16}|d|$. For $e > HL^{40}$, a fixed small extra positive length tilt in the common analytic neighborhood gives an upper bound

$$\exp[-cHL^{40} + O(H)].$$

For $|d| > \sqrt{H}L^{40}$, splitting the two signs and adding $\theta_0 = cL^{40}/\sqrt{H}$ gives an upper bound

$$\exp[-cL^{80} + C_R c^2 L^{80} + O(L^{57})].$$

Choose $c > 0$ fixed and sufficiently small after R ; this is at most $\exp[-c'L^{80}]$ for large n . All cubic discriminant errors still tend to zero, since every relevant tilt is $n^{-1/3}$ or $n^{-2/3}$ times a fixed power of L . These bounds use unrestricted destination sums and hence cover all omitted steps. They are $o(d_n)$.

Consequently, for some fixed $b_0 > 0$, every complete transformed row has mass at most

$$1 - b_0 d_n, \quad \text{and one may take } b_0 = (1 - m_*)/2 \quad (26.29)$$

for all sufficiently large n . This establishes the required quantitative contraction rather than relying on a qualitative finite-height limit.

26.7 Terminal, renewal, and first-hit errors

The positive transformed Neumann sum is bounded by $(b_0 d_n)^{-1}$. Its logarithm is $O(\log L) = o(F/L)$.

The terminal representation is exactly the one in the exact positive macro-kernel: after elapsed macro-length k , its factor is $\rho^{n-1-k}t^{1-h}$, apart from the fixed initial factor. Since $g_n(1) = 0$, (26.11) gives

$$f_n(s, h/H) \leq p_n(1)h/H + CL^{16}(1 - s).$$

Writing $r_0 = n - 1 - k \geq 0$, the terminal product is bounded by a fixed constant times the exponential of

$$r_0[\log \rho + O(\eta L^{16})] + h[-\log t + \beta p_n(1)] + O(\eta L^{16}).$$

The two small corrections tend to zero, while $\log \rho < 0$ and $-\log t < 0$. The terminal product is therefore uniformly bounded independently of n, k, h . The initial-height adjustment is

$$F[f_n(0, 1/H) - f_n(0, 0)] = \beta p_n(0) = o(1).$$

Thus the confined coefficients satisfy

$$a_n^{[\leq RH]} \rho^n \leq C_2 d_n^{-1} e^{-F f_n(0,0) + o(1)}. \quad (26.30)$$

The imported first-hit estimate is

$$a_n^{[> RH]} \rho^n \leq C_3 \exp[-R^2 F / (4D_0)] \quad (26.31)$$

for a fixed $D_0 > 0$. Choose the fixed R , in addition to the Legendre-domain requirement above, so that $R^2 / (4D_0) > C + 1$. Equation (26.23) shows $f_n(0, 0) \rightarrow C$, so (26.31) is exponentially smaller than (26.30). Taking logarithms after adding the two positive contributions gives

$$D_n \geq F f_n(0, 0) - O(\log L). \quad (26.32)$$

Combining (26.23) and (26.32) proves (26.1). More explicitly, this construction yields

$$D_n \geq CF \left[1 + \frac{7\ell}{3L} + \frac{\kappa}{L} \right] - 2CFL^{-5/4} - O\left(F \frac{(1+\ell)^2}{L^2} + FL^{-4} + \log L \right). \quad (26.33)$$

Every error in (26.33) is $o(F/L)$. No endpoint, terminal, or geometric-series term remains at the target scale.

27 A matching coefficient lower bound

We prove

$$D_n \leq C_* F \left[1 + \frac{7 \log L}{3L} + \frac{\kappa}{L} \right] + o(F/L). \quad (27.1)$$

The complete row is essential at this precision. At its threshold, critical small- q transitions retain mass m_* , while the remaining mass $1 - m_*$ concentrates near the height cutoff. Discarding the former would change the desired constant.

Our construction samples independent tilted rows along a deterministic arch, except for a short central interval. A deterministic patch in that interval enforces both exact physical degree and exact endpoint height. Its cost is $o(F/L)$, so no lattice local limit theorem for an exact global constraint is needed.

27.1 An interpolated legal cutoff

Set $h_0 = L^{21}$ and $\eta = 2/L^2$. For real $h \geq h_0$, put $u = (1 - \eta)h$ and define

$$\chi_h(q) = \begin{cases} 1, & q \leq \lfloor u \rfloor, \\ u - \lfloor u \rfloor, & q = \lfloor u \rfloor + 1, \\ 0, & q > \lfloor u \rfloor + 1. \end{cases}$$

All weights lie in $[0, 1]$. They therefore define a lower subkernel of the exact positive combinatorial kernel whenever its retained q values are legal. The interpolation is a proof device; it does not assert fractional combinatorial multiplicities.

Throughout this section e is the *total* macro degree, including the leading $x/(1 - x)$ multiplier. Define

$$R_h(s, \theta) = \sum_{e, q, d} \chi_h(q) w(e, q, d) e^{se + \theta d}. \quad (27.2)$$

Let $V(h) > 0$ be the small positive solution of $R_h(V(h), 0) = 1$, and put

$$N(h) = \sum_{e, q, d} q \chi_h(q) w(e, q, d) e^{V(h)e}.$$

The row is below 1 at $s = 0$, strictly increases in s , and exceeds 1 in the established endpoint regime. The solution is unique.

The exact local-row result from the pinned second-order report gives, first in terms of the actual cutoff u ,

$$V(h) = \frac{\alpha}{u} \left[\frac{1}{2} \log u + \log \log u + c_* + \epsilon(u) \right], \quad \sup_{u \geq (1-\eta)h_0} |\epsilon(u)| \longrightarrow 0.$$

Hence, for each fixed R , uniformly on $h_0 \leq h \leq RH$,

$$V(h) = \frac{\alpha}{h} \left[\frac{1}{2} \log h + \log \log h + c_* + o(1) \right], \quad (27.3)$$

$$N(h) = (1 - m_*)h[1 + o(1)]. \quad (27.4)$$

The cutoff shrink changes the bracket in (27.3) by $O(L^{-1})$. For arbitrarily large h , we use only $V(h) \asymp \log h/h$ and the derivative bounds below.

The functions are continuous and piecewise continuously differentiable. Implicit differentiation on each interpolation interval gives

$$V'(h) = - \frac{(1 - \eta)W_{\lfloor u \rfloor + 1}(V(h), 0)}{(R_h)_s(V(h), 0)}.$$

The differentiated square-root transfer and the positive endpoint amplitude imply

$$c \frac{\log h}{h^2} \leq -V'(h) \leq C \frac{\log h}{h^2}, \quad |N'(h)| \leq C \log h, \quad ch \leq N(h) \leq Ch. \quad (27.5)$$

Indeed, the endpoint weight is $\Theta(\log h/h)$, the length mean $(R_h)_s$ is $\Theta(h)$, the q -weighted boundary term is $O(\log h)$, and the mixed q -length moment is $O(h^2)$. These estimates also hold as one-sided bounds at the junctions.

27.2 An elementary energy minimum constructs the arch

For $Y > h_0$ define

$$\mathcal{J}_n(Y) = nV(Y) + 2\sqrt{\frac{2}{v}} \int_{h_0}^Y \sqrt{V(h) - V(Y)} dh. \quad (27.6)$$

It is continuous and tends to infinity as $Y \rightarrow \infty$, because the integral over $[Y/4, Y/2]$ is at least $c\sqrt{Y \log Y}$. Also $\mathcal{J}_n(h_0) = nV(h_0) \gg F$.

For any absolutely continuous physical-time trial path $h(t) \geq h_0$ with endpoints h_0 and peak Y , the square inequality gives

$$\frac{\dot{h}^2}{2v} + V(h) \geq V(Y) + |\dot{h}| \sqrt{\frac{2(V(h) - V(Y))}{v}}.$$

Its integral is at least $\mathcal{J}_n(Y)$. The trial obtained by adding h_0 to the leading Kepler arch has action $O(F)$. Consequently $\mathcal{J}_n$ has an interior global minimum with value $O(F)$.

At a smooth point, differentiating the integral gives

$$\mathcal{J}'_n(Y) = V'(Y) \left[n - \sqrt{\frac{2}{v}} \int_{h_0}^Y (V(h) - V(Y))^{-1/2} dh \right].$$

The derivative under the integral is dominated locally by a constant times $(Y - h)^{-1/2}$. At a cutoff junction, the two one-sided derivatives have the same bracket and strictly negative factors V' . An interior minimum therefore satisfies, in either case,

$$n = \sqrt{\frac{2}{v}} \int_{h_0}^Y (V(h) - V(Y))^{-1/2} dh. \quad (27.7)$$

Choose such a minimizing Y , and write $\lambda = V(Y)$.

Construct the ascending physical-time path by $\dot{h} = \sqrt{2v(V(h) - \lambda)}$ and reflect it about its peak. Equation (27.7) makes its duration exactly n . It attains equality in the square inequality, so its action is $\mathcal{J}_n(Y)$ and is no greater than any trial action. No nonsmooth Euler–Lagrange theorem or derivative of the $o(1)$ remainder in (27.3) is being invoked.

Its peak satisfies $Y = \Theta(H)$. The upper bound follows from kinetic coercivity and action $O(F)$. If $Y \leq \sqrt{n}$, then $nV(Y) \gg F$, a contradiction. Thus $\log Y \geq L/2$ eventually, and $nV(Y) = O(F)$ then gives $Y \geq cH$.

Along this path define the signed momentum

$$\theta = \pm \sqrt{\frac{2(V(h) - \lambda)}{v}}, \quad \frac{dt}{dj} = \frac{N(h)}{\alpha}.$$

Its macro-coordinate velocity is

$$\frac{dh}{dj} = \frac{v}{\alpha} N(h) \theta. \quad (27.8)$$

The continuous macro duration is

$$m = 2 \int_{h_0}^Y \frac{dh}{(v/\alpha) N(h) \sqrt{2(V(h) - V(Y))/v}} = \Theta\left(\sqrt{Y/\log Y}\right) = \Theta(F/L).$$

For this estimate, below $Y/2$ the potential difference is comparable to $\log h/h$; above $Y/2$ it is comparable to $(L/Y^2)(Y - h)$, by (27.5).

Round m to an integer M and reparameterize to $j \in [0, M]$, changing the speed by $1 + O(M^{-1})$. Denote its sampled heights and momenta by h_j and θ_j . The profile is symmetric, $h_j = h_{M-j}$, and

$$|\theta_j| \leq C\sqrt{L/h_j}, \quad |\theta'(j)| \leq CL/h_j, \quad |h''(j)| \leq CL^2. \quad (27.9)$$

Before reparameterization, $\theta' = N(h)V'(h)/\alpha$. Furthermore, with $f = (v/\alpha)N\theta$,

$$ff' = (v/\alpha)^2[NN'\theta^2 + N^2V'/v] = O(L^2).$$

The apparent square-root singularity at the peak cancels in these macro-coordinate expressions.

27.3 Independent complete row laws

At position j , take the probability law on (e, q, d) proportional to

$$\chi_{h_j}(q)w(e, q, d)e^{\lambda e + \theta_j d},$$

and call its row mass R_j . Uniform differentiated transfer gives

$$R_j = 1 + O(L^{3/2}/\sqrt{h_j}) = 1 + o(1), \quad (27.10)$$

$$\mathbb{E}d_j = (v/\alpha)N(h_j)\theta_j + O(L^2), \quad (27.11)$$

$$\mathbb{E}e_j = N(h_j)/\alpha + O(L^{3/2}\sqrt{h_j}). \quad (27.12)$$

Here is the comparison underlying these statements. Uniformly for all q , after including the finitely many small indices, the reduced analytic factor $W_q e^{-q\Psi}$ has bounded logarithmic first derivatives. Since $\lambda + v\theta_j^2/2 = V(h_j)$,

$$\Psi(\lambda, \theta_j) - \Psi(V(h_j), 0) = O((L/h_j)^{3/2}).$$

Multiplication by $q \leq h_j$ yields (27.10). The derivative $\Psi_\theta = \nu\theta + O(\lambda + \theta^2)$, with $\nu = v/\alpha$, gives (27.11); the change of the q mean costs at most $O(\theta_j h_j L^{3/2}/\sqrt{h_j}) = O(L^2)$. Likewise $\Psi_s = \alpha^{-1} + O(|\theta| + \lambda)$ gives (27.12).

The centered logarithmic moment generating functions satisfy

$$\log \mathbb{E}e^{z(d_j - \mathbb{E}d_j)} \leq Cz^2 h_j L, \quad |z| \leq c/\sqrt{h_j L}, \quad (27.13)$$

$$\log \mathbb{E}e^{z(e_j - \mathbb{E}e_j)} \leq Cz^2 h_j^2, \quad |z| \leq c/h_j. \quad (27.14)$$

To verify these bounds, differentiate the exact normalized row twice throughout the stated extra-tilt intervals. At fixed q , increment variance is $O(q)$ and conditional mean is $O(q\sqrt{L/h_j})$; variation of this mean over $q \leq h_j$ adds at most $O(h_j L)$. Length has conditional variance $O(q)$ and conditional mean $O(q)$, so its total variance is $O(h_j^2)$. The same bounds persist under the extra tilt. Taylor's theorem for the logarithmic row proves (27.13)–(27.14).

27.4 Sample both ends and leave a central gap

Let r be the smallest integer at least $M^{3/4}$ with $r \equiv M \pmod{2}$, and put $a = (M - r)/2$, $b = (M + r)/2$. Leave positions $a, \dots, b - 1$ unsampled. Sample all others independently from the

row laws. View the left part forward from seed height $\lfloor h_0 \rfloor$ and the right part backward from that same terminal seed height. With probability tending to one, every sampled boundary has error

$$|E_j| \leq h_j/(10L^2). \quad (27.15)$$

We give quantitative bounds for this assertion.

On the ascending half while $h \leq Y/2$, $f(h) \geq c\sqrt{h \log h}$. The number of preceding positions is $O(\sqrt{h/\log h})$, and

$$\sum_{i < j} h_i L \leq CLh^{3/2}/\sqrt{\log h}.$$

By (27.11) and (27.9), the accumulated mean mismatch from the deterministic profile is bounded by

$$CL^2 \sqrt{h/\log h} + O(h/M) = o(h/L^2), \quad h \geq h_0 = L^{21}.$$

Bernstein's inequality from (27.13), at threshold $h/(20L^2)$, has exponent at least

$$c\sqrt{h \log h}/L^5 \geq cL^{11/2} \sqrt{\log L}.$$

Near the peak, use total variance $O(nL)$ and threshold cH/L^2 ; the exponent is at least $cF/L^5 \gg L$. A union bound over $M = \exp(O(L))$ boundaries still tends to zero. The right half uses its complementary suffix from the fixed terminal height, with the same estimates. No conditioning is used.

Every retained q satisfies $\lceil (1 - 2/L^2)h_j \rceil \leq h_j - h_j/L^2$ eventually, so (27.15) makes it legal at the actual starting height. Positivity of the boundary heights follows as well.

Two further estimates hold together with probability tending to one:

$$\left| \sum e_j - \sum \mathbb{E} e_j \right| \leq LH\sqrt{M}, \quad |\Delta| \leq L\sqrt{nL}. \quad (27.16)$$

The sums run over sampled positions and Δ is the exact height difference between the two central endpoints. Chebyshev's inequality already suffices: the degree variance is $O(MH^2)$ and height variance is $O(nL)$. The total drift bias $O(ML^2)$ is $o(\sqrt{nL})$. The parity choice makes the deterministic heights at a and b exactly equal.

27.5 Repair physical degree and height exactly

Let X be the sampled total degree. The ascent seed, final descent seed and original first letter have total degree $2\lfloor h_0 \rfloor + 1$. Define the required patch degree

$$D = n - (2\lfloor h_0 \rfloor + 1) - X.$$

The complete sum of the model means $N(h_j)/\alpha$ differs from its physical-time integral n by $O(HL)$: total variation of N is $O(HL)$ by (27.5). The cumulative error in (27.12) is also $O(HL)$, since $\sum_j \sqrt{h_j} = O(M\sqrt{H})$. The unsampled middle has model degree $r(1 - m_*)Y/\alpha[1 + o(1)]$. Therefore

$$\frac{D}{rY} \longrightarrow \frac{1 - m_*}{\alpha} \quad (27.17)$$

uniformly on the event (27.16), because $r \gg L\sqrt{M}$ and $r \gg L$. In particular $D > 0$.

Choose integer patch total degrees L_i , differing by at most one, with sum D . For the local positive Gaussian estimate use internal block degrees $\ell_i = L_i - 1$ and retain its leading x term. Thus

the sampled rows and the patch both count full physical degree exactly; there is no unaccounted one-unit shift per sampled block. Choose integer increments d_i , differing by at most one, with sum Δ . Linear interpolation keeps all patch heights at $Y[1 + o(1)]$.

By (27.17), every shifted q saddle is $(1 - m_*)Y[1 + o(1)]$, with a fixed positive legality margin below those heights. Moreover $|d_i| \leq |\Delta|/r + 1 = o(\sqrt{YL})$. The audited local lower estimate therefore supplies a patch whose critical logarithmic cost is at most

$$CrL + C\frac{\Delta^2}{rY} \leq CrL + C\frac{ML^3}{r} = o(M). \quad (27.18)$$

Its exact degree and endpoint height are the prescribed ones for every retained outside realization.

There is no overcounting. The total number M and the patch positions are fixed; deleting those positions from a completed marked macro-path recovers its outside data. Selecting the patch degree and increment pattern, then summing its legal q windows, retains a subset of the exact positive macro-path expansion. Every outside fractional χ factor is at most one. Thus these weighted contributions are genuine lower bounds for the original coefficient. The exact seeds cost $O(h_0) = o(M)$.

27.6 Remove the tilts

The sampled product of subkernel critical weights $\chi_{h_j}(q_j)w(e_j, q_j, d_j)$ equals its product probability multiplied by

$$\exp\left\{-\lambda \sum e_j - \sum \theta_j d_j\right\} \prod R_j.$$

Since every χ_{h_j} lies in $[0, 1]$, this product is a lower bound on the corresponding product of original critical weights. By (27.10), $\sum \log R_j = o(M)$. The difference between $\lambda \sum e_j$ and λn is $O(rL) + \text{polylog}(n) = o(M)$. Put $g_j = h_{j+1} - h_j$. On each sampled interval, summation by parts and (27.15) give

$$\sum \theta_j (d_j - g_j) = o(M). \quad (27.19)$$

The interior bound is $C \sum (L/h_j)(h_j/L^2) = O(M/L)$. At the central edges, $\theta = O((r/M)\sqrt{L/H})$ and $|E| \leq CH/L^2$, giving $O(r/L)$. Seed-boundary terms are polylogarithmic.

The omitted middle deterministic action is $O(rL) = o(M)$. A Stieltjes quadrature estimate gives

$$\sum_{j=0}^{M-1} \theta_j g_j = 2\sqrt{\frac{2}{v}} \int_{h_0}^Y \sqrt{V(h) - V(Y)} dh + o(M). \quad (27.20)$$

Its error is bounded by $C \int |\theta'| |h'| dj \leq C \int_{h_0}^Y (L/h) dh = O(L^2)$, plus endpoint roundings. Summing the retained independent realizations, whose probability is $1 - o(1)$, and inserting the patch and seeds proves

$$D_n \leq \mathcal{J}_n(Y) + o(M), \quad M = \Theta(F/L). \quad (27.21)$$

27.7 Evaluate a fixed trial action

No perturbed-peak expansion is required. By construction, $\mathcal{J}_n(Y)$ is no greater than the action of

$$h_{\text{trial}}(t) = h_0 + Hy_*(t/n), \quad y_*(s) = Y_0 \sin^2 u, \quad s = \frac{u - \sin u \cos u}{\pi},$$

where $Y_0 = (2v\alpha/(3\pi^2))^{1/3}$. Uniformly on $L^{-6} \leq y \leq R$,

$$\frac{n}{F} V(Hy) = \frac{\alpha}{3y} + \frac{\alpha}{2Ly} \left[\frac{7}{3} \log L + \log y + 2 \log(2/3) + 2c_* + o(1) \right]. \quad (27.22)$$

The row-threshold error is uniform, and the elementary log-log expansion has an additional bracket error $O((\log L + |\log y|)/L) = o(1)$ in this range. Adding h_0 changes the resulting bulk integral by $o(F/L)$.

The trial portions $y \leq L^{-6}$ cost only $O(F/L^3) + \text{polylog}(n) = o(F/L)$. For the potential, use $V(h) \leq C \log h/h$ and $y_*(s) \asymp s^{2/3}$; the portion with $Hy_* \leq h_0$ costs only a polynomial in L . The kinetic endpoint integral has the same $O(F/L^3)$ bound. Thus (27.22) may be integrated along the full leading arch at $o(F/L)$ error.

Its leading kinetic-plus-potential action is C_*F , and

$$\frac{\alpha}{2} \int_0^1 \frac{ds}{y_*(s)} = C_*, \quad \frac{\int_0^1 (\log y_*) ds/y_*}{\int_0^1 ds/y_*} = \log Y_0 - 2 \log 2.$$

Indeed $ds/y_* = 2 du/(\pi Y_0)$. The trial action is consequently

$$C_*F \left[1 + \frac{7 \log L}{3L} + \frac{\log Y_0 - 2 \log 3 + 2c_*}{L} \right] + o(F/L).$$

Combined with (27.21), this proves (27.1). The $o(F/L)$ error includes all endpoint, normalization, fluctuation and exact-degree repair costs.

28 The inverse constant

Let $\lambda_0 = \log \mu$ and define the integer threshold $N(Y) = \min\{n : a_n \geq Y\}$. The frozen foundation proves that a_n is nondecreasing, by appending a copy of its last letter. Put

$$B_* = C_* \lambda_0^{-4/3}, \quad \kappa_{\text{inv}} = \kappa - \frac{2}{3} \log \lambda_0.$$

Numerically,

$$B_* = 0.893568257650894909124642189453 \dots, \\ \kappa_{\text{inv}} = -2.398003540515347428750752039232 \dots$$

Corollary 28.1. *With $T = \log Y$,*

$$\begin{aligned} N(Y) &= \frac{T}{\lambda_0} + B_* T^{1/3} (\log T)^{2/3} \\ &\quad + \frac{7B_*}{3} T^{1/3} (\log T)^{-1/3} \log \log T + B_* \kappa_{\text{inv}} T^{1/3} (\log T)^{-1/3} \\ &\quad + o\left(T^{1/3} (\log T)^{-1/3}\right). \end{aligned} \tag{28.1}$$

Proof. Write $F_T = T^{1/3} (\log T)^{2/3}$ and $G_T = T^{1/3} (\log T)^{-1/3}$. Uniformly for $x = T/\lambda_0 + O(F_T)$,

$$x^{1/3} (\log x)^{2/3} = \lambda_0^{-1/3} \left[F_T - \frac{2}{3} (\log \lambda_0) G_T + o(G_T) \right],$$

$$x^{1/3} (\log x)^{-1/3} \log \log x = \lambda_0^{-1/3} G_T \log \log T + o(G_T), \quad x^{1/3} (\log x)^{-1/3} = \lambda_0^{-1/3} G_T + o(G_T).$$

The feedback from changing x by $O(F_T)$ is $O(F_T^2/T) = O(T^{-1/3} (\log T)^{4/3}) = o(1)$. Thus substituting the real expression on the right of (28.1), without its remainder, into the smooth asymptotic formula for $\log a_n$ gives $T + o(G_T)$. For every fixed $\varepsilon > 0$, round that expression minus εG_T downward and plus εG_T upward. Theorem 24.1 puts the corresponding logarithmic coefficients strictly below and above T for large T . Monotonicity brackets $N(Y)$ between them. Letting $\varepsilon \downarrow 0$ proves the corollary. The integer rounding error is $O(1) = o(G_T)$. $\square$

The shift $-(2/3)\log\lambda_0$ is necessary: it comes from $\log(T/\lambda_0) = \log T - \log\lambda_0$. It was hidden inside the previous $O(G_T)$ inverse remainder and becomes visible at the present precision.

[write, 2 October 2026] Here $T = \log Y$ (Part V writes $T = \log p$). This corollary is the case $J = 1$ of Theorem 35.1, with the same constant κ_{inv} ; the previous inverse result is Corollary 20.1.

29 Meaning and limits of the result

The lower construction gives a variational interpretation at the precision proved here. With the interpolated cutoff potential V from Section 27, endpoints $h_0 = L^{21}$, and physical time interval $[0, n]$, the energy minimum satisfies

$$D_n = \min_{h(0)=h(n)=h_0, h \geq h_0} \int_0^n \left[\frac{\dot{h}^2}{2v} + V(h) \right] dt + o(F/L).$$

Indeed, the coefficient lower construction bounds D_n above by this minimum plus $o(F/L)$; the fixed Kepler trial bounds the minimum by the right side of Theorem 24.1; and the coefficient upper bound matches that value. This is a finite-precision action statement, not an all-orders reduction.

The exact local constant c_* enters through the mass needed to fill the defective row $m_* < 1$. The remaining term $\log Y_0 - 2\log 3$ comes from the weighted logarithmic height along the Kepler arch. The weight makes its angular parameter uniform, so $\log \sin^2 u$ contributes $-2\log 2$.

The theorem still leaves an unquantified $o(F/L)$ error in the logarithm, which can diverge. It does not imply a multiplicative equivalent for a_n , a power-law prefactor, a fluctuation determinant, or an exact integer inverse rule. Higher inverse-log terms are a possible next research direction, but no all-orders expansion or transseries is asserted in this report. No unreviewed formal hierarchy is included in the proved theorem or its reproducibility certificates.

[write, 2 October 2026] The higher inverse-log terms are established, for every fixed order, in Part V. No all-orders expansion uniform in the order, and no transseries, is asserted there either.

Part V

All finite inverse logarithmic orders for A202061

Source. Batch-77 manuscript 38, archive `oeis-a202061-allorders-report.zip`, main file `a202061-allorders.tex` (258 lines) with four `\input` files (`proofs/reduction-section.tex`, `action-section.tex`, `inverse-section.tex`, `checks-scope-section.tex`), inlined; dated 2 October 2026. Author line: “A proof and reproducible research companion”. Printed in full except for its table of contents; label prefix `a61:ao:.`

Abstract of the source.

Let a_n count classical 120-avoiding ascent sequences. We establish the entire dominant inverse-logarithmic sector of $n \log \mu - \log a_n$: for every fixed nonnegative integer J ,

$$n \log \mu - \log a_n = C n^{1/3} (\log n)^{2/3} \left\{ \sum_{j=0}^J \frac{P_j(\kappa + \frac{7}{3} \log \log n)}{(\log n)^j} + o((\log n)^{-J}) \right\}.$$

The universal polynomial P_j has degree j . We give a finite beta-moment recurrence for every coefficient, explicit initial polynomials, a matching monotone inverse expansion, and reproducible symbolic and numerical checks. The coefficient proof reduces the positive macro-kernel to a smooth classical action at any prescribed fixed inverse-log order. Its lower construction retains the full critical row mass and repairs both integer constraints; its upper construction uses an exact convex-potential calibration. Endpoint cutoffs justify the action expansion without divergent integrals. The result does not determine a multiplicative equivalent, a power-law prefactor, or an exponentially complete transseries.

[write, 2 October 2026] “The entire dominant inverse-logarithmic sector” means every *fixed finite* order J : the theorem is not uniform in J , gives no $o(1)$ error in $\log a_n$, and is not an exponentially complete transseries, as the Part itself states after Theorem 30.1 and in Section 37.

30 The sequence and the main theorem

An ascent sequence is a word of nonnegative integers with $w_1 = 0$ and

$$w_i \leq 1 + \#\{j < i - 1 : w_j < w_{j+1}\} \quad (i \geq 2).$$

It avoids classical 120 if there are no indices $i < j < k$ with $w_k < w_i < w_j$. Let a_n count these words, with $a_0 = 1$; this is OEIS A202061 [1]. All logarithms in this report are natural.

Let z_* be the smallest positive root of $z^3 - 5z^2 + 6z - 1 = 0$. Define the critical constants

$$\begin{aligned} \rho &= 3z_*^2 - 10z_* + 2, & \mu &= \rho^{-1}, & t_* &= z_*^{-1}, \\ \alpha &= \frac{8z_*^2 - 29z_* + 9}{7}, & v &= \frac{2z_*^2 - 7z_* + 2}{2}, \\ m_* &= \frac{\rho}{(1 - \rho)z_* + \rho}, & C_q &= \frac{\sqrt{-16z_*^2 + 55z_* - 10}}{2\sqrt{\pi}}, \\ a_* &= \frac{\rho t_* C_q}{1 - \rho}, & c_* &= \log \frac{1 - m_*}{2a_*}. \end{aligned}$$

Here $\rho, \alpha, v, a_* > 0$, $\mu > 1$, and $0 < m_* < 1$. The constants in the answer are

$$Y_0 = \left(\frac{2v\alpha}{3\pi^2} \right)^{1/3}, \quad C = \frac{\alpha}{Y_0}, \quad \kappa = \log Y_0 - 2 \log 3 + 2c_*. \quad (30.1)$$

Their numerical values include

$$\begin{aligned} Y_0 &= 0.228432139006429054268160391628 \dots, \\ C &= 2.232625307612844922179764700526 \dots, \\ c_* &= 0.866796245805539952988634807649 \dots, \\ \kappa &= -1.940148183136270392093282071371 \dots \end{aligned}$$

For the rest of the paper put

$$\begin{aligned} L &= \log n, & H &= n^{2/3} L^{1/3}, & F &= H^2/n = n^{1/3} L^{2/3}, & M_0 &= F/L, \\ D_n &= n \log \mu - \log a_n, & B &= \kappa + \frac{7}{3} \log L. \end{aligned}$$

Theorem 30.1 (All finite inverse-logarithmic orders). *There are universal polynomials $P_j(B)$, independent of all the critical constants, such that for each fixed integer $J \geq 0$,*

$$\frac{D_n}{CF} = \sum_{j=0}^J \frac{P_j(B)}{L^j} + o(L^{-J}) \quad (n \rightarrow \infty \text{ through integers}). \quad (30.2)$$

Here $P_0 = 1$, $\deg P_j = j$, and

$$[B^j]P_j(B) = \binom{2/3}{j} (3/2)^j \quad (j \geq 1).$$

A finite recurrence generating all P_j is given in Section 34. In particular,

$$P_1(B) = B, \quad P_2(B) = -\frac{B^2}{4} + \frac{7B}{2} - 10.$$

The previous approved report proves the terms through P_1/L . The first additional complete inverse-log layer is therefore $CFP_2(B)/L^2$, with an $o(F/L^2)$ remainder. Numbering by individual logarithmic scales would split one polynomial layer into several terms; we consistently index whole powers of L^{-1} .

The quantifiers are essential: J is chosen first, and all auxiliary cutoffs and sufficiently-large- n thresholds may depend on it. Equation (30.2) makes no assertion for $J = J(n)$ tending to infinity. Since $F/L^J \rightarrow \infty$ for every fixed J , it does not provide an $o(1)$ error in $\log a_n$.

[write, 2 October 2026] “The previous approved report” is Part IV. Taking $J = 0$ and $J = 1$ gives the coefficient theorems of Parts I–IV. This theorem does *not* imply the non-D-finiteness of Corollary 1.2(3), the exact height-walk representation of Part I (which Section 31 imports), or the finite-height radius theorems (10.3) and (21.1) (whose constant c_* it uses).

31 Positive kernel and imported foundation

We state the enumerative and analytic inputs explicitly. Their proofs, exact identities, and earlier mathematical reviews are retained in the unchanged third-order dependency (Part IV of this report, and through it Parts I–III). This report proves the stronger reduction and subsequent all-finite-orders calculation from those inputs; it does not rederive the entire enumerative foundation. The audits in the companion package are research verification, not journal peer review or proof-assistant certification.

The exact positive height-walk representation has suffix series

$$e_h = \frac{1}{1-x} + \frac{x}{1-x} \sum_{q=1}^h (Q_q(x, T)e)_{h+1-q}, \quad (Tf)_h = f_{h+1}, \quad A(x) = 1 + xe_1(x).$$

Here $Q(x; u, T) = \sum_{q \geq 1} Q_q(x, T)u^q$ is the root vanishing at $u = 0$ of

$$xT(1-u)Q^2 + [(1-x)(u - (1-x)) + xT(x-u)]Q + (1-x)u = 0.$$

A macro-step with indices q, r changes height by $d = 1 - q + r$, has weight $x[T_0^r]Q_q(x, T_0)/(1 - x)$, and is legal exactly when q does not exceed its starting height. Its destination is then positive. If $k(e, q, d)$ is its multiplicity at total macro degree e , put

$$w(e, q, d) = k(e, q, d)\rho^e t_*^d, \quad W_q(s, \theta) = \sum_{e, d} w(e, q, d)e^{se+\theta d}.$$

The degree e includes the leading multiplier. The critical row mass is $\sum_q W_q(0, 0) = m_* < 1$. This missing mass is important; it must not be lost by replacing complete rows with only their large- q tails.

The quadratic branch supplies the uniformly differentiated analytic square-root transfer

$$W_q(s, \theta) = a(s, \theta)q^{-3/2}e^{q\Psi(s, \theta)}[1 + E_q(s, \theta)], \quad a(0, 0) = a_*, \quad (31.1)$$

where a and Ψ are analytic near $(0, 0)$, and $E_q = O(q^{-1})$ with the parameter derivatives used below. In particular,

$$\Psi(s, \theta) = \frac{s + v\theta^2/2}{\alpha} + O(s^2 + |s\theta| + |\theta|^3). \quad (31.2)$$

The imported transfer controls the first four parameter derivatives. Together with finitely many positive small- q terms, it gives uniformly bounded first and second logarithmic derivatives of the reduced weights $W_q e^{-q\Psi}$ for every $q \geq 1$. Higher finite inverse-log orders do not require arbitrarily many derivatives of this exact discrete row: only the scalar endpoint Laplace expansion is taken to higher orders.

The exact repair uses the local positive Gaussian lower bound. For fixed $K_0 > 3$, internal block degree $\ell \geq L^{K_0}$, and $|d| \leq C_0\sqrt{\ell L}$, summing a fixed sufficiently wide $O(\sqrt{\ell})$ conditional q window gives

$$\sum_q \rho^\ell t_*^d B(\ell, q, q + d - 1) \geq c\ell^{-2} \exp\left[-\frac{d^2}{2v\ell} - o(1)\right]. \quad (31.3)$$

Its center is $q = \alpha\ell + \beta_0(d - 1)$, where $\beta_0 = -4(z_* - 3)(2z_* - 1)/7$; all retained q differ from $\alpha\ell$ by $O(\sqrt{\ell L})$. The leading x term makes the whole macro degree $\ell + 1$. The bound follows from a positive summand, uniform Stirling estimates, and the exact Hessian identity; it does not assume a full coefficient equivalent.

At elapsed macro degree k and height h , the exact residual terminal factor is $\rho^{n-1-k}t_*^{1-h}$, apart from the fixed original leading ρ . Finally, for some fixed $D_0 > 0$, the first-hit estimate is

$$a_n^{[>RH]} \rho^n \leq C_0 \exp[-R^2 F/(4D_0)] \quad (31.4)$$

for fixed R and sufficiently large n . These are the full imported positive-kernel dependencies used below. No unproved finite-height spectral equivalence is invoked.

32 Reduction to a classical action

Write $\mathcal{S}(r) = \sum_{j \geq 0} (3/2)_j r^j$ as a formal series. The symbol $(3/2)_j$ is the rising factorial. For T large set $A = \log T + c_*$ and define the explicit smooth truncation

$$k_N(T) = \frac{T}{2} + \log T + c_* + \sum_{j=1}^N \frac{p_j(\log T + c_*)}{T^j}, \quad \phi_N(h) = \frac{\alpha k_N(\log h)}{h}. \quad (32.1)$$

The polynomials p_j are uniquely determined by substitution into

$$k - \frac{T}{2} - \log k + \log \mathcal{S}(1/k) + \log \frac{a_*}{1 - m_*} = 0. \quad (32.2)$$

For example,

$$\begin{aligned} p_1(A) &= 2A - 3, \\ p_2(A) &= -2A^2 + 10A - \frac{33}{2}, \\ p_3(A) &= \frac{8}{3}A^3 - 24A^2 + 86A - 120. \end{aligned}$$

[**write, 2 October 2026**] Equation (32.2) is the linear-logarithmic equation $aX + b \log X = L$ with $a = 1$, $b = -1$, perturbed by $\log \mathcal{S}(1/k)$; its large solution is the Lambert-core theorem `p0:thm:lambert-core` of the transseries volume (see the Guide). No novelty is claimed for this step.

Choose a fixed $h_b > 1$, after N , so that $\phi_N > 0$, $\phi'_N < 0$, and $\phi''_N > 0$ on $[h_b, \infty)$. Define

$$\mathcal{A}_N(n) = \min_{\substack{h(0)=h(n)=h_b \\ h \geq h_b}} \int_0^n \left[\frac{\dot{h}^2}{2v} + \phi_N(h) \right] dt, \quad (32.3)$$

over absolutely continuous paths with finite action. The logarithmic formula is never used down to height zero. Different sufficiently large fixed h_b change the minimum by $O(1)$.

Theorem 32.1 (Tunable action reduction). *For every fixed integer $K \geq 0$, choosing $N = K + 3$ gives*

$$D_n = \mathcal{A}_N(n) + o(M_0/L^K).$$

The following subsections prove the theorem with explicit error rates. Letters used as local cutoff exponents are subscripted to distinguish them from the polynomial argument B .

[**write, 2 October 2026**] *A tempting false reading.* $B = \kappa + \frac{7}{3} \log L = \kappa + \frac{7}{3} \log \log n$ is not a constant: it grows without bound, and each $P_j(B)/L^j$ is a polynomial in $\log \log n$ divided by $(\log n)^j$. It is unrelated to the block coefficient $B(\ell, q, q + d - 1)$ in (31.3) (Part I's $B(\ell, q, r)$), to the calibration bound B of Part II, to B_n of Part III, and to the constant $B_* = 0.8935 \dots$ of Part IV.

32.1 Quantitative local Watson expansion

Let $R_r(s, \theta)$ denote the exact row with all destinations retained and fractional q cutoff at real r : the cutoff $\chi_r(q)$ is one through $\lfloor r \rfloor$, equals $r - \lfloor r \rfloor$ at $\lfloor r \rfloor + 1$, and is zero thereafter. The tilt s counts the entire macro degree. For $T = \log r$ and $k = r\Psi(s, \theta) = T/2 + \log T + O(1)$, every fixed j_0 gives

$$\begin{aligned} R_r(s, \theta) &= m_* + a_* \frac{e^{k-T/2}}{k} \left[\sum_{j=0}^{j_0} (3/2)_j k^{-j} + O_{j_0}(k^{-j_0-1}) \right] \\ &\quad + O_{j_0}(r^{-1/4} T^{c_{j_0}} + |s| + |\theta|). \end{aligned} \quad (32.4)$$

The same estimate holds uniformly within a fixed distance of this band, with any fixed number of k derivatives of the normalized scalar endpoint expansion. Only the first two parameter derivatives of the exact row are needed in the other arguments.

Here is a proof that controls the critical head. The reduced-factor comparison following (31.1), on a row of bounded total mass, permits replacement of the reduced weights by critical weights with error $O(|s| + |\theta|)$. Split the scalar sum at r/k^2 and $r/2$. On the first part, $e^{kq/r} - 1 \leq C_0 kq/r$; summing the critical $q^{-3/2}$ tail gives a perturbation $O(r^{-1/2})$, while the missing critical tail is $O(kr^{-1/2})$. The middle part is at most

$$C_0 \frac{k}{\sqrt{r}} e^{k/2} = O(r^{-1/4} T^{c_0}).$$

On the last half, the Riemann-sum and fractional-last-term error is relative $O(k/r)$, and the transfer remainder costs relative $O(1/r)$. The resulting endpoint integral is

$$r^{-1/2} e^k \int_0^{1/2} (1-u)^{-3/2} e^{-ku} du.$$

Taylor expansion of $(1-u)^{-3/2}$ and integration of each u^j give $(3/2)_j k^{-j-1}$. The integrated remainder is $O(k^{-j_0-2})$; extending the moment integrals to infinity adds an exponentially small term. Differentiating the normalized integral in k inserts powers of u , giving the claimed differentiated remainders. All estimates are uniform under fractional cutoff interpolation.

There is a unique small positive root s_r with $R_r(s_r, 0) = 1$. The imported first approximation places its k in the stated band. Taking logarithms in (32.4), and using the implicit k derivative $1 + O(1/T)$, proves

$$\frac{rs_r}{\alpha} = k_N(T) + O_N \left(\frac{(1 + \log T)^{N+1}}{T^{N+1}} \right) + O_N(r^{-1/4} T^{c_N}). \quad (32.5)$$

Indeed finite substitution leaves this residual and the implicit derivative is bounded below by $1/2$. The discrepancy $r\Psi(s_r, 0) - rs_r/\alpha = O(T^2/r)$ is absorbed in the algebraic error. Direct differentiation of the explicit truncation gives, for fixed j ,

$$\phi_N^{(j)}(h) = \frac{\alpha}{2} (-1)^j j! \frac{\log h}{h^{j+1}} [1 + O_{N,j}(\log \log h / \log h)]. \quad (32.6)$$

This proves the asserted eventual positivity, decrease, and convexity.

The exact fractional-cutoff root is continuous and piecewise C^1 ; it need not be C^2 at integer knots. The lower construction uses only its audited one-sided first derivative bounds. Smooth higher derivatives belong to (32.1) or a finite scalar surrogate, never to an unjustifiably differentiated discrete remainder.

We also need a quantitative row gap. Fix K and E ; let $N = K + 3$, $\delta = L^{-K-2}$, and $HL^{-E} \leq r \leq (R+1)H$. If the tilts are a negative fixed power of n times a fixed power of L and

$$r\Psi(s, \theta) \leq k_N(\log r) - \delta + o(\delta),$$

then, uniformly,

$$R_r(s, \theta) \leq 1 - c\delta. \quad (32.7)$$

To see this even if the actual k is far below the endpoint band, first dominate the row by $1 + O(|s| + |\theta|)$ times the positive, increasing scalar envelope

$$G_r(k) = \sum_q \chi_r(q) W_q(0, 0) e^{kq/r}.$$

Monotonicity permits replacement of k by its upper bound in the endpoint band. There, (32.5) has error $o(\delta)$, the envelope mass is bounded, and the k derivative tends to $1 - m_* > 0$. The envelope root differs from the exact row root by an algebraically small error. The mean-value theorem now proves (32.7).

32.2 Action facts and endpoint bounds

The following estimates apply to $V = \phi_N$, and their coarse versions also apply to the exact fractional-cutoff potential. For endpoint a and proposed peak Y , put

$$J_a(Y) = nV(Y) + 2\sqrt{2/v} \int_a^Y \sqrt{V(h) - V(Y)} dh.$$

The pointwise square inequality shows that any path with this peak has action at least $J_a(Y)$. At an interior minimum of J_a ,

$$n = \sqrt{2/v} \int_a^Y [V(h) - V(Y)]^{-1/2} dh. \quad (32.8)$$

The rising solution $\dot{h} = \sqrt{2v[V(h) - V(Y)]}$, reflected at its peak, has this exact duration and action. The minimum exists: $J_a(a) = nV(a)$ is too large, while $J_a(Y) \rightarrow \infty$ as $Y \rightarrow \infty$. At a fractional-cutoff junction the one-sided derivatives of J_a have strictly negative potential factors multiplying the same continuous duration bracket; thus the argument still supplies an exact-duration arch.

For fixed a , for a a fixed power of L , or for $a = H\epsilon$ with $\epsilon = L^{-E}$, a shifted or regularized Kepler trial has action $O(F)$. Kinetic coercivity bounds the peak above by $O(H)$; the lower bound $nV(Y)$ bounds it below by a positive multiple of H . Thus $Y \asymp H$. Convexity of ϕ_N gives a unique smooth minimizing arch, symmetric about its midpoint and strictly concave in its interior.

For $h \leq Y/2$, $V(h) - V(Y) \asymp \log h/h$, once h_b is large. For $Y/2 \leq h \leq Y$ the difference is comparable to $(L/Y^2)(Y - h)$. The energy formula therefore gives

$$\int_{\{h \leq a\}} [\dot{h}^2/(2v) + V(h)] dt \leq C_0 \sqrt{a \log a}, \quad h_b \leq a \leq Y/2, \quad (32.9)$$

$$\int_0^n \frac{dt}{h(t)} \leq C_0 n/H = C_0 M_0. \quad (32.10)$$

For the second bound, split the height integral at Y/L^{C_1} , $C_1 > 2$. Above this height, $\log h \asymp L$ and the beta-integral comparison gives $O(\sqrt{Y/L})$. Below it, $\log h \geq \log h_b > 0$ gives $O(\sqrt{Y} L^{-C_1/2})$. The peak singularity is integrable.

Comparing the peak minima for $h_b \leq a_1 \leq a_2 \ll H$ yields

$$0 \leq \mathcal{A}_{a_1}(n) - \mathcal{A}_{a_2}(n) \leq 2\sqrt{2/v} \int_{a_1}^{a_2} \sqrt{V(h)} dh \leq C_0 \sqrt{a_2 \log a_2}. \quad (32.11)$$

Both minimizing peaks exceed a_2 in the ranges used here. Raising the endpoint to a fixed power of L costs only a fixed power of L ; raising it to $H\epsilon$ costs $O(F\sqrt{\epsilon})$.

32.3 Normalized rows and exact integer repair

Fix K and choose

$$B_0 = K + 3, \quad \eta = L^{-B_0}, \quad A_0 = 4B_0 + 10, \quad h_0 = L^{A_0}, \quad N = K + 3. \quad (32.12)$$

Let $V_\eta(h)$ be the exact row root at fractional cutoff $(1 - 2\eta)h$, and let $N_\eta(h)$ be its q moment. Constants in the pinned normalized-row proof are uniform as $\eta \rightarrow 0$, since the cutoff factor eventually lies in $[1/2, 1]$. In particular,

$$V_\eta \asymp \frac{\log h}{h}, \quad -V'_\eta \asymp \frac{\log h}{h^2}, \quad N_\eta \asymp h, \quad |N'_\eta| \leq C_0 \log h,$$

with one-sided derivatives at the knots. Use the exact duration- n energy arch, with peak $Y \asymp H$, energy $\lambda_0 = V_\eta(Y)$, and signed momentum θ . Change time by $dt/dj = N_\eta(h)/\alpha$. Its real macro duration is $\asymp M_0$. Round that duration to an integer M , and sample the symmetric arch at the M slots as in the pinned lower construction. The estimates needed below are

$$|\theta_j| \leq C_0 \sqrt{L/h_j}, \quad |\theta'_j| \leq C_0 L/h_j, \quad |h''_j| \leq C_0 L^2.$$

At each nonpatch slot independently sample the complete row proportional to

$$\chi_{(1-2\eta)h_j}(q)w(e, q, d)e^{\lambda_0 e + \theta_j d}.$$

The normalization and means satisfy

$$\begin{aligned} R_j &= 1 + O(L^{3/2}/\sqrt{h_j}), \\ \mathbb{E}d_j &= (v/\alpha)N_\eta(h_j)\theta_j + O(L^2), \\ \mathbb{E}e_j &= N_\eta(h_j)/\alpha + O(L^{3/2}\sqrt{h_j}). \end{aligned}$$

Centered log moment-generating functions are bounded by $C_0 z^2 h_j L$ for $|z| \leq c/\sqrt{h_j L}$, and, for length, by $C_0 z^2 h_j^2$ for $|z| \leq c/h_j$. These use only the preceding one-sided bounds, small tilts, and differentiated analytic transfer.

Remove $r_0 = M^{3/4} + O(1)$ central slots, choosing parity so their deterministic edge heights agree. The left prefix and independent right complementary suffix start from the two fixed seeds. Require the stronger error tube $|E_j| \leq \eta h_j/10$. Below $Y/2$, cumulative variance is at most $C_0 L h^{3/2}/\sqrt{\log h}$, while deterministic mismatch is at most

$$C_0 L^2 \sqrt{h/\log h} + C_0 h/M = o(\eta h).$$

Bernstein's exponent at threshold ηh is at least

$$c\eta^2 \frac{\sqrt{h \log h}}{L} \geq cL^{A_0/2-2B_0-1} \sqrt{\log L} = cL^4 \sqrt{\log L}.$$

The quadratic regime is valid because threshold times maximal subexponential scale divided by variance is $O(\eta \sqrt{\log h/L})$. Near the peak the exponent is at least $c\eta^2 F/L$. Both dominate $\log M = O(L)$, so the union over all boundaries has probability $1 - o(1)$. Each retained $q \leq (1 - 2\eta)h + 1$ is legal on this tube.

Retain also the original high-probability bounds

$$|X - \mathbb{E}X| \leq LH\sqrt{M}, \quad |\Delta| \leq L\sqrt{nL},$$

where X is outside degree and Δ the patch height gap. Total degree bias remains $O(HL)$. The missing degree divided by $r_0 Y$ tends to $(1 - m_*)/\alpha$; rounding, seeds, and fluctuations are negligible relative to $r_0 Y$. The exact integer patch in the pinned proof therefore applies with its fixed legality

margin around $q = (1 - m_*)Y[1 + o(1)]$. It imposes exact total degree and exact final height using (31.3), at logarithmic cost

$$O(r_0L + ML^3/r_0) = o(M_0/L^K).$$

The patch locations are fixed, deletion recovers the outside marked data, and fractional weights are at most one; this is the required weighted injection. The seeds cost only a fixed power of L .

For completeness, every other lower-construction error has a tunable rate:

$$\begin{aligned} \text{row normalizations: } & O(ML^{3/2-A_0/2}), \\ \text{interior summation by parts: } & O(ML\eta), \\ \text{central boundary terms: } & O(r_0L\eta), \\ \text{omitted central tilt and action: } & O(r_0L), \\ \text{Stieltjes quadrature: } & O(L^2). \end{aligned}$$

The first two are $o(M_0/L^K)$ by (32.12); the remaining ones have either a power saving in M or only fixed powers of L . Macro-duration rounding and degree-mean bias are included in the exact patch, and their deterministic action corrections are fixed powers of L . The negative logarithm of retained probability is $o(1)$. Consequently

$$D_n \leq \mathcal{A}_{V_\eta, h_0}(n) + o(M_0/L^K).$$

Compare this minimum with the ϕ_N arch with endpoints h_0 . Set $a = HL^{-C_1}$, $C_1 = 2K + 8$. On $h \geq a$, the row-root expansion and cutoff displacement give

$$|V_\eta(h) - \phi_N(h)| \leq \frac{C_0}{h} \left[\eta L + \frac{(1 + \log L)^{N+1}}{L^{N+1}} + n^{-c} L^{c'} \right].$$

Its integral along the smooth arch is $o(M_0/L^K)$ by (32.10). Below a , both potentials are at most $C_0 \log h/h$, so (32.9) bounds the comparison by $O(FL^{-C_1/2}) = o(M_0/L^K)$. Finally (32.11) replaces h_0 by h_b at a fixed-power-of- L cost. Hence

$$D_n \leq \mathcal{A}_N(n) + o(M_0/L^K). \quad (32.13)$$

32.4 Exact smooth-potential calibration

Keep $N = K + 3$, and put

$$\begin{aligned} \epsilon &= L^{-E}, \quad E = 2K + 8, \quad \delta = L^{-K-2}, \quad u = y + \epsilon, \\ U_n(y) &= \frac{n}{F} \left[\phi_N(Hu) - \frac{\alpha\delta}{Hu} \right] = \frac{\alpha}{Lu} [k_N(\log(Hu)) - \delta]. \end{aligned} \quad (32.14)$$

On any fixed $0 \leq y \leq R$, (32.6) gives

$$U_n \asymp u^{-1}, \quad -U'_n \asymp u^{-2}, \quad U''_n \asymp u^{-3}, \quad |U_n^{(j)}| \leq C_j u^{-j-1}.$$

The potential is positive, decreasing, and convex on the whole half-line for large n , because $H\epsilon \rightarrow \infty$.

Let y_n minimize $\int_0^1 [y'^2/(2v) + U_n(y)] dt$ over nonnegative paths with endpoints zero. A regularized Kepler trial bounds the action by a constant. Kinetic coercivity gives a uniform H^1 bound and uniform compactness, proving existence; convexity proves uniqueness. The first variation against nonnegative tests gives $y_n'' \leq vU'_n(y_n) < 0$ in distributions. Concavity and the endpoint values imply

strict interior positivity; the Euler equation then is $y_n'' = vU_n'(y_n) < 0$. The arch is symmetric and has a single peak in a fixed positive compact interval. Choose R above it.

The energy relation implies

$$\int_0^1 \frac{dt}{y_n(t) + \epsilon} \leq C_0. \quad (32.15)$$

Indeed, with Y its peak, $U_n(y) - U_n(Y) \geq c[(y + \epsilon)^{-1} - (Y + \epsilon)^{-1}]$; the rising reciprocal-height integral is at most a constant times

$$\sqrt{Y + \epsilon} \int_0^Y (y + \epsilon)^{-1/2} (Y - y)^{-1/2} dy \leq C_0.$$

All required derivatives grow only as fixed powers of L ; for example,

$$|y_n'| \leq C_0 \epsilon^{-1/2}, \quad |y_n''| \leq C_0 \epsilon^{-2}, \quad |y_n'''| \leq C_0 \epsilon^{-7/2}.$$

Define

$$\begin{aligned} p &= -y_n'/v, \quad A_1 = p' = -U_n'(y_n) > 0, \quad J_1(A_1) = \min_{0 \leq y \leq R} \{A_1 y + U_n(y)\}, \\ g(1) &= 0, \quad g' = vp^2/2 - J_1(p'), \quad f(t, y) = g(t) + p(t)y. \end{aligned}$$

Strict convexity places the minimizer in $J_1(p'(t))$ exactly at $y_n(t)$, including its boundary values. Thus, for all $0 \leq y \leq R$,

$$-f_t + v f_y^2/2 = J_1(A_1) - A_1 y \leq U_n(y). \quad (32.16)$$

Integration by parts using $y_n(0) = y_n(1) = 0$ gives the exact identity

$$f(0, 0) = \int_0^1 [y_n'^2/(2v) + U_n(y_n)] dt. \quad (32.17)$$

There is no perturbative Legendre-minimum error. Since $0 \leq J_1'(A_1) \leq R$, the displayed path bounds give $\|f\|_{C^2} \leq L^{P_0}$ for some fixed $P_0 = P_0(K)$. The constrained Legendre value is C^1 at its boundary transition, which suffices for g'' and this C^2 Taylor bound.

Let $\mathcal{A}_\epsilon$ denote the unperturbed physical ϕ_N action with endpoints $H\epsilon$. Evaluating the two functionals at each other's minimizers and using (32.15) gives

$$\mathcal{A}_\epsilon - C_0 M_0 \delta \leq Ff(0, 0) \leq \mathcal{A}_\epsilon.$$

Together with (32.11), this proves

$$Ff(0, 0) = \mathcal{A}_N(n) + O(F\sqrt{\epsilon} + M_0 \delta) = \mathcal{A}_N(n) + o(M_0/L^K). \quad (32.18)$$

32.5 Transformed-row contraction and upper coefficient bound

On legal states $1 \leq h \leq RH$, $q \leq h$, and $k + e \leq n - 1$, use the exact transformed weight

$$w(e, q, d) \exp\{F[f(k/n, h/H) - f((k + e)/n, (h + d)/H)]\}.$$

Set $\gamma = F/n$, $\beta = F/H$, so $\beta^2 = \gamma$ and $\gamma H = L$. Choose a fixed $T_0 > P_0 + 2$. On the good-step region $e \leq H L^{T_0}$, $|d| \leq \sqrt{H} L^{T_0}$, the C^2 Taylor error in the exponent is $n^{-1/3} L^{P_1}$ for fixed P_1 . Its linear tilts are $s = -\gamma f_t$ and $\theta = -\beta p$. By (31.2) and (32.16), at $r = h + H\epsilon$,

$$r\Psi(s, \theta) \leq \frac{r\gamma}{\alpha} U_n(h/H) + n^{-1/3} L^{P_2} = k_N(\log r) - \delta + n^{-1/3} L^{P_2}.$$

Enlarge the legal cutoff $q \leq h$ to the fractional cutoff r . Every analytic tilt and amplitude error is a negative fixed power of n times a fixed power of L . Since $r \geq H\epsilon$, (32.7) applies, and the good-step row is at most $1 - c\delta$, after the Taylor factor is included.

The discarded steps contribute $o(\delta)$ uniformly. The entire transformed exponent is bounded by $C_0\gamma L^{P_0}e + C_0\beta L^{P_0}|d|$. For $e > HL^{T_0}$, an extra fixed small positive degree tilt in the common analytic neighborhood gives $\exp[-cHL^{T_0} + C_0H]$. For $|d| > \sqrt{HL}^{T_0}$, use both signs of the extra height tilt $\theta_0 = cL^{T_0}/\sqrt{H}$. Its logarithmic bound, from the root expansion, is

$$-cL^{2T_0} + C_R c^2 L^{2T_0} + O(L^{P_0+T_0+1/2} + L^{P_0+1} + L).$$

The base-height quadratic term $O(L^{2P_0+1})$ is absorbed here since $T_0 > P_0 + 2$. Choose c small after R . Cubic errors are $n^{-1/3}$ times a fixed power of L , leaving $\exp[-c'L^{2T_0}]$. These unrestricted-destination bounds cover all omitted legal steps. The complete row is therefore at most $1 - c_0\delta$.

The positive Neumann sum is bounded by $1/(c_0\delta)$, with logarithm $O(\log L)$. The exact terminal factor is $\rho^{n-1-k}t_*^{1-h}$. Because $g(1) = 0$, terminal calibration only changes the coefficients of $n - 1 - k$ and $h - 1$ by $O(\gamma L^{P_0})$ and $O(\beta L^{P_0})$; these are negligible relative to the fixed negative coefficients $\log \rho$ and $-\log t_*$. The terminal factor is uniformly bounded, and the initial-height correction $\beta p(0)$ is $o(1)$. Hence

$$D_n^{[\leq RH]} \geq Ff(0, 0) - O(\log L).$$

Choose R also large enough that the exponent in (31.4) exceeds a fixed trial-action upper bound. The outside contribution is then exponentially smaller than the confined upper coefficient bound. Equations (32.18) and (32.13) complete the proof of Theorem 32.1.

32.6 An explicit error ledger

Let $Q_K = M_0/L^K$. With the choices above, the errors divided by Q_K are bounded as follows:

$$\begin{aligned} \text{normalization: } & O(L^{-K-19/2}), \\ \text{interior summation: } & O(L^{-2}), \\ \text{central boundaries: } & O(M^{-1/4}L^{-2}), \\ \text{patch and omitted action: } & O(M^{-1/4}L^{K+1} + M^{-3/4}L^{K+3}), \\ \text{bulk root truncation: } & O((1 + \log L)^{K+4}/L^4), \\ \text{bulk cutoff displacement: } & O(L^{-2}), \\ \text{small heights and softened endpoints: } & O(L^{-3}), \\ \text{upper potential slack: } & O(L^{-2}). \end{aligned}$$

Seeds, deterministic rounding, and Stieltjes quadrature are fixed powers of L , hence $o(Q_K)$. The renewal logarithm is $O(\log L)$, and retained-event logarithm is $o(1)$. Taylor and amplitude errors are algebraically small in n and are $o(\delta)$ at row level. No step silently promotes an unspecified $o(M_0)$ bound to arbitrary inverse-log precision.

33 Endpoint-safe expansion of the action

This part is a deterministic calculation. It applies to any fixed $\alpha > 0$, $v > 0$, and real c , with c_* in (32.1) replaced by c . Define Y_0, C as before and $\kappa = \log Y_0 - 2\log 3 + 2c$. Write $\mathcal{A}_{N,b}$ for the action with fixed lower endpoint b .

Theorem 33.1 (Finite-order action expansion). *For each fixed $J \geq 0$ and $N \geq J + 2$, choose a sufficiently large fixed b so that $V_N(h) = \alpha k_N(\log h)/h$ is positive, decreasing, and strictly convex on $[b, \infty)$. Then*

$$\frac{\mathcal{A}_{N,b}(n)}{CF} = \sum_{j=0}^J \frac{P_j(B)}{L^j} + O_{J,N} \left(\frac{(1 + \log L)^{m_{J,N}}}{L^{J+1}} \right) \quad (33.1)$$

for some finite $m_{J,N}$. The polynomials are independent of N, b, α, v, c and satisfy Theorem 30.1.

33.1 Exact peak parameterization

The direct method gives a minimizing path: a constant trial bounds the kinetic energy, which bounds its maximum; weak H^1 compactness and uniform convergence then give a minimizer. Strict convexity gives uniqueness. The first variation shows it lies above b in the interior and solves $h'' = vV'_N(h) < 0$. It is symmetric with one peak $Y > b$, and conservation of energy gives

$$\frac{h'^2}{2v} = V_N(h) - V_N(Y).$$

Conversely, the resulting arch at each peak is stationary and hence minimizing. Its duration is continuous and one-to-one in Y , ranges from zero to infinity, and is therefore increasing. Smooth dependence follows from the linearized boundary-value operator, whose quadratic form is

$$\int [\eta'^2/v + V''_N(h)\eta^2] dt > 0$$

for nonzero endpoint-vanishing variations.

Put $T = \log Y$, $z_0 = b/Y$, $k = k_N(T)$, and

$$q_T(z) = \frac{k_N(T + \log z)}{k}, \quad I_{\pm}(T) = \int_{z_0}^1 [q_T(z)/z - 1]^{\pm 1/2} dz.$$

Then the exact duration and action are

$$n(Y) = \sqrt{\frac{2}{v\alpha}} Y^{3/2} k^{-1/2} I_-(T), \quad (33.2)$$

$$\mathcal{A}(Y) = \sqrt{\frac{2\alpha}{v}} \sqrt{Yk} [I_-(T) + 2I_+(T)]. \quad (33.3)$$

The second formula follows also from $n\alpha k/Y + 2\sqrt{2\alpha/v}\sqrt{Yk}I_+$. Both limiting integrals are $\pi/2$, giving $Y \sim Y_0 H$ and $\mathcal{A} \sim CF$.

33.2 The lower endpoint must be removed before expansion

Fix a requested order and take $\delta_0 = T^{-C_2}$ with C_2 large enough after that order. The function k_N is positive and increasing on $[\log b, T]$ when b is sufficiently large. Thus, on $z_0 \leq z \leq \delta_0$,

$$c_0/T \leq q_T(z) \leq 1, \quad q_T(z)/z - 1 \geq q_T(z)/(2z).$$

The actual, unexpanded endpoint integrals satisfy

$$\int_{z_0}^{\delta_0} [q_T(z)/z - 1]^{1/2} dz = O(\sqrt{\delta_0}), \quad (33.4)$$

$$\int_{z_0}^{\delta_0} [q_T(z)/z - 1]^{-1/2} dz = O(\sqrt{T} \delta_0^{3/2}). \quad (33.5)$$

Every coefficient integrand extended from δ_0 down to zero has a tail bounded by a fixed finite sum of

$$\int_0^{\delta_0} z^{-1/2}(1 + |\log z|)^m dz = O(\sqrt{\delta_0}(1 + |\log \delta_0|)^m).$$

Increasing C_2 makes these tails smaller than any prescribed finite inverse power of T . In particular, no uniform large-height expansion at heights of order b has been assumed.

33.3 Cancellation at the peak and integrable moments

Factor

$$\frac{q_T(z)}{z} - 1 = \frac{1-z}{z}(1 + r_T(z)), \quad r_T(z) = \frac{k_N(T + \log z) - k_N(T)}{k_N(T)(1-z)}. \quad (33.6)$$

The apparent singularity at $z = 1$ is removable, with limit $-k'_N(T)/k_N(T)$. On $\delta_0 \leq z \leq 1$,

$$|r_T(z)| \leq \frac{C_0 |\log z|}{T(1-z)} \leq \frac{C_1(1 + |\log z|)}{T} = O(\log T/T).$$

For the second inequality, use $1-z \geq 1/2$ when $z \leq 1/2$ and $-z \log z \leq 1-z$ when $z \geq 1/2$. Finite Taylor expansion of the explicit smooth k_N about T and expansion of $1/k_N(T)$ are therefore uniform on this interval. Each numerator polynomial in $w = \log z$ has zero constant term. Taylor remainders retain a factor w , so division by $1-z$ adds no nonintegrable singularity. The binomial expansions of $(1 + r_T)^{\pm 1/2}$ have remainders T^{-J-1} times fixed powers of $1 + \log T$ at a requested order.

The leading weights are

$$w_-(z) = z^{1/2}(1-z)^{-1/2}, \quad w_+(z) = z^{-1/2}(1-z)^{1/2}.$$

All coefficient terms are linear combinations of

$$w_\sigma(z) \frac{(\log z)^m}{(1-z)^r}, \quad m \geq r. \quad (33.7)$$

They are integrable at $z = 1$ since $\log z = -(1-z) + O((1-z)^2)$, and at $z = 0$ since the worst power there is $z^{-1/2}$. Together with (33.4)–(33.5), this proves legitimate finite-order termwise integration with the stated uniform remainders.

The beta-moment rule is

$$\begin{aligned} M_-(r, m) &= \partial_a^m B(a, 1/2 - r)|_{a=3/2}, \\ M_+(r, m) &= \partial_a^m B(a, 3/2 - r)|_{a=1/2}. \end{aligned} \quad (33.8)$$

For a negative second beta argument this denotes analytic continuation of the derivative. The differentiated integrand itself is convergent by $m \geq r$; the formula does not assign values to divergent integrals. One may continue the convergent differentiated integral in the second argument, or integrate by parts using the order- m zero at $z = 1$. For exact computation without spurious Gamma poles, first use

$$B(a, b_0 - r) = B(a, b_0) \prod_{j=0}^{r-1} \frac{a + b_0 - r + j}{b_0 - r + j}, \quad b_0 \in \{1/2, 3/2\}, \quad (33.9)$$

and only then differentiate.

A duration derivative is needed for inversion, so it is useful to check it directly. On $z \leq \delta_0$, $q_T \geq c/T$ and $|\partial_T q_T| \leq C_0/T$ imply another $O(\sqrt{T}z^{1/2})$ integrand bound after differentiating

I_- . The moving lower-limit term is $O(\sqrt{T}(b/Y)^{3/2})$, exponentially small. The artificial cut adds no larger error. On the central interval, differentiating the removable expression (33.6) gives $O(T^{-2}(1 + \log T)^m)$ against the same integrable weight. Thus

$$I'_-(T) = O(T^{-2}(1 + \log T)^m) \quad \text{and} \quad \frac{d \log n}{dT} = \frac{3}{2} + O(1/T).$$

This proves an inverse derivative bounded away from zero; it does not differentiate an unspecified asymptotic remainder.

33.4 Fixed cores and moving cutoffs

For a positive decreasing convex potential, differentiation with respect to the common endpoint a gives the boundary-momentum identity

$$\frac{\partial \mathcal{A}_a(n)}{\partial a} = -2\sqrt{2/v}\sqrt{V(a) - V(Y_a(n))}.$$

Consequently,

$$0 \leq \mathcal{A}_b(n) - \mathcal{A}_a(n) \leq 2\sqrt{2/v} \int_b^a \sqrt{V(h)} dh = O(\sqrt{a \log a})$$

for the present potential. At $a = H\epsilon$ this is $O(F\sqrt{\epsilon})$. A fixed positive decreasing C^2 convex extension to $[0, b]$ exists, for example the quadratic matching V, V', V'' at b . Changing from endpoint b to endpoint zero for such an extension costs $O(1)$. Two fixed admissible core extensions therefore differ by $O(1)$ at fixed duration. A negative well or an n -dependent core is not an admissible substitute.

Uniformly for minimizing arches with $b \leq a = o(Y)$,

$$\int_0^n \frac{dt}{h(t)} = O(n/Y).$$

One may verify this by dividing by n/Y : the resulting ratio is $R(T)/I_-(T)$, where

$$R(T) = \int_{a/Y}^1 z^{-1} [q_T(z)/z - 1]^{-1/2} dz.$$

On $z \geq T^{-C_2}$, use a constant multiple of $z^{-1/2}(1 - z)^{-1/2}$; below it use $C_0\sqrt{T}z^{-1/2}$. Taking $C_2 > 1$ bounds $R(T)$ uniformly while I_- stays positive. The estimate remains uniform for $V_N - \alpha\eta_1/h$ with $\eta_1 = o(1)$, because positivity, decrease, and convexity persist. Comparison at each other's minimizers bounds the action perturbation by $O(M_0\eta_1)$.

34 A universal polynomial recurrence

The peak variable k makes the coefficient algorithm especially simple. Replace $\mathcal{S}$ by a sufficiently long finite truncation $\mathcal{S}_m$, and set

$$T(k) = 2k - 2\log k + 2\log \mathcal{S}_m(1/k) - 2c - 2\log 2. \tag{34.1}$$

Its derivative is $2 + O(1/k) > 0$. Its smooth inverse agrees with the explicit $k_N(T)$ to every order that can enter (33.1). More quantitatively, with $m \geq N + 2$, substitution of k_N in the implicit equation leaves residual $O(T^{-N-1}(1 + \log T)^{N+1})$. The implicit derivative is $1 + O(1/k)$, so the same bound holds for the difference of the two solutions; their first derivative difference is

$O(T^{-N-2}(1 + \log T)^{N+1})$. On the central interval the difference of their removable r_T expressions is $O(T^{-N-3}(1 + \log T)^{N+2})$, while peak normalizations differ relatively by $O(T^{-N-2}(1 + \log T)^{N+1})$. The lower tails are already controlled. Thus the finite smooth surrogate may be used for coefficient extraction without assuming differentiability of an exact discrete row root.

Let $r = 1/k$ be formal, $w = \log z$, and $\Delta(r, w) = k(T + w) - k(T)$. Subtracting two copies of (34.1) gives

$$\Delta = \frac{w}{2} + \log(1 + r\Delta) - \log \mathcal{S} \left(\frac{r}{1 + r\Delta} \right) + \log \mathcal{S}(r). \quad (34.2)$$

Iteration from $\Delta = w/2$ determines every coefficient. They are polynomials in w vanishing at $w = 0$. At every fixed order this is also a uniform analytic expansion for $|w| \leq C_2 \log k$, because its right side has Δ derivative $O(1/k)$.

Define formal series with constant term one by coefficientwise finite beta-moment integration:

$$J_\sigma(r) = \frac{2}{\pi} \int_0^1 w_\sigma(z) \left[1 + \frac{r\Delta(r, \log z)}{1 - z} \right]^{\sigma/2} dz, \quad \sigma = -1, +1. \quad (34.3)$$

Section 33 supplies the analytic justification; this notation does not assert convergence of an infinite integrated series. With $g = \log 2$, the first terms are

$$\begin{aligned} J_-(r) &= 1 + (1 - g)r + \frac{3}{2}(1 - g)r^2 + O(r^3), \\ J_+(r) &= 1 - gr - \frac{3}{2}gr^2 + O(r^3). \end{aligned}$$

Now put $x = 1/L$ and $q = 3k/L$. Taking logarithms of (33.2) and using (34.1) gives the universal implicit equation

$$q = 1 + x \left[\frac{3}{2}B + 3 \log 2 + \frac{7}{2} \log q - 3 \log \mathcal{S}(3x/q) - \log J_-(3x/q) \right]. \quad (34.4)$$

Eliminating Y between duration and action gives the final recurrence

$$\sum_{j \geq 0} P_j(B) x^j = q^{2/3} J_-(3x/q)^{-1/3} \frac{J_-(3x/q) + 2J_+(3x/q)}{3}. \quad (34.5)$$

To compute through x^J , truncate each series after degree J and perform finitely many iterations starting at $q = 1$. No summation of a divergent infinite Watson series is involved.

34.1 Why the recurrence is asymptotic rather than only formal

The leading duration relation reads

$$\log n = \frac{3}{2}T - \frac{1}{2} \log T + \text{constant} + O(\log T/T),$$

so the actual peak satisfies

$$T = \frac{2}{3}L + \frac{1}{3} \log L + \log Y_0 + O(\log L/L).$$

Finite substitution determines its inverse expansion. The direct derivative bound $d \log n / dT = 3/2 + O(1/T)$ implies, by the mean-value theorem, that an approximate inverse with residual $O(L^{-J-1}(1 + \log L)^m)$ has an error of the same order. Equivalently the derivative in k is $3 + O(1/k)$. The endpoint-safe integration estimates are uniform at the fixed order, and all finite algebraic operations preserve their remainders. This proves (33.1).

In (34.4), dependence on q beyond the constant term is multiplied by x . The coefficient of x has degree one in B ; for $j \geq 2$, the coefficient of x^j in q has degree at most $j - 1$. Equation (34.5) therefore gives $\deg P_j \leq j$. Its B^j term arises only from the j uses of $(3B/2)x$ in $q^{2/3}$, proving

$$[B^j]P_j = \binom{2/3}{j}(3/2)^j \neq 0.$$

For $J \geq 1$, apply Theorem 32.1 with $K = J - 1$ and $N = J + 2$, exactly meeting Theorem 33.1's requirement. Its error is $o(F/L^J)$. The leading-only case follows at once. This proves Theorem 30.1.

34.2 Explicit checked polynomials

The first five polynomials are

$$\begin{aligned} P_0(B) &= 1, \\ P_1(B) &= B, \\ P_2(B) &= -\frac{B^2}{4} + \frac{7B}{2} - 10, \\ P_3(B) &= \frac{4B^3 - 105B^2 + 774B - 2050 - 27\pi^2}{24}, \\ P_4(B) &= -\frac{1}{48}[7B^4 - 273B^3 + 3444B^2 - (19768 + 189\pi^2)B \\ &\quad + 1944\zeta(3) + 999\pi^2 + 46290]. \end{aligned}$$

They are exact expressions. The companion includes the recurrence generator and an independent direct-substitution verification of P_0 through P_4 in the original $T = \log Y$ duration and action integrals, without importing the generator. The second calculation derives its own local row polynomials and Gamma-jet beta moments; all duration residuals and coefficient differences vanish through L^{-4} , and the auxiliary constants cancel.

For a short transparent check of P_2 , the four convergent moments needed are

$$\begin{aligned} \int_0^1 z^{1/2}(1-z)^{-3/2} \log z \, dz &= -2\pi(1-g), \\ \int_0^1 z^{1/2}(1-z)^{-5/2} \log^2 z \, dz &= \frac{8\pi}{3}(1-g), \\ \int_0^1 z^{-1/2}(1-z)^{-1/2} \log z \, dz &= -2\pi g, \\ \int_0^1 z^{-1/2}(1-z)^{-3/2} \log^2 z \, dz &= 8\pi g. \end{aligned}$$

Set $a = 3B/2 + 3g$. Then

$$q = 1 + ax + \left[\frac{7}{2}a - \frac{33}{2} + 3g\right]x^2 + O(x^3),$$

while $W(r) = J_-(r)^{-1/3}[J_-(r) + 2J_+(r)]/3$ obeys

$$W(r) = 1 - \frac{2g}{3}r + \left(\frac{1}{9} - g - \frac{g^2}{9}\right)r^2 + O(r^3).$$

Substitution of $r = 3x/q$ in $q^{2/3}W(r)$ cancels every $\log 2$ term and yields $P_1 = B$ and $P_2 = -(B - 10)(B - 4)/4$.

35 The inverse threshold expansion

For $p > 1$ define

$$N(p) = \min\{n \geq 0 : a_n \geq p\}, \quad \lambda = \log \mu > 0, \quad T = \log p, \quad S = \log T.$$

The sequence is nondecreasing: append a copy of the last letter to an avoiding ascent sequence. This preserves the ascent condition and creates no new 120 occurrence, since any new occurrence using the appended letter would already use its identical predecessor. Deleting the final letter recovers the original word, so the map is injective. The asymptotic growth makes $N(p)$ finite for sufficiently large p .

Theorem 35.1 (All finite inverse orders). *Put*

$$\kappa_{\text{inv}} = \kappa - \frac{2}{3} \log \lambda, \quad B_{\text{inv}} = \kappa_{\text{inv}} + \frac{7}{3} \log S.$$

For every fixed integer $J \geq 0$,

$$\begin{aligned} N(p) &= \frac{T}{\lambda} + \frac{C}{\lambda^{4/3}} T^{1/3} S^{2/3} \sum_{j=0}^J \frac{P_j(B_{\text{inv}})}{S^j} \\ &\quad + o(T^{1/3} S^{2/3-J}) \quad (p \rightarrow \infty). \end{aligned} \tag{35.1}$$

The polynomials are exactly those of Theorem 30.1.

The power $\lambda^{-4/3}$ includes one factor λ^{-1} from inversion and one factor $\lambda^{-1/3}$ from the argument $n = T/\lambda$. The constant shift is $\kappa_{\text{inv}} = -2.39800354051534742875 \dots$

35.1 Covariance of the finite polynomial hierarchy

In the peak variable k , the logarithmic duration equation is

$$\begin{aligned} L &= 3k - \frac{7}{2} \log k - \frac{3}{2} \kappa - \frac{7}{2} \log 3 - 3 \log 2 \\ &\quad + 3 \log \mathcal{S}(1/k) + \log J_-(1/k). \end{aligned} \tag{35.2}$$

The parameters L and κ enter only through $L + 3\kappa/2$.

[**write, 2 October 2026**] Equation (35.2) has the same linear-logarithmic form, with $a = 3$, $b = -\frac{7}{2}$. The inverse theorem above reverses $\lambda n - D_n$ about $n = T/\lambda$ and brackets $N(p)$ by monotonicity, as in theorems `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the transseries volume. At $J = 1$ it is Corollary 28.1.

Elimination of the peak height gives

$$\frac{\mathcal{A}(n)}{C n^{1/3}} = (3k)^{2/3} J_-(1/k)^{-1/3} \frac{J_-(1/k) + 2J_+(1/k)}{3},$$

with no other L or κ dependence. These equations are interpreted at each finite order using the smooth surrogate and uniform remainders proved above.

Define the finite expression

$$f_J(L, \kappa) = L^{2/3} \sum_{j=0}^J \frac{P_j(\kappa + \frac{7}{3} \log L)}{L^j}.$$

For every fixed J, Q and κ in a compact interval,

$$f_J(L - \sigma, \kappa) - f_J(L, \kappa - 2\sigma/3) = O_{J,Q}(L^{2/3-J-1}(1 + \log L)^{m_J}) \quad (35.3)$$

uniformly for $|\sigma| \leq Q$. One proof expands the finite smooth peak equations one order further; their dependence on $L + 3\kappa/2$ gives the formal identity, and all action estimates are uniform on compact parameter sets. Alternatively, Taylor-expand the two finite polynomial-log expressions in σ/L . The peak identity matches every coefficient through degree J , and ordinary finite Taylor remainders give (35.3). No convergent infinite series is needed.

Equivalently, the polynomials satisfy

$$P'_{j+1}(B) = (1 - 3j/2)P_j(B) + \frac{7}{2}P'_j(B). \quad (35.4)$$

Indeed the formal function f depends on $L + 3\kappa/2$, hence $f_L = (2/3)f_\kappa$; coefficient comparison gives the identity. It determines P_{j+1} only up to an additive constant. The beta moments supply the remaining information.

35.2 A squeeze using smooth truncations only

Fix J and define

$$d_J(x) = Cx^{1/3}f_J(\log x, \kappa), \quad R_J(x) = x^{1/3}(\log x)^{2/3-J}.$$

The coefficient theorem says, at every sufficiently large integer,

$$\log a_n = \lambda n - d_J(n) + r_J(n), \quad r_J(n) = o(R_J(n)). \quad (35.5)$$

Put

$$x_0 = T/\lambda, \quad x_* = x_0 + d_J(x_0)/\lambda, \quad F_T = T^{1/3}S^{2/3}, \quad E_J = T^{1/3}S^{2/3-J}.$$

The explicit finite sum is smooth and, uniformly for $x = x_0 + O(F_T)$,

$$d_J(x) = O(F_T), \quad d'_J(x) = O(T^{-2/3}S^{2/3}).$$

The degree bounds on P_j prove these directly by differentiating finitely many terms. Therefore

$$d_J(x_*) - d_J(x_0) = O(T^{-1/3}S^{4/3}) = o(1), \quad \lambda x_* - d_J(x_*) = T + o(1).$$

Fix $\varepsilon > 0$ and set

$$n_- = \lfloor x_* - \varepsilon E_J \rfloor, \quad n_+ = \lceil x_* + \varepsilon E_J \rceil.$$

Both lie in $x_0 + O(F_T)$, and $E_J \rightarrow \infty$ at every fixed order, so integer rounding is $o(E_J)$. The derivative estimate gives $d_J(n_\pm) - d_J(x_*) = o(1)$. Also $r_J(n_\pm) = o(E_J)$: the defining eventual all-integer bound for r_J applies at these two indices, and $R_J(n_\pm)/E_J$ stays between fixed positive constants. No local regularity of the exact remainder is assumed. Substituting in (35.5),

$$\log a_{n_-} = T - \lambda \varepsilon E_J + o(E_J) < T, \quad \log a_{n_+} = T + \lambda \varepsilon E_J + o(E_J) > T$$

for large T . Monotonicity puts $N(p)$ between these indices; then letting $\varepsilon \downarrow 0$ proves $N(p) = x_* + o(E_J)$.

Finally apply (35.3) with $L = S$ and $\sigma = \log \lambda$:

$$\frac{d_J(T/\lambda)}{\lambda} = \frac{C}{\lambda^{4/3}}T^{1/3}f_J(S, \kappa_{\text{inv}}) + O(T^{1/3}S^{2/3-J-1}(1 + \log S)^{m_J}).$$

The last error is $o(E_J)$, proving (35.1). Only the smooth finite approximant was differentiated; the discrete deficit was never treated as a differentiable function. The proof gives no exact integer rounding rule and no $o(1)$ absolute inverse error.

36 Examples and reproducible checks

The first new layer can be written without polynomial notation:

$$\begin{aligned} \frac{D_n}{CF} = 1 + \frac{\frac{7}{3} \log L + \kappa}{L} \\ + \frac{-\frac{49}{36}(\log L)^2 + (\frac{49}{6} - \frac{7\kappa}{6}) \log L - \frac{\kappa^2}{4} + \frac{7\kappa}{2} - 10}{L^2} + o(L^{-2}). \end{aligned}$$

It contains three individually ordered logarithmic scales. The same expanded expression gives the inverse correction after replacing L by S and κ by κ_{inv} , with the prefactor in (35.1).

As a simple check of covariance through this layer, let $q_0 = \log \lambda$. Expansion of

$$(1 - q_0/S)^{2/3} \left[1 + \frac{\kappa + \frac{7}{3} \log(S - q_0)}{S - q_0} + \frac{P_2(\kappa + \frac{7}{3} \log(S - q_0))}{(S - q_0)^2} \right]$$

has first coefficient $B - 2q_0/3$ and second coefficient

$$P_2(B) + \frac{q_0}{3} B - \frac{7q_0}{3} - \frac{q_0^2}{9} = P_2(B - 2q_0/3).$$

Here $B = \kappa + 7 \log S/3$. The separate inverse-scaling script checks this identity exactly.

The independent analytic audit also evaluates the unexpanded fixed-core action integrals. With the scalar test choice $\alpha = v = 1$, $c = 0$, fixed $b = e^{20}$, and the local truncation through p_3 , let

$$\mathcal{R} = \left[\frac{\mathcal{A}}{CF} - 1 - \frac{B}{L} - \frac{P_2(B)}{L^2} \right] \frac{L^3}{1 + |B|^3}.$$

The quadrature output is

$T = \log Y$	$L = \log n(Y)$	$\mathcal{R}$
100	148.803293801692	-0.0561634466134
200	298.472530837447	-0.0420190156067
400	598.135684186915	-0.0304070996191
800	1197.79487846036	-0.0197229706770

This is a diagnostic for the action calculation, not coefficient data or a proof of the asymptotic limit. The analytic endpoint estimates and error bounds establish the theorem. The audit additionally checks the four beta-log moments by direct 65-digit quadrature.

The companion archive supplies four complementary verification routes:

1. The universal peak- k recurrence generates P_0 through P_4 with exact symbolic arithmetic and checks degree and leading coefficient identities.
2. Direct substitution into the original duration and action formulas independently checks P_2 , and a further independent implementation derives P_0 through P_4 using Gamma-jet moments.
3. An independent audit script checks local p_1, p_2, p_3 , global P_1, P_2 , convergent moment quadrature, and unexpanded action values.
4. Exact dependency manifests pin the unchanged third-order report and its complete second-order, sharp-deficit, and positive-walk foundations. Their inherited mathematical checks are replayed.

The README gives the commands for verification and rebuilding. The final integrated review binds the article sources and PDF by SHA-256; page rendering is checked separately for layout. A fresh extraction replay checks that the release is self-contained. These finite computational checks supplement the proofs rather than turning asymptotic statements into certified finite thresholds.

37 What remains open

Theorem 30.1 establishes all finite orders in one dominant sector of the logarithm. In this restricted sense it is an inverse-logarithmic asymptotic hierarchy, with both L^{-1} and $\log L$ present. The term “transseries” is sometimes used broadly for such structures, but an exponentially complete transseries has not been obtained here.

A multiplicative equivalent for a_n would need control of the logarithm down to $o(1)$ after every relevant subleading term is included. The remainder after any fixed order here may diverge. Thus even knowing all P_j does not identify a multiplicative amplitude or a power-law prefactor. For example, adding $\log n$ or a scale n^ε with fixed $0 < \varepsilon < 1/3$ to a logarithmic model changes none of the present finite-order statements, while changing its exponential substantially.

The following are concrete further questions.

1. Can the row-to-action reduction be sharpened to smaller powers of n , then ultimately to $O(1)$ or $o(1)$ precision? This would require finer discrete normalization, repair, endpoint, and fluctuation estimates than the tunable bounds used here.
2. What additional sectors occur below $n^{1/3}$ times inverse powers of $\log n$? A fluctuation determinant, a logarithmic term, and a multiplicative amplitude cannot be inferred from the present action hierarchy.
3. How do P_j and the hidden remainder constants grow as $j \rightarrow \infty$? The factorial Watson coefficients suggest examining optimal truncation or resummation, but no such result follows from fixed-order estimates alone.
4. Can effective constants and computable thresholds be proved? The present asymptotics do not certify finite integer values of $N(p)$ or guarantee an accuracy for a given numerical input.
5. Which other positive macro-kernels share the same universal polynomials after their own amplitude and action reductions are established? The deterministic calculation is universal in α, v, c , while the discrete reduction remains a separate proof obligation for each model.

Earlier dependencies retain their original historical scope and open-question statements. The present report supersedes only their question about this all-finite-orders dominant sector. It makes no additional literature-priority claim.

[write, 2 October 2026] In this report the “earlier dependencies” are Parts I–IV. The question this Part supersedes is the one about lower-order corrections and an all-orders inverse expansion (Part I, Section 9, and the closing sections of Parts II–IV). Their other open questions remain open: a multiplicative equivalent, amplitude or prefactor; an exponentially complete transseries; explicit thresholds.

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